

CHAPTER 1

Numbers and Functions

The subject of this course is “functions of one real variable” so we begin by wondering what a real number “really” is, and then, in the next section, what a function is.

1. What is a number?

1.1. Different kinds of numbers. The simplest numbers are the *positive integers*

$$1, 2, 3, 4, \dots$$

the number *zero*

$$0,$$

and the *negative integers*

$$\dots, -4, -3, -2, -1.$$

Together these form the integers or “whole numbers.”

Next, there are the numbers you get by dividing one whole number by another (nonzero) whole number. These are the so called fractions or *rational numbers* such as

$$\frac{1}{2}, \frac{1}{3}, \frac{2}{3}, \frac{1}{4}, \frac{2}{4}, \frac{3}{4}, \frac{4}{3}, \dots$$

or

$$-\frac{1}{2}, -\frac{1}{3}, -\frac{2}{3}, -\frac{1}{4}, -\frac{2}{4}, -\frac{3}{4}, -\frac{4}{3}, \dots$$

By definition, any whole number is a rational number (in particular zero is a rational number.)

You can add, subtract, multiply and divide any pair of rational numbers and the result will again be a rational number (provided you don’t try to divide by zero).

One day in middle school you were told that there are other numbers besides the rational numbers, and the first example of such a number is the square root of two. It has been known ever since the time of the greeks that no rational number exists whose square is exactly 2, i.e. you can’t find a fraction $\frac{m}{n}$ such that

$$\left(\frac{m}{n}\right)^2 = 2, \text{ i.e. } m^2 = 2n^2.$$

Nevertheless, if you compute x^2 for some values of x between 1 and 2, and check if you get more or less than 2, then it looks like there should be some number x between 1.4 and 1.5 whose square is exactly 2. So, we *assume* that there is such a number, and we call it the square root of 2, written as $\sqrt{2}$. This raises several questions. How do we know there really is a number between 1.4 and 1.5 for which $x^2 = 2$? How many other such numbers are we going to assume into existence? Do these new numbers obey the same algebra rules (like $a + b = b + a$) as the rational numbers? If we knew precisely what these numbers (like $\sqrt{2}$) were then we could perhaps answer such questions. It turns out to be rather difficult to give a precise description of what a number is, and in this course we won’t try to get anywhere near the bottom of this issue. Instead we will think of numbers as “infinite decimal expansions” as follows.

One can represent certain fractions as decimal fractions, e.g.

$$\frac{279}{25} = \frac{1116}{100} = 11.16.$$

x	x^2
1.2	1.44
1.3	1.69
1.4	$1.96 < 2$
1.5	$2.25 > 2$
1.6	2.56

Not all fractions can be represented as decimal fractions. For instance, expanding $\frac{1}{3}$ into a decimal fraction leads to an unending decimal fraction

$$\frac{1}{3} = 0.333\ 333\ 333\ 333\ 333\ \dots$$

It is impossible to write the complete decimal expansion of $\frac{1}{3}$ because it contains infinitely many digits. But we can describe the expansion: each digit is a three. An electronic calculator, which always represents numbers as **finite** decimal numbers, can never hold the number $\frac{1}{3}$ exactly.

Every fraction can be written as a decimal fraction which may or may not be finite. If the decimal expansion doesn't end, then it must repeat. For instance,

$$\frac{1}{7} = 0.142857\ 142857\ 142857\ 142857\ \dots$$

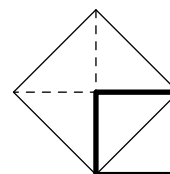
Conversely, any infinite repeating decimal expansion represents a rational number.

A **real number** is specified by a possibly unending decimal expansion. For instance,

$$\sqrt{2} = 1.414\ 213\ 562\ 373\ 095\ 048\ 801\ 688\ 724\ 209\ 698\ 078\ 569\ 671\ 875\ 376\ 9\ \dots$$

Of course you can never write *all* the digits in the decimal expansion, so you only write the first few digits and hide the others behind dots. To give a precise description of a real number (such as $\sqrt{2}$) you have to explain how you could *in principle* compute as many digits in the expansion as you would like. During the next three semesters of calculus we will not go into the details of how this should be done.

1.2. A reason to believe in $\sqrt{2}$. The Pythagorean theorem says that the hypotenuse of a right triangle with sides 1 and 1 must be a line segment of length $\sqrt{2}$. In middle or high school you learned something similar to the following geometric construction of a line segment whose length is $\sqrt{2}$. Take a square with side of length 1, and construct a new square one of whose sides is the diagonal of the first square. The figure you get consists of 5 triangles of equal area and by counting triangles you see that the larger square has exactly twice the area of the smaller square. Therefore the diagonal of the smaller square, being the side of the larger square, is $\sqrt{2}$ as long as the side of the smaller square.



Why are real numbers called real? All the numbers we will use in this first semester of calculus are “real numbers.” At some point (in 2nd semester calculus) it becomes useful to assume that there is a number whose square is -1 . No real number has this property since the square of any real number is positive, so it was decided to call this new imagined number “imaginary” and to refer to the numbers we already have (rationals, $\sqrt{2}$ -like things) as “real.”

1.3. The real number line and intervals. It is customary to visualize the real numbers as points on a straight line. We imagine a line, and choose one point on this line, which we call the **origin**. We also decide which direction we call “left” and hence which we call “right.” Some draw the number line vertically and use the words “up” and “down.”

To plot any real number x one marks off a distance x from the origin, to the right (up) if $x > 0$, to the left (down) if $x < 0$.

The **distance along the number line** between two numbers x and y is $|x - y|$. In particular, the distance is never a negative number.

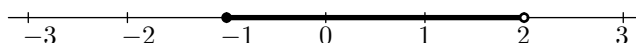


Figure 1. To draw the half open interval $[-1, 2)$ use a filled dot to mark the endpoint which is included and an open dot for an excluded endpoint.

Similarly, the *intersection of two sets $\mathcal{A}$ and $\mathcal{B}$* is the set of numbers which belong to both sets. This notation is used:

$$\mathcal{A} \cap \mathcal{B} = \{x \mid x \text{ belongs to both } \mathcal{A} \text{ and } \mathcal{B}\}$$

2. Exercises

1. What is the 2007th digit after the period in the expansion of $\frac{1}{7}$?

2. Which of the following fractions have finite decimal expansions?

$$a = \frac{2}{3}, \quad b = \frac{3}{25}, \quad c = \frac{276937}{15625}.$$

3. Draw the following sets of real numbers. Each of these sets is the union of one or more intervals. Find those intervals. Which of these sets are finite?

$$\mathcal{A} = \{x \mid x^2 - 3x + 2 \leq 0\}$$

$$\mathcal{B} = \{x \mid x^2 - 3x + 2 \geq 0\}$$

$$\mathcal{C} = \{x \mid x^2 - 3x > 3\}$$

$$\mathcal{D} = \{x \mid x^2 - 5 > 2x\}$$

$$\mathcal{E} = \{t \mid t^2 - 3t + 2 \leq 0\}$$

$$\mathcal{F} = \{\alpha \mid \alpha^2 - 3\alpha + 2 \geq 0\}$$

$$\mathcal{G} = (0, 1) \cup (5, 7]$$

$$\mathcal{H} = (\{1\} \cup \{2, 3\}) \cap (0, 2\sqrt{2})$$

$$\mathcal{Q} = \{\theta \mid \sin \theta = \frac{1}{2}\}$$

$$\mathcal{R} = \{\varphi \mid \cos \varphi > 0\}$$

4. Suppose $\mathcal{A}$ and $\mathcal{B}$ are intervals. Is it always true that $\mathcal{A} \cap \mathcal{B}$ is an interval? How about $\mathcal{A} \cup \mathcal{B}$?

5. Consider the sets

$$\mathcal{M} = \{x \mid x > 0\} \text{ and } \mathcal{N} = \{y \mid y > 0\}.$$

Are these sets the same?

6. **Group Problem.**

Write the numbers

$$x = 0.3131313131\dots, \quad y = 0.273273273273\dots$$

$$\text{and } z = 0.21541541541541541\dots$$

as fractions (i.e. write them as $\frac{m}{n}$, specifying m and n .)

(Hint: show that $100x = x + 31$. A similar trick works for y , but z is a little harder.)

7. **Group Problem.**

Is the number whose decimal expansion after the period consists only of nines, i.e.

$$x = 0.9999999999999999\dots$$

an integer?

3. Functions

Wherein we meet the main characters of this semester

3.1. Definition. To specify a *function* f you must

- (1) give a *rule* which tells you how to compute the value $f(x)$ of the function for a given real number x , and:
- (2) say for which real numbers x the rule may be applied.

The set of numbers for which a function is defined is called its *domain*. The set of all possible numbers $f(x)$ as x runs over the domain is called the *range* of the function. The rule must be *unambiguous*: the same x must always lead to the same $f(x)$.

For instance, one can define a function f by putting $f(x) = \sqrt{x}$ for all $x \geq 0$. Here the rule defining f is “take the square root of whatever number you’re given”, and the function f will accept all nonnegative real numbers.

The rule which specifies a function can come in many different forms. Most often it is a formula, as in the square root example of the previous paragraph. Sometimes you need a few formulas, as in

$$g(x) = \begin{cases} 2x & \text{for } x < 0 \\ x^2 & \text{for } x \geq 0 \end{cases} \quad \text{domain of } g = \text{all real numbers.}$$

Functions which are defined by different formulas on different intervals are sometimes called *piecewise defined functions*.

3.2. Graphing a function. You get the *graph of a function* f by drawing all points whose coordinates are (x, y) where x must be in the domain of f and $y = f(x)$.

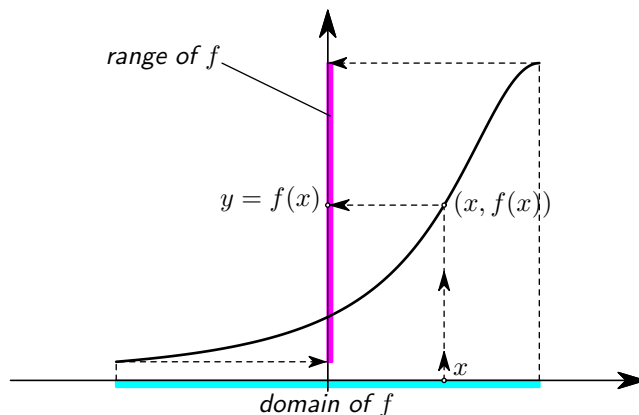


Figure 3. The graph of a function f . The domain of f consists of all x values at which the function is defined, and the range consists of all possible values f can have.

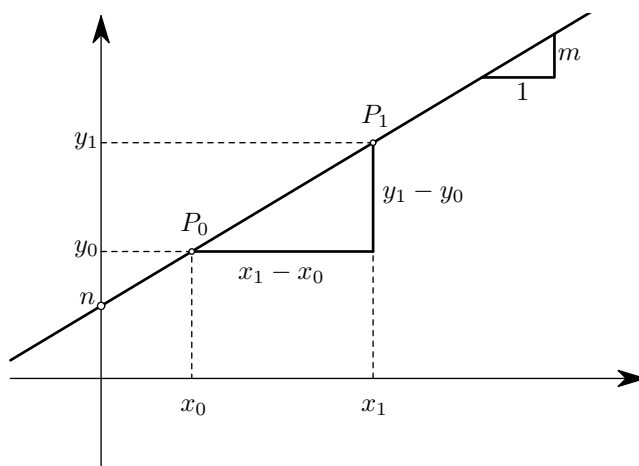


Figure 4. A straight line and its slope. The line is the graph of $f(x) = mx + n$. It intersects the y -axis at height n , and the ratio between the amounts by which y and x increase as you move from one point to another on the line is $\frac{y_1 - y_0}{x_1 - x_0} = m$.

3.3. Linear functions. A function which is given by the formula

$$f(x) = mx + n$$

where m and n are constants is called a **linear function**. Its graph is a straight line. The constants m and n are the **slope** and **y -intercept** of the line. Conversely, any straight line which is not vertical (i.e. not parallel to the y -axis) is the graph of a linear function. If you know two points (x_0, y_0) and (x_1, y_1) on the line, then then one can compute the slope m from the “rise-over-run” formula

$$m = \frac{y_1 - y_0}{x_1 - x_0}.$$

This formula actually contains a theorem from Euclidean geometry, namely it says that the ratio $(y_1 - y_0) : (x_1 - x_0)$ is the same for every pair of points (x_0, y_0) and (x_1, y_1) that you could pick on the line.

3.4. Domain and “biggest possible domain.” In this course we will usually not be careful about specifying the domain of the function. When this happens the domain is understood to be the set of all x for which the rule which tells you how to compute $f(x)$ is meaningful. For instance, if we say that h is the function

$$h(x) = \sqrt{x}$$

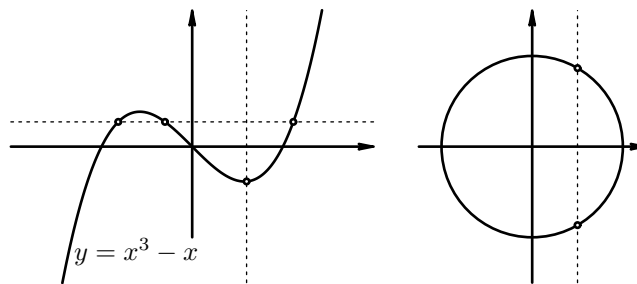


Figure 5. The graph of $y = x^3 - x$ fails the “horizontal line test,” but it passes the “vertical line test.” The circle fails both tests.

then the domain of h is understood to be the set of all nonnegative real numbers

$$\text{domain of } h = [0, \infty)$$

since $\sqrt{x}$ is well-defined for all $x \geq 0$ and undefined for $x < 0$.

A systematic way of finding the domain and range of a function for which you are only given a formula is as follows:

- The domain of f consists of all x for which $f(x)$ is well-defined (“makes sense”)
- The range of f consists of all y for which you can solve the equation $f(x) = y$.

3.5. Example – find the domain and range of $f(x) = 1/x^2$. The expression $1/x^2$ can be computed for all real numbers x except $x = 0$ since this leads to division by zero. Hence the domain of the function $f(x) = 1/x^2$ is

$$\text{“all real numbers except 0”} = \{x \mid x \neq 0\} = (-\infty, 0) \cup (0, \infty).$$

To find the range we ask “for which y can we solve the equation $y = f(x)$ for x ,” i.e. we for which y can you solve $y = 1/x^2$ for x ?

If $y = 1/x^2$ then we must have $x^2 = 1/y$, so first of all, since we have to divide by y , y can’t be zero. Furthermore, $1/y = x^2$ says that y must be positive. On the other hand, if $y > 0$ then $y = 1/x^2$ has a solution (in fact two solutions), namely $x = \pm 1/\sqrt{y}$. This shows that the range of f is

$$\text{“all positive real numbers”} = \{x \mid x > 0\} = (0, \infty).$$

3.6. Functions in “real life.” One can describe the motion of an object using a function. If some object is moving along a straight line, then you can define the following function: Let $x(t)$ be the distance from the object to a fixed marker on the line, at the time t . Here the domain of the function is the set of all times t for which we know the position of the object, and the rule is

Given t , measure the distance between the object and the marker at time t .

There are many examples of this kind. For instance, a biologist could describe the growth of a cell by defining $m(t)$ to be the mass of the cell at time t (measured since the birth of the cell). Here the domain is the interval $[0, T]$, where T is the life time of the cell, and the rule that describes the function is

Given t , weigh the cell at time t .

3.7. The Vertical Line Property. Generally speaking graphs of functions are curves in the plane but they distinguish themselves from arbitrary curves by the way they intersect vertical lines: **The graph of a function cannot intersect a vertical line “ $x = \text{constant}$ ” in more than one point.** The reason why this is true is very simple: if two points lie on a vertical line, then they have the same x coordinate, so if they also lie on the graph of a function f , then their y -coordinates must also be equal, namely $f(x)$.

3.8. Examples. The graph of $f(x) = x^3 - x$ “goes up and down,” and, even though it intersects several horizontal lines in more than one point, it intersects *every* vertical line in exactly one point.

The collection of points determined by the equation $x^2 + y^2 = 1$ is a circle. It is not the graph of a function since the vertical line $x = 0$ (the y -axis) intersects the graph in two points $P_1(0, 1)$ and $P_2(0, -1)$. See Figure 6.

4. Inverse functions and Implicit functions

For many functions the rule which tells you how to compute it is not an explicit formula, but instead an equation which you still must solve. A function which is defined in this way is called an “implicit function.”

4.1. Example. One can define a function f by saying that for each x the value of $f(x)$ is the solution y of the equation

$$x^2 + 2y - 3 = 0.$$

In this example you can solve the equation for y ,

$$y = \frac{3 - x^2}{2}.$$

Thus we see that the function we have defined is $f(x) = (3 - x^2)/2$.

Here we have two definitions of the same function, namely

- (i) “ $y = f(x)$ is defined by $x^2 + 2y - 3 = 0$,” and
- (ii) “ f is defined by $f(x) = (3 - x^2)/2$.”

The first definition is the implicit definition, the second is explicit. You see that with an “implicit function” it isn’t the function itself, but rather the way it was defined that’s implicit.

4.2. Another example: domain of an implicitly defined function. Define g by saying that for any x the value $y = g(x)$ is the solution of

$$x^2 + xy - 3 = 0.$$

Just as in the previous example one can then solve for y , and one finds that

$$g(x) = y = \frac{3 - x^2}{x}.$$

Unlike the previous example this formula does not make sense when $x = 0$, and indeed, for $x = 0$ our rule for g says that $g(0) = y$ is the solution of

$$0^2 + 0 \cdot y - 3 = 0, \text{ i.e. } y \text{ is the solution of } 3 = 0.$$

That equation has no solution and hence $x = 0$ does not belong to the domain of our function g .

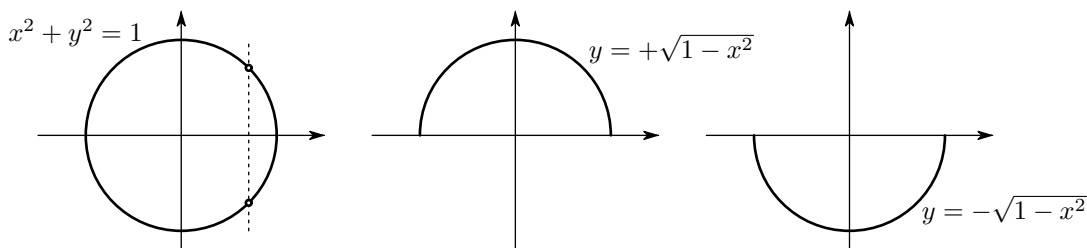


Figure 6. The circle determined by $x^2 + y^2 = 1$ is not the graph of a function, but it contains the graphs of the two functions $h_1(x) = \sqrt{1 - x^2}$ and $h_2(x) = -\sqrt{1 - x^2}$.

4.3. Example: the equation alone does not determine the function. Define $y = h(x)$ to be the solution of

$$x^2 + y^2 = 1.$$

If $x > 1$ or $x < -1$ then $x^2 > 1$ and there is no solution, so $h(x)$ is at most defined when $-1 \leq x \leq 1$. But when $-1 < x < 1$ there is another problem: not only does the equation have a solution, but it even has two solutions:

$$x^2 + y^2 = 1 \iff y = \sqrt{1 - x^2} \text{ or } y = -\sqrt{1 - x^2}.$$

The rule which defines a function must be unambiguous, and since we have not specified which of these two solutions is $h(x)$ the function is not defined for $-1 < x < 1$.

One can fix this by making a choice, but there are many possible choices. Here are three possibilities:

$$h_1(x) = \text{the nonnegative solution } y \text{ of } x^2 + y^2 = 1$$

$$h_2(x) = \text{the nonpositive solution } y \text{ of } x^2 + y^2 = 1$$

$$h_3(x) = \begin{cases} h_1(x) & \text{when } x < 0 \\ h_2(x) & \text{when } x \geq 0 \end{cases}$$

4.4. Why use implicit functions? In all the examples we have done so far we could replace the implicit description of the function with an explicit formula. This is not always possible or if it is possible the implicit description is much simpler than the explicit formula. For instance, you can define a function f by saying that $y = f(x)$ if and only if

$$(1) \quad y^3 + 3y + 2x = 0.$$

This means that the recipe for computing $f(x)$ for any given x is “solve the equation $y^3 + 3y + 2x = 0$.” E.g. to compute $f(0)$ you set $x = 0$ and solve $y^3 + 3y = 0$. The only solution is $y = 0$, so $f(0) = 0$. To compute $f(1)$ you have to solve $y^3 + 3y + 2 \cdot 1 = 0$, and if you’re lucky you see that $y = -1$ is the solution, and $f(1) = -1$.

In general, no matter what x is, the equation (1) turns out to have exactly one solution y (which depends on x , this is how you get the function f). Solving (1) is not easy. In the early 1500s Cardano and Tartaglia discovered a formula¹ for the solution. Here it is:

$$y = f(x) = \sqrt[3]{-x + \sqrt{1 + x^2}} - \sqrt[3]{x + \sqrt{1 + x^2}}.$$

The implicit description looks a lot simpler, and when we try to differentiate this function later on, it will be much easier to use “implicit differentiation” than to use the Cardano-Tartaglia formula directly.

4.5. Inverse functions. If you have a function f , then you can try to define a new function f^{-1} , the so-called **inverse function of f** , by the following prescription:

$$(2) \quad \text{For any given } x \text{ we say that } y = f^{-1}(x) \text{ if } y \text{ is the solution to the equation } f(y) = x.$$

So to find $y = f^{-1}(x)$ you solve the equation $x = f(y)$. If this is to define a function then the prescription (2) must be unambiguous and the equation $f(y) = x$ has to have a solution and cannot have more than one solution.

¹To see the solution and its history visit

http://www.gap-system.org/~history/HistTopics/Quadratic_etc_equations.html

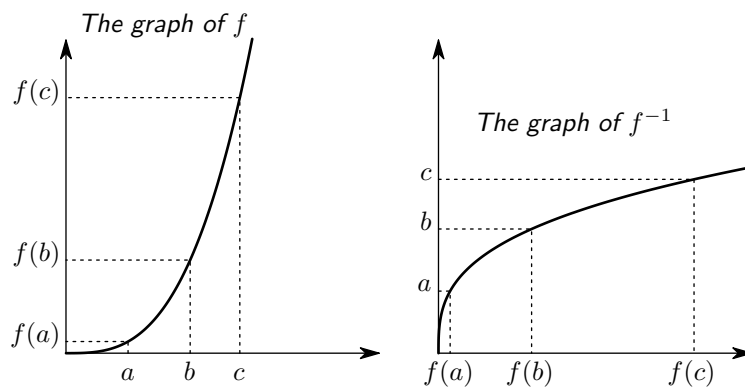


Figure 7. The graph of a function and its inverse are mirror images of each other.

4.6. Examples. Consider the function f with $f(x) = 2x + 3$. Then the equation $f(y) = x$ works out to be

$$2y + 3 = x$$

and this has the solution

$$y = \frac{x - 3}{2}.$$

So $f^{-1}(x)$ is defined for all x , and it is given by $f^{-1}(x) = (x - 3)/2$.

Next we consider the function $g(x) = x^2$ with domain all positive real numbers. To see for which x the inverse $g^{-1}(x)$ is defined we try to solve the equation $g(y) = x$, i.e. we try to solve $y^2 = x$. If $x < 0$ then this equation has no solutions since $y \geq 0$ for all y . But if $x \geq 0$ then $y = \sqrt{x}$ does have a solution, namely $y = \sqrt{x}$.

So we see that $g^{-1}(x)$ is defined for all nonnegative real numbers x , and that it is given by $g^{-1}(x) = \sqrt{x}$.

4.7. Inverse trigonometric functions. The familiar trigonometric functions Sine, Cosine and Tangent have inverses which are called arcsine, arccosine and arctangent.

$y = f(x)$		$x = f^{-1}(y)$	
$y = \sin x$	$(-\pi/2 \leq x \leq \pi/2)$	$x = \arcsin(y)$	$(-1 \leq y \leq 1)$
$y = \cos x$	$(0 \leq x \leq \pi)$	$x = \arccos(y)$	$(-1 \leq y \leq 1)$
$y = \tan x$	$(-\pi/2 < x < \pi/2)$	$x = \arctan(y)$	

The notations $\arcsin y = \sin^{-1} y$, $\arccos x = \cos^{-1} x$, and $\arctan u = \tan^{-1} u$ are also commonly used for the inverse trigonometric functions. We will avoid the $\sin^{-1} y$ notation because it is ambiguous. Namely, everybody writes the square of $\sin y$ as

$$(\sin y)^2 = \sin^2 y.$$

Replacing the 2's by -1 's would lead to

$$\arcsin y = \sin^{-1} y \stackrel{??}{=} (\sin y)^{-1} = \frac{1}{\sin y}, \quad \text{which is not true!}$$

5. Exercises

- 8.** The functions f and g are defined by

$$f(x) = x^2 \text{ and } g(s) = s^2.$$

Are f and g the same functions or are they different?

- 9.** Find a formula for the function f which is defined by

$$y = f(x) \iff x^2 y + y = 7.$$

What is the domain of f ?

- 10.** Find a formula for the function f which is defined by

$$y = f(x) \iff x^2 y - y = 6.$$

What is the domain of f ?

- 11.** Let f be the function defined by $y = f(x) \iff y$ is the largest solution of

$$y^2 = 3x^2 - 2xy.$$

Find a formula for f . What are the domain and range of f ?

12. Find a formula for the function f which is defined by

$$y = f(x) \iff 2x + 2xy + y^2 = 5 \text{ and } y > -x.$$

Find the domain of f .

13. Use a calculator to compute $f(1.2)$ in three decimals where f is the implicitly defined function from §4.4. (There are (at least) two different ways of finding $f(1.2)$)

14. **Group Problem.**

(a) True or false:

for all x one has $\sin(\arcsin x) = x$?

(b) True or false:

for all x one has $\arcsin(\sin x) = x$?

15. On a graphing calculator plot the graphs of the following functions, and explain the results. (Hint: first do the previous exercise.)

$$f(x) = \arcsin(\sin x), \quad -2\pi \leq x \leq 2\pi$$

$$g(x) = \arcsin(x) + \arccos(x), \quad 0 \leq x \leq 1$$

$$h(x) = \arctan \frac{\sin x}{\cos x}, \quad |x| < \pi/2$$

$$k(x) = \arctan \frac{\cos x}{\sin x}, \quad |x| < \pi/2$$

$$l(x) = \arcsin(\cos x), \quad -\pi \leq x \leq \pi$$

$$m(x) = \cos(\arcsin x), \quad -1 \leq x \leq 1$$

16. Find the inverse of the function f which is given by $f(x) = \sin x$ and **whose domain is** $\pi \leq x \leq 2\pi$. Sketch the graphs of both f and f^{-1} .

17. Find a number a such that the function $f(x) = \sin(x + \pi/4)$ with domain $a \leq x \leq a + \pi$ has an inverse. Give a formula for $f^{-1}(x)$ using the arcsine function.

18. Draw the graph of the function h_3 from §4.3.

19. A function f is given which satisfies

$$f(2x + 3) = x^2$$

for all real numbers x .

Compute

$$(a) f(0) \quad (b) f(3) \quad (c) f(x)$$

$$(d) f(y) \quad (e) f(f(2))$$

where x and y are arbitrary real numbers.

What are the range and domain of f ?

20. A function f is given which satisfies

$$f\left(\frac{1}{x+1}\right) = 2x - 12.$$

for all real numbers x .

Compute

$$(a) f(1) \quad (b) f(0) \quad (c) f(x)$$

$$(d) f(t) \quad (e) f(f(2))$$

where x and t are arbitrary real numbers.

What are the range and domain of f ?

21. Does there exist a function f which satisfies

$$f(x^2) = x + 1$$

for **all** real numbers x ?

* * *

The following exercises review precalculus material involving quadratic expressions $ax^2 + bx + c$ in one way or another.

22. Explain how you "complete the square" in a quadratic expression like $ax^2 + bx$.

23. Find the range of the following functions:

$$f(x) = 2x^2 + 3$$

$$g(x) = -2x^2 + 4x$$

$$h(x) = 4x + x^2$$

$$k(x) = 4 \sin x + \sin^2 x$$

$$\ell(x) = 1/(1 + x^2)$$

$$m(x) = 1/(3 + 2x + x^2).$$

24. **Group Problem.**

For each real number a we define a line ℓ_a with equation $y = ax + a^2$.

(a) Draw the lines corresponding to $a = -2, -1, -\frac{1}{2}, 0, \frac{1}{2}, 1, 2$.

(b) Does the point with coordinates $(3, 2)$ lie on one or more of the lines ℓ_a (where a can be any number, not just the five values from part (a))? If so, for which values of a does $(3, 2)$ lie on ℓ_a ?

(c) Which points in the plane lie on at least one of the lines ℓ_a ?

25. For which values of m and n does the graph of $f(x) = mx + n$ intersect the graph of $g(x) = 1/x$ in exactly one point and also contain the point $(-1, 1)$?

26. For which values of m and n does the graph of $f(x) = mx + n$ **not** intersect the graph of $g(x) = 1/x$?

Derivatives (1)

To work with derivatives you have to know what a limit is, but to motivate why we are going to study limits let's first look at the two classical problems that gave rise to the notion of a derivative: the tangent to a curve, and the instantaneous velocity of a moving object.

1. The tangent to a curve

Suppose you have a function $y = f(x)$ and you draw its graph. If you want to find the tangent to the graph of f at some given point on the graph of f , how would you do that?

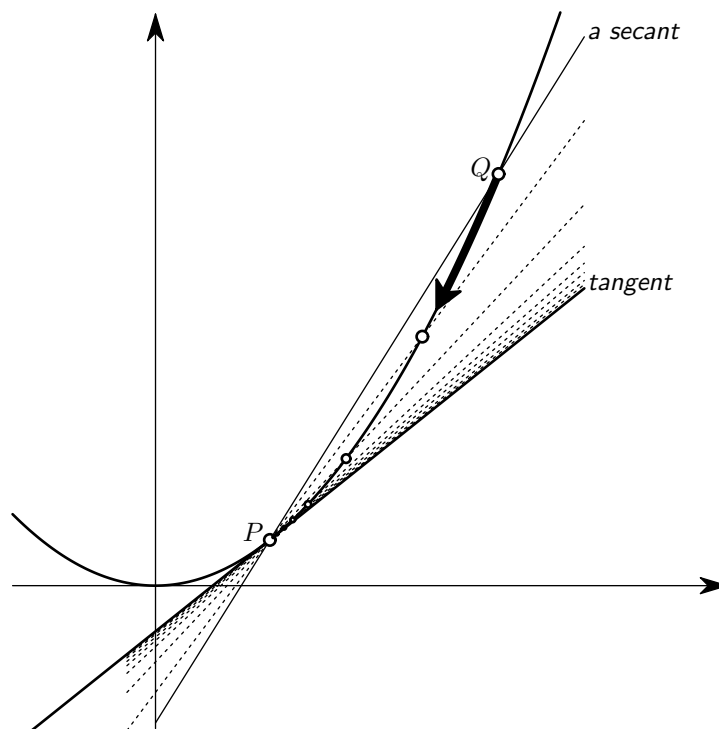


Figure 1. Constructing the tangent by letting $Q \rightarrow P$

Let P be the point on the graph at which you want to draw the tangent. If you are making a real paper and ink drawing you would take a ruler, make sure it goes through P and then turn it until it doesn't cross the graph anywhere else.

If you are using equations to describe the curve and lines, then you could pick a point Q on the graph and construct the line through P and Q ("construct" means "find an equation for"). This line is called a "secant," and it is of course not the tangent that you're looking for. But if you choose Q to be very close to P then the secant will be close to the tangent.

So this is our recipe for constructing the tangent through P : pick another point Q on the graph, find the line through P and Q , and see what happens to this line as you take Q closer and closer to P . The resulting secants will then get closer and closer to some line, and that line is the tangent.

We'll write this in formulas in a moment, but first let's worry about how close Q should be to P . We can't set Q equal to P , because then P and Q don't determine a line (you need *two* points to determine a line). If you choose Q different from P then you don't get the tangent, but at best something that is "close" to it. Some people have suggested that one should take Q "infinitely close" to P , but it isn't clear what that would mean. The concept of a limit is meant to solve this confusing problem.

2. An example – tangent to a parabola

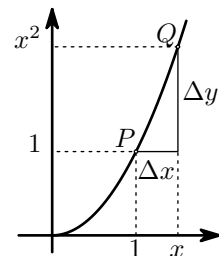
To make things more concrete, suppose that the function we had was $f(x) = x^2$, and that the point was $(1, 1)$. The graph of f is of course a parabola.

Any line through the point $P(1, 1)$ has equation

$$y - 1 = m(x - 1)$$

where m is the slope of the line. So instead of finding the equation of the secant and tangent lines we will find their slopes.

Let Q be the other point on the parabola, with coordinates (x, x^2) . We can "move Q around on the graph" by changing x . Whatever x we choose, it must be different from 1, for otherwise P and Q would be the same point. What we want to find out is how the line through P and Q changes if x is changed (and in particular, if x is chosen very close to a). Now, as one changes x one thing stays the same, namely, the secant still goes through P . So to describe the secant we only need to know its slope. By the "rise over run" formula, the slope of the secant line joining P and Q is



$$m_{PQ} = \frac{\Delta y}{\Delta x} \quad \text{where} \quad \Delta y = x^2 - 1 \quad \text{and} \quad \Delta x = x - 1.$$

By factoring $x^2 - 1$ we can rewrite the formula for the slope as follows

$$(3) \quad m_{PQ} = \frac{\Delta y}{\Delta x} = \frac{x^2 - 1}{x - 1} = \frac{(x - 1)(x + 1)}{x - 1} = x + 1.$$

As x gets closer to 1, the slope m_{PQ} , being $x + 1$, gets closer to the value $1 + 1 = 2$. We say that

the limit of the slope m_{PQ} as Q approaches P is 2.

In symbols,

$$\lim_{Q \rightarrow P} m_{PQ} = 2,$$

or, since Q approaching P is the same as x approaching 1,

$$(4) \quad \lim_{x \rightarrow 1} m_{PQ} = 2.$$

So we find that the tangent line to the parabola $y = x^2$ at the point $(1, 1)$ has equation

$$y - 1 = 2(x - 1), \text{ i.e. } y = 2x - 1.$$

A warning: you cannot substitute $x = 1$ in equation (3) to get (4) even though it looks like that's what we did. The reason why you can't do that is that when $x = 1$ the point Q coincides with the point P so "the line through P and Q " is not defined; also, if $x = 1$ then $\Delta x = \Delta y = 0$ so that the rise-over-run formula for the slope gives

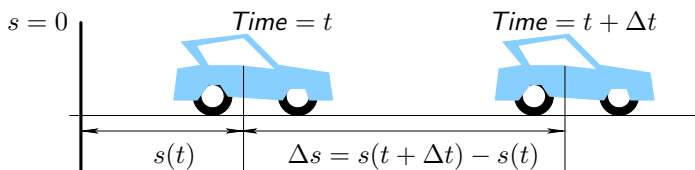
$$m_{PQ} = \frac{\Delta x}{\Delta y} = \frac{0}{0} = \text{undefined}.$$

It is only after the algebra trick in (3) that setting $x = 1$ gives something that is well defined. But if the intermediate steps leading to $m_{PQ} = x + 1$ aren't valid for $x = 1$ why should the final result mean anything for $x = 1$?

Something more complicated has happened. We did a calculation which is valid for all $x \neq 1$, and later looked at what happens if x gets “very close to 1.” This is the concept of a limit and we’ll study it in more detail later in this section, but first another example.

3. Instantaneous velocity

If you try to define “instantaneous velocity” you will again end up trying to divide zero by zero. Here is how it goes: When you are driving in your car the speedometer tells you how fast you are going, i.e. what your velocity is. What is this velocity? What does it mean if the speedometer says “50mph”?



We all know what **average velocity** is. Namely, if it takes you two hours to cover 100 miles, then your average velocity was

$$\frac{\text{distance traveled}}{\text{time it took}} = 50 \text{ miles per hour.}$$

This is not the number the speedometer provides you – it doesn’t wait two hours, measure how far you went and compute distance/time. If the speedometer in your car tells you that you are driving 50mph, then that should be your velocity **at the moment** that you look at your speedometer, i.e. “distance traveled over time it took” at the moment you look at the speedometer. But during the moment you look at your speedometer no time goes by (because a moment has no length) and you didn’t cover any distance, so your velocity at that moment is $\frac{0}{0}$, i.e. undefined. Your velocity at **any** moment is undefined. But then what is the speedometer telling you?

To put all this into formulas we need to introduce some notation. Let t be the time (in hours) that has passed since we got onto the road, and let $s(t)$ be the distance we have covered since then.

Instead of trying to find the velocity exactly at time t , we find a formula for the average velocity during some (short) time interval beginning at time t . We’ll write Δt for the length of the time interval.

At time t we have traveled $s(t)$ miles. A little later, at time $t + \Delta t$ we have traveled $s(t + \Delta t)$. Therefore during the time interval from t to $t + \Delta t$ we have moved $s(t + \Delta t) - s(t)$ miles. Our average velocity in that time interval is therefore

$$\frac{s(t + \Delta t) - s(t)}{\Delta t} \text{ miles per hour.}$$

The shorter you make the time interval, i.e. the smaller you choose Δt , the closer this number should be to the instantaneous velocity at time t .

So we have the following formula (definition, really) for the velocity at time t

$$(5) \quad v(t) = \lim_{\Delta t \rightarrow 0} \frac{s(t + \Delta t) - s(t)}{\Delta t}.$$

4. Rates of change

The two previous examples have much in common. If we ignore all the details about geometry, graphs, highways and motion, the following happened in both examples:

We had a function $y = f(x)$, and we wanted to know how much $f(x)$ changes if x changes. If you change x to $x + \Delta x$, then y will change from $f(x)$ to $f(x + \Delta x)$. The change in y is therefore

$$\Delta y = f(x + \Delta x) - f(x),$$

and the average rate of change is

$$(6) \quad \frac{\Delta y}{\Delta x} = \frac{f(x + \Delta x) - f(x)}{\Delta x}.$$

This is the average rate of change of f over the interval from x to $x + \Delta x$. To define **the rate of change of the function f at x** we let the length Δx of the interval become smaller and smaller, in the hope that the average rate of change over the shorter and shorter time intervals will get closer and closer to some number. If that happens then that “limiting number” is called the rate of change of f at x , or, the **derivative** of f at x . It is written as

$$(7) \quad f'(x) = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x}.$$

Derivatives and what you can do with them are what the first half of this semester is about. The description we just went through shows that to understand what a derivative is you need to know what a limit is. In the next chapter we’ll study limits so that we get a less vague understanding of formulas like (7).

5. Examples of rates of change

5.1. Acceleration as the rate at which velocity changes. As you are driving in your car your velocity does not stay constant, it changes with time. Suppose $v(t)$ is your velocity at time t (measured in miles per hour). You could try to figure out how fast your velocity is changing by measuring it at one moment in time (you get $v(t)$), then measuring it a little later (you get $v(t + \Delta t)$). You conclude that your velocity increased by $\Delta v = v(t + \Delta t) - v(t)$ during a time interval of length Δt , and hence

$$\left\{ \begin{array}{l} \text{average rate at which} \\ \text{your velocity changed} \end{array} \right\} = \frac{\Delta v}{\Delta t} = \frac{v(t + \Delta t) - v(t)}{\Delta t}.$$

This rate of change is called your *average acceleration* (over the time interval from t to $t + \Delta t$). Your *instantaneous acceleration* at time t is the limit of your average acceleration as you make the time interval shorter and shorter:

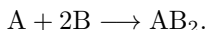
$$\{\text{acceleration at time } t\} = a = \lim_{\Delta t \rightarrow 0} \frac{v(t + \Delta t) - v(t)}{\Delta t}.$$

th the average and instantaneous accelerations are measured in “miles per hour per hour,” i.e. in

$$(\text{mi/h})/\text{h} = \text{mi/h}^2.$$

Or, if you had measured distances in meters and time in seconds then velocities would be measured in meters per second, and acceleration in meters per second per second, which is the same as meters per second², i.e. “meters per squared second.”

5.2. Reaction rates. Think of a chemical reaction in which two substances A and B react to form AB₂ according to the reaction



If the reaction is taking place in a closed reactor, then the “amounts” of A and B will be decreasing, while the amount of AB₂ will increase. Chemists write $[A]$ for the amount of “A” in the chemical reactor (measured in moles). Clearly $[A]$ changes with time so it defines a *function*. We’re mathematicians so we will write “ $[A](t)$ ” for the number of moles of A present at time t .

To describe how fast the amount of A is changing we consider the derivative of $[A]$ with respect to time, i.e.

$$[A]'(t) = \lim_{\Delta t \rightarrow 0} \frac{[A](t + \Delta t) - [A](t)}{\Delta t}.$$

This quantity is the rate of change of $[A]$. The notation “ $[A]'(t)$ ” is really only used by calculus professors. If you open a paper on chemistry you will find that the derivative is written in LEIBNIZ notation:

$$\frac{d[A]}{dt}$$

More on this in §1.2

How fast does the reaction take place? If you add more A or more B to the reactor then you would expect that the reaction would go faster, i.e. that more AB₂ is being produced per second. The law of *mass-action*

kinetics from chemistry states this more precisely. For our particular reaction it would say that the rate at which A is consumed is given by

$$\frac{d[A]}{dt} = k[A][B]^2,$$

in which the constant k is called the *reaction constant*. It's a constant that you could try to measure by timing how fast the reaction goes.

6. Exercises

27. Repeat the reasoning in §2 to find the slope at the point $(\frac{1}{2}, \frac{1}{4})$, or more generally at any point (a, a^2) on the parabola with equation $y = x^2$.

28. Repeat the reasoning in §2 to find the slope at the point $(\frac{1}{2}, \frac{1}{8})$, or more generally at any point (a, a^3) on the curve with equation $y = x^3$.

29. Group Problem.

Should you trust your calculator?

Find the slope of the tangent to the parabola $y = x^2$ at the point $(\frac{1}{3}, \frac{1}{9})$ (You have already done this: see exercise 27).

Instead of doing the algebra you could try to compute the slope by using a calculator. This exercise is about how you do that and what happens if you try (too hard).

Compute $\frac{\Delta y}{\Delta x}$ for various values of Δx :

$$\Delta x = 0.1, 0.01, 0.001, 10^{-6}, 10^{-12}.$$

As you choose Δx smaller your computed $\frac{\Delta y}{\Delta x}$ ought to get closer to the actual slope. Use at least 10 decimals and organize your results in a table like this:

Δx	$f(a)$	$f(a + \Delta x)$	Δy	$\Delta y / \Delta x$
0.1	...	...	...	...
0.01	...	...	...	...
0.001	...	...	...	...
10^{-6}	...	...	...	...
10^{-12}	...	...	...	...

Look carefully at the ratios $\Delta y / \Delta x$. Do they look like they are converging to some number? Compare the values of $\frac{\Delta y}{\Delta x}$ with the true value you got in the beginning of this problem.

30. Simplify the algebraic expressions you get when you compute Δy and $\Delta y / \Delta x$ for the following functions

(a) $y = x^2 - 2x + 1$

(b) $y = \frac{1}{x}$

(c) $y = 2^x$

31. Look ahead at Figure 3 in the next chapter. What is the derivative of $f(x) = x \cos \frac{\pi}{x}$ at the points A and B on the graph?

32. Suppose that some quantity y is a function of some other quantity x , and suppose that y is a mass, i.e. y is measured in pounds, and x is a length, measured in feet. What units do the increments Δy and Δx , and the derivative dy/dx have?

33. A tank is filling with water. The volume (in gallons) of water in the tank at time t (seconds) is $V(t)$. What units does the derivative $V'(t)$ have?

34. Group Problem.

Let $A(x)$ be the area of an equilateral triangle whose sides measure x inches.

(a) Show that $\frac{dA}{dx}$ has the units of a length.

(b) Which length does $\frac{dA}{dx}$ represent geometrically? [Hint: draw two equilateral triangles, one with side x and another with side $x + \Delta x$. Arrange the triangles so that they both have the origin as their lower left hand corner, and so their base is on the x-axis.]

35. Group Problem.

Let $A(x)$ be the area of a square with side x , and let $L(x)$ be the perimeter of the square (sum of the lengths of all its sides). Using the familiar formulas for $A(x)$ and $L(x)$ show that $A'(x) = \frac{1}{2}L(x)$.

Give a geometric interpretation that explains why $\Delta A \approx \frac{1}{2}L(x)\Delta x$ for small Δx .

36. Let $A(r)$ be the area enclosed by a circle of radius r , and let $L(r)$ be the length of the circle. Show that $A'(r) = L(r)$. (Use the familiar formulas from geometry for the area and perimeter of a circle.)

37. Let $V(r)$ be the volume enclosed by a sphere of radius r , and let $S(r)$ be its surface area. Show that $V'(r) = S(r)$. (Use the formulas $V(r) = \frac{4}{3}\pi r^3$ and $S(r) = 4\pi r^2$.)

Limits and Continuous Functions

1. Informal definition of limits

While it is easy to define precisely in a few words what a square root is ($\sqrt{a}$ is the positive number whose square is a) the definition of the limit of a function runs over several terse lines, and most people don't find it very enlightening when they first see it. (See §2.) So we postpone this for a while and fine tune our intuition for another page.

1.1. Definition of limit (1st attempt). If f is some function then

$$\lim_{x \rightarrow a} f(x) = L$$

is read “the limit of $f(x)$ as x approaches a is L .” It means that if you choose values of x which are close **but not equal** to a , then $f(x)$ will be close to the value L ; moreover, $f(x)$ gets closer and closer to L as x gets closer and closer to a .

The following alternative notation is sometimes used

$$f(x) \rightarrow L \quad \text{as} \quad x \rightarrow a;$$

(read “ $f(x)$ approaches L as x approaches a ” or “ $f(x)$ goes to L as x goes to a ”.)

1.2. Example. If $f(x) = x + 3$ then

$$\lim_{x \rightarrow 4} f(x) = 7,$$

is true, because if you substitute numbers x close to 4 in $f(x) = x + 3$ the result will be close to 7.

1.3. Example: substituting numbers to guess a limit. What (if anything) is

$$\lim_{x \rightarrow 2} \frac{x^2 - 2x}{x^2 - 4}?$$

Here $f(x) = (x^2 - 2x)/(x^2 - 4)$ and $a = 2$.

We first try to substitute $x = 2$, but this leads to

$$f(2) = \frac{2^2 - 2 \cdot 2}{2^2 - 4} = \frac{0}{0}$$

which does not exist. Next we try to substitute values of x close but not equal to 2. Table 1 suggests that $f(x)$ approaches 0.5.

x	$f(x)$	x	$g(x)$
3.000000	0.600000	1.000000	1.009990
2.500000	0.555556	0.500000	1.009980
2.100000	0.512195	0.100000	1.009899
2.010000	0.501247	0.010000	1.008991
2.001000	0.500125	0.001000	1.000000

Table 1. Finding limits by substituting values of x “close to a .” (Values of $f(x)$ and $g(x)$ rounded to six decimals.)

1.4. Example: Substituting numbers can suggest the wrong answer. The previous example shows that our first definition of “limit” is not very precise, because it says “ x close to a ,” but how close is close enough? Suppose we had taken the function

$$g(x) = \frac{101\,000x}{100\,000x + 1}$$

and we had asked for the limit $\lim_{x \rightarrow 0} g(x)$.

Then substitution of some “small values of x ” could lead us to believe that the limit is $1.000\dots$. Only when you substitute even smaller values do you find that the limit is 0 (zero)!

See also problem [29](#).

2. The formal, authoritative, definition of limit

The informal description of the limit uses phrases like “closer and closer” and “really very small.” In the end we don’t really know what they mean, although they are suggestive. “Fortunately” there is a good definition, i.e. one which is unambiguous and can be used to settle any dispute about the question of whether $\lim_{x \rightarrow a} f(x)$ equals some number L or not. Here is the definition. It takes a while to digest, so read it once, look at the examples, do a few exercises, read the definition again. Go on to the next sections. Throughout the semester come back to this section and read it again.

2.1. Definition of $\lim_{x \rightarrow a} f(x) = L$. We say that L is the limit of $f(x)$ as $x \rightarrow a$, if

- (1) $f(x)$ need not be defined at $x = a$, but it must be defined for all other x in some interval which contains a .
- (2) for every $\varepsilon > 0$ one can find a $\delta > 0$ such that for all x in the domain of f one has

$$(8) \quad |x - a| < \delta \text{ implies } |f(x) - L| < \varepsilon.$$

Why the absolute values? The quantity $|x - y|$ is the distance between the points x and y on the number line, and one can measure how close x is to y by calculating $|x - y|$. The inequality $|x - y| < \delta$ says that “the distance between x and y is less than δ ,” or that “ x and y are closer than δ .”

What are ε and δ ? The quantity ε is how close you would like $f(x)$ to be to its limit L ; the quantity δ is how close you have to choose x to a to achieve this. To prove that $\lim_{x \rightarrow a} f(x) = L$ you must assume that someone has given you an unknown $\varepsilon > 0$, and then find a positive δ for which (8) holds. The δ you find will depend on ε .

2.2. Show that $\lim_{x \rightarrow 5} 2x + 1 = 11$. We have $f(x) = 2x + 1$, $a = 5$ and $L = 11$, and the question we must answer is “how close should x be to 5 if want to be sure that $f(x) = 2x + 1$ differs less than ε from $L = 11$?”

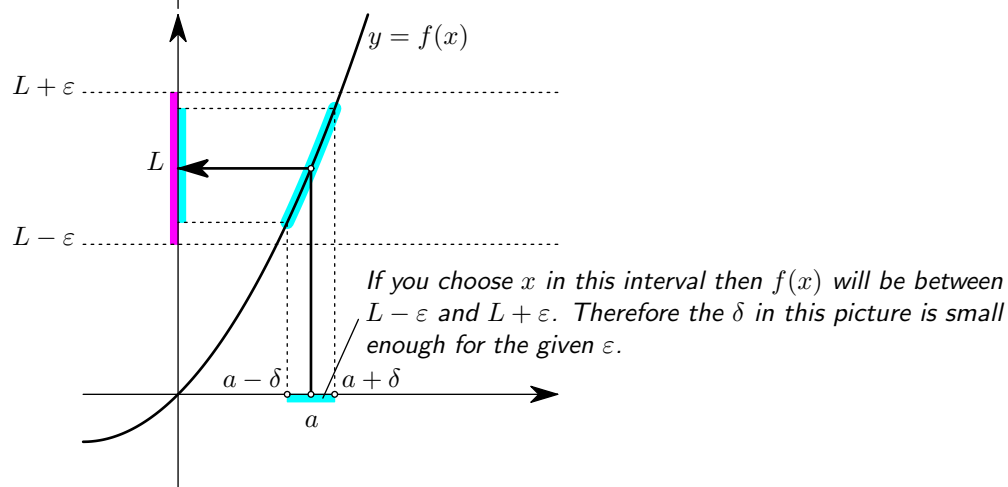
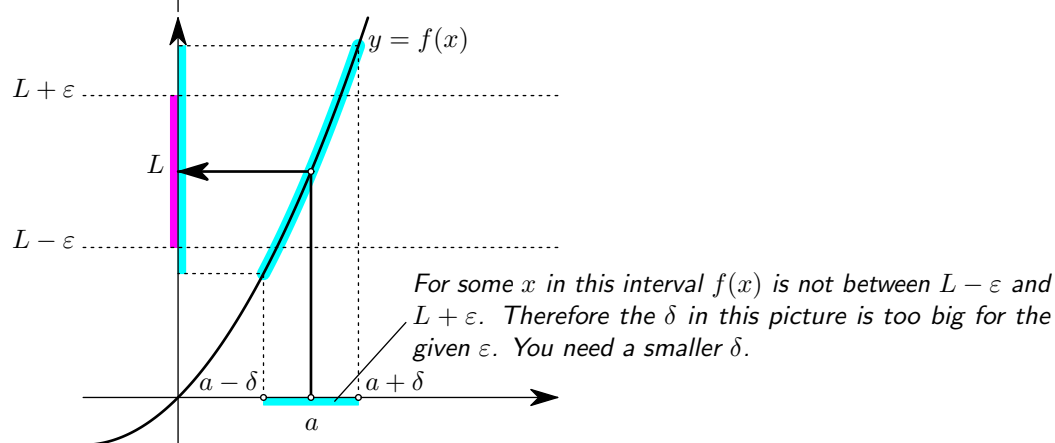
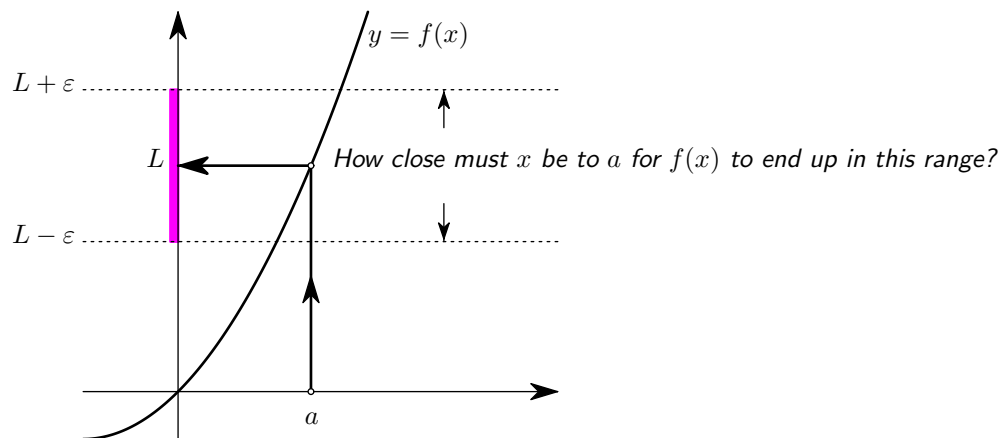
To figure this out we try to get an idea of how big $|f(x) - L|$ is:

$$|f(x) - L| = |(2x + 1) - 11| = |2x - 10| = 2 \cdot |x - 5| = 2 \cdot |x - a|.$$

So, if $2|x - a| < \varepsilon$ then we have $|f(x) - L| < \varepsilon$, i.e.

$$\text{if } |x - a| < \frac{1}{2}\varepsilon \text{ then } |f(x) - L| < \varepsilon.$$

We can therefore choose $\delta = \frac{1}{2}\varepsilon$. No matter what $\varepsilon > 0$ we are given our δ will also be positive, and if $|x - 5| < \delta$ then we can guarantee $|(2x + 1) - 11| < \varepsilon$. That shows that $\lim_{x \rightarrow 5} 2x + 1 = 11$.



2.3. The limit $\lim_{x \rightarrow 1} x^2 = 1$ and the “don’t choose $\delta > 1$ ” trick. We have $f(x) = x^2$, $a = 1$, $L = 1$, and again the question is, “how small should $|x - 1|$ be to guarantee $|x^2 - 1| < \varepsilon$?”

We begin by estimating the difference $|x^2 - 1|$

$$|x^2 - 1| = |(x - 1)(x + 1)| = |x + 1| \cdot |x - 1|.$$

Propagation of errors – another interpretation of ε and δ

According to the limit definition “ $\lim_{x \rightarrow R} \pi x^2 = A$ ” is true if *for every $\varepsilon > 0$ you can find a $\delta > 0$ such that $|x - R| < \delta$ implies $|\pi x^2 - A| < \varepsilon$* . Here's a more concrete situation in which ε and δ appear in exactly the same roles:

Suppose you are given a circle drawn on a piece of paper, and you want to know its area. You decide to measure its radius, R , and then compute the area of the circle by calculating

$$\text{Area} = \pi R^2.$$

The area is a function of the radius, and we'll call that function f :

$$f(x) = \pi x^2.$$

When you measure the radius R you will make an error, simply because you can never measure anything with infinite precision. Suppose that R is the real value of the radius, and that x is the number you measured. Then the size of the error you made is

$$\text{error in radius measurement} = |x - R|.$$

When you compute the area you also won't get the exact value: you would get $f(x) = \pi x^2$ instead of $A = f(R) = \pi R^2$. The error in your computed value of the area is

$$\text{error in area} = |f(x) - f(R)| = |f(x) - A|.$$

Now you can ask the following question:

Suppose you want to know the area with an error of at most ε , then what is the largest error that you can afford to make when you measure the radius?

The answer will be something like this: if you want the computed area to have an error of at most $|f(x) - A| < \varepsilon$, then the error in your radius measurement should satisfy $|x - R| < \delta$. You have to do the algebra with inequalities to compute δ when you know ε , as in the examples in this section.

You would expect that if your measured radius x is close enough to the real value R , then your computed area $f(x) = \pi x^2$ will be close to the real area A .

In terms of ε and δ this means that you would expect that no matter how accurately you want to know the area (i.e. how small you make ε) you can always achieve that precision by making the error in your radius measurement small enough (i.e. by making δ sufficiently small).

As x approaches 1 the factor $|x - 1|$ becomes small, and if the other factor $|x + 1|$ were a constant (e.g. 2 as in the previous example) then we could find δ as before, by dividing ε by that constant.

Here is a trick that allows you to replace the factor $|x + 1|$ with a constant. We hereby agree *that we always choose our δ so that $\delta \leq 1$* . If we do that, then we will always have

$$|x - 1| < \delta \leq 1, \text{ i.e. } |x - 1| < 1,$$

and x will always be between 0 and 2. Therefore

$$|x^2 - 1| = |x + 1| \cdot |x - 1| < 3|x - 1|.$$

If we now want to be sure that $|x^2 - 1| < \varepsilon$, then this calculation shows that we should require $3|x - 1| < \varepsilon$, i.e. $|x - 1| < \frac{1}{3}\varepsilon$. So we should choose $\delta \leq \frac{1}{3}\varepsilon$. We must also live up to our promise never to choose $\delta > 1$, so if we are handed an ε for which $\frac{1}{3}\varepsilon > 1$, then we choose $\delta = 1$ instead of $\delta = \frac{1}{3}\varepsilon$. To summarize, we are going to choose

$$\delta = \text{the smaller of } 1 \text{ and } \frac{1}{3}\varepsilon.$$

We have shown that if you choose δ this way, then $|x - 1| < \delta$ implies $|x^2 - 1| < \varepsilon$, no matter what $\varepsilon > 0$ is.

The expression “the smaller of a and b ” shows up often, and is abbreviated to $\min(a, b)$. We could therefore say that in this problem we will choose δ to be

$$\delta = \min(1, \frac{1}{3}\varepsilon).$$

2.4. Show that $\lim_{x \rightarrow 4} 1/x = 1/4$. Solution: We apply the definition with $a = 4$, $L = 1/4$ and $f(x) = 1/x$. Thus, for any $\varepsilon > 0$ we try to show that if $|x - 4|$ is small enough then one has $|f(x) - 1/4| < \varepsilon$.

We begin by estimating $|f(x) - 1/4|$ in terms of $|x - 4|$:

$$|f(x) - 1/4| = \left| \frac{1}{x} - \frac{1}{4} \right| = \left| \frac{4 - x}{4x} \right| = \frac{|x - 4|}{|4x|} = \frac{1}{|4x|} |x - 4|.$$

As before, things would be easier if $1/|4x|$ were a constant. To achieve that we again agree not to take $\delta > 1$. If we always have $\delta \leq 1$, then we will always have $|x - 4| < 1$, and hence $3 < x < 5$. How large can $1/|4x|$ be in this situation? Answer: the quantity $1/|4x|$ increases as you decrease x , so if $3 < x < 5$ then it will never be larger than $1/(4 \cdot 3) = \frac{1}{12}$.

We see that if we never choose $\delta > 1$, we will always have

$$|f(x) - \frac{1}{4}| \leq \frac{1}{12} |x - 4| \quad \text{for } |x - 4| < \delta.$$

To guarantee that $|f(x) - \frac{1}{4}| < \varepsilon$ we could therefore require

$$\frac{1}{12} |x - 4| < \varepsilon, \quad \text{i.e. } |x - 4| < 12\varepsilon.$$

Hence if we choose $\delta = 12\varepsilon$ or any smaller number, then $|x - 4| < \delta$ implies $|f(x) - 1/4| < \varepsilon$. Of course we have to honor our agreement never to choose $\delta > 1$, so our choice of δ is

$$\delta = \text{the smaller of } 1 \text{ and } 12\varepsilon = \min(1, 12\varepsilon).$$

3. Exercises

38. Group Problem.

Joe offers to make square sheets of paper for Bruce. Given $x > 0$ Joe plans to mark off a length x and cut out a square of side x . Bruce asks Joe for a square with area 4 square foot. Joe tells Bruce that he can't measure **exactly** 2 foot and the area of the square he produces will only be approximately 4 square foot. Bruce doesn't mind as long as the area of the square doesn't differ more than 0.01 square foot from what he really asked for (namely, 4 square foot).

(a) What is the biggest error Joe can afford to make when he marks off the length x ?

(b) Jen also wants square sheets, with area 4 square feet. However, she needs the error in the area to be less than 0.00001 square foot. (She's paying).

How accurate must Joe measure the side of the squares he's going to cut for Jen?

Use the ε - δ definition to prove the following limits

39. $\lim_{x \rightarrow 1} 2x - 4 = 6$

40. $\lim_{x \rightarrow 2} x^2 = 4$.

41. $\lim_{x \rightarrow 2} x^2 - 7x + 3 = -7$

42. $\lim_{x \rightarrow 3} x^3 = 27$

43. $\lim_{x \rightarrow 2} x^3 + 6x^2 = 32$.

44. $\lim_{x \rightarrow 4} \sqrt{x} = 2$.

45. $\lim_{x \rightarrow 3} \sqrt{x+6} = 9$.

46. $\lim_{x \rightarrow 2} \frac{1+x}{4+x} = \frac{1}{2}$.

47. $\lim_{x \rightarrow 1} \frac{2-x}{4-x} = \frac{1}{3}$.

48. $\lim_{x \rightarrow 3} \frac{x}{6-x} = 1$.

49. $\lim_{x \rightarrow 0} \sqrt{|x|} = 0$

50. Group Problem.

(Joe goes cubic.) Joe is offering to build cubes of side x . Airline regulations allow you take a cube on board provided its volume and surface area add up to less than 33 (everything measured in feet). For instance, a cube with 2 foot sides has volume+area equal to $2^3 + 6 \times 2^2 = 32$.

If you ask Joe to build a cube whose volume plus total surface area is 32 cubic feet with an error of at most ε , then what error can he afford to make when he measures the side of the cube he's making?

51. Our definition of a derivative in (7) contains a limit. What is the function " f " there, and what is the variable?

4. Variations on the limit theme

Not all limits are "for $x \rightarrow a$." here we describe some possible variations on the concept of limit.

4.1. Left and right limits. When we let “ x approach a ” we allow x to be both larger or smaller than a , as long as x gets close to a . If we explicitly want to study the behaviour of $f(x)$ as x approaches a through values larger than a , then we write

$$\lim_{x \searrow a} f(x) \text{ or } \lim_{x \rightarrow a+} f(x) \text{ or } \lim_{x \rightarrow a+0} f(x) \text{ or } \lim_{x \rightarrow a, x > a} f(x).$$

All four notations are in use. Similarly, to designate the value which $f(x)$ approaches as x approaches a through values below a one writes

$$\lim_{x \nearrow a} f(x) \text{ or } \lim_{x \rightarrow a-} f(x) \text{ or } \lim_{x \rightarrow a-0} f(x) \text{ or } \lim_{x \rightarrow a, x < a} f(x).$$

The precise definition of right limits goes like this:

4.2. Definition of right-limits. *Let f be a function. Then*

$$(9) \quad \lim_{x \searrow a} f(x) = L.$$

means that for every $\varepsilon > 0$ one can find a $\delta > 0$ such that

$$a < x < a + \delta \implies |f(x) - L| < \varepsilon$$

holds for all x in the domain of f .

The left-limit, i.e. the one-sided limit in which x approaches a through values less than a is defined in a similar way. The following theorem tells you how to use one-sided limits to decide if a function $f(x)$ has a limit at $x = a$.

4.3. Theorem. *If both one-sided limits*

$$\lim_{x \searrow a} f(x) = L_+, \text{ and } \lim_{x \nearrow a} f(x) = L_-$$

exist, then

$$\lim_{x \rightarrow a} f(x) \text{ exists } \iff L_+ = L_-.$$

In other words, if a function has both left- and right-limits at some $x = a$, then that function has a limit at $x = a$ if the left- and right-limits are equal.

4.4. Limits at infinity. Instead of letting x approach some finite number, one can let x become “larger and larger” and ask what happens to $f(x)$. If there is a number L such that $f(x)$ gets arbitrarily close to L if one chooses x sufficiently large, then we write

$$\lim_{x \rightarrow \infty} f(x) = L, \text{ or } \lim_{x \uparrow \infty} f(x) = L, \text{ or } \lim_{x \nearrow \infty} f(x) = L.$$

(“The limit for x going to infinity is L .”)

4.5. Example – Limit of $1/x$. The larger you choose x , the smaller its reciprocal $1/x$ becomes. Therefore, it seems reasonable to say

$$\lim_{x \rightarrow \infty} \frac{1}{x} = 0.$$

Here is the precise definition:

4.6. Definition of limit at ∞ . *Let f be some function which is defined on some interval $x_0 < x < \infty$. If there is a number L such that for every $\varepsilon > 0$ one can find an A such that*

$$x > A \implies |f(x) - L| < \varepsilon$$

for all x , then we say that the limit of $f(x)$ for $x \rightarrow \infty$ is L .

The definition is very similar to the original definition of the limit. Instead of δ which specifies how close x should be to a , we now have a number A which says how large x should be, which is a way of saying “how close x should be to infinity.”

4.7. Example – Limit of $1/x$ (again) . To *prove* that $\lim_{x \rightarrow \infty} 1/x = 0$ we apply the definition to $f(x) = 1/x$, $L = 0$.

For given $\varepsilon > 0$ we need to show that

$$(10) \quad \left| \frac{1}{x} - L \right| < \varepsilon \text{ for all } x > A$$

provided we choose the right A .

How do we choose A ? A is not allowed to depend on x , but it may depend on ε .

If we assume for now that we will only consider positive values of x , then (10) simplifies to

$$\frac{1}{x} < \varepsilon$$

which is equivalent to

$$x > \frac{1}{\varepsilon}.$$

This tells us how to choose A . Given any positive ε , we will simply choose

$$A = \frac{1}{\varepsilon}$$

Then one has $|\frac{1}{x} - 0| = \frac{1}{x} < \varepsilon$ for all $x > A$. Hence we have proved that $\lim_{x \rightarrow \infty} 1/x = 0$.

5. Properties of the Limit

The precise definition of the limit is not easy to use, and fortunately we won't use it very often in this class. Instead, there are a number of properties that limits have which allow you to compute them without having to resort to "epsilon-ness."

The following properties also apply to the variations on the limit from 4. I.e. the following statements remain true if one replaces each limit by a one-sided limit, or a limit for $x \rightarrow \infty$.

Limits of constants and of x . If a and c are constants, then

$$(P_1) \quad \lim_{x \rightarrow a} c = c$$

and

$$(P_2) \quad \lim_{x \rightarrow a} x = a.$$

Limits of sums, products and quotients. Let F_1 and F_2 be two given functions whose limits for $x \rightarrow a$ we know,

$$\lim_{x \rightarrow a} F_1(x) = L_1, \quad \lim_{x \rightarrow a} F_2(x) = L_2.$$

Then

$$(P_3) \quad \lim_{x \rightarrow a} (F_1(x) + F_2(x)) = L_1 + L_2,$$

$$(P_4) \quad \lim_{x \rightarrow a} (F_1(x) - F_2(x)) = L_1 - L_2,$$

$$(P_5) \quad \lim_{x \rightarrow a} (F_1(x) \cdot F_2(x)) = L_1 \cdot L_2$$

Finally, if $\lim_{x \rightarrow a} F_2(x) \neq 0$,

$$(P_6) \quad \lim_{x \rightarrow a} \frac{F_1(x)}{F_2(x)} = \frac{L_1}{L_2}.$$

In other words the limit of the sum is the sum of the limits, etc. One can prove these laws using the definition of limit in §2 but we will not do this here. However, I hope these laws seem like common sense: if, for x close to a , the quantity $F_1(x)$ is close to L_1 and $F_2(x)$ is close to L_2 , then certainly $F_1(x) + F_2(x)$ should be close to $L_1 + L_2$.

There are two more properties of limits which we will add to this list later on. They are the “Sandwich Theorem” (§9) and the substitution theorem (§10).

6. Examples of limit computations

6.1. Find $\lim_{x \rightarrow 2} x^2$. One has

$$\begin{aligned}\lim_{x \rightarrow 2} x^2 &= \lim_{x \rightarrow 2} x \cdot x \\ &= \left(\lim_{x \rightarrow 2} x \right) \cdot \left(\lim_{x \rightarrow 2} x \right) && \text{by } (P_5) \\ &= 2 \cdot 2 = 4.\end{aligned}$$

Similarly,

$$\begin{aligned}\lim_{x \rightarrow 2} x^3 &= \lim_{x \rightarrow 2} x \cdot x^2 \\ &= \left(\lim_{x \rightarrow 2} x \right) \cdot \left(\lim_{x \rightarrow 2} x^2 \right) && (P_5) \text{ again} \\ &= 2 \cdot 4 = 8,\end{aligned}$$

and, by (P_4)

$$\lim_{x \rightarrow 2} x^2 - 1 = \lim_{x \rightarrow 2} x^2 - \lim_{x \rightarrow 2} 1 = 4 - 1 = 3,$$

and, by (P_4) again,

$$\lim_{x \rightarrow 2} x^3 - 1 = \lim_{x \rightarrow 2} x^3 - \lim_{x \rightarrow 2} 1 = 8 - 1 = 7,$$

Putting all this together, one gets

$$\lim_{x \rightarrow 2} \frac{x^3 - 1}{x^2 - 1} = \frac{2^3 - 1}{2^2 - 1} = \frac{8 - 1}{4 - 1} = \frac{7}{3}$$

because of (P_6) . To apply (P_6) we must check that the denominator (“ L_2 ”) is not zero. Since the denominator is 3 everything is OK, and we were allowed to use (P_6) .

6.2. Try the examples 1.3 and 1.4 using the limit properties. To compute $\lim_{x \rightarrow 2} (x^2 - 2x)/(x^2 - 4)$ we first use the limit properties to find

$$\lim_{x \rightarrow 2} x^2 - 2x = 0 \text{ and } \lim_{x \rightarrow 2} x^2 - 4 = 0.$$

to complete the computation we would like to apply the last property (P_6) about quotients, but this would give us

$$\lim_{x \rightarrow 2} f(x) = \frac{0}{0}.$$

The denominator is zero, so we were not allowed to use (P_6) (and the result doesn’t mean anything anyway). We have to do something else.

The function we are dealing with is a **rational function**, which means that it is the quotient of two polynomials. For such functions there is an algebra trick which always allows you to compute the limit even if you first get $\frac{0}{0}$. The thing to do is to divide numerator and denominator by $x - 2$. In our case we have

$$x^2 - 2x = (x - 2) \cdot x, \quad x^2 - 4 = (x - 2) \cdot (x + 2)$$

so that

$$\lim_{x \rightarrow 2} f(x) = \lim_{x \rightarrow 2} \frac{(x - 2) \cdot x}{(x - 2) \cdot (x + 2)} = \lim_{x \rightarrow 2} \frac{x}{x + 2}.$$

After this simplification we **can** use the properties $(P_{\dots})$ to compute

$$\lim_{x \rightarrow 2} f(x) = \frac{2}{2 + 2} = \frac{1}{2}.$$

6.3. Example – Find $\lim_{x \rightarrow 2} \sqrt{x}$. Of course, you would think that $\lim_{x \rightarrow 2} \sqrt{x} = \sqrt{2}$ and you can indeed prove this using δ & ε (See problem 44.) But is there an easier way? There is nothing in the limit properties which tells us how to deal with a square root, and using them we can't even prove that there is a limit. However, if you *assume* that the limit exists then the limit properties allow us to find this limit.

The argument goes like this: suppose that there is a number L with

$$\lim_{x \rightarrow 2} \sqrt{x} = L.$$

Then property (P_5) implies that

$$L^2 = \left(\lim_{x \rightarrow 2} \sqrt{x}\right) \cdot \left(\lim_{x \rightarrow 2} \sqrt{x}\right) = \lim_{x \rightarrow 2} \sqrt{x} \cdot \sqrt{x} = \lim_{x \rightarrow 2} x = 2.$$

In other words, $L^2 = 2$, and hence L must be either $\sqrt{2}$ or $-\sqrt{2}$. We can reject the latter because whatever x does, its squareroot is always a positive number, and hence it can never “get close to” a negative number like $-\sqrt{2}$.

Our conclusion: if the limit exists, then

$$\lim_{x \rightarrow 2} \sqrt{x} = \sqrt{2}.$$

The result is not surprising: if x gets close to 2 then $\sqrt{x}$ gets close to $\sqrt{2}$.

6.4. Example – The derivative of $\sqrt{x}$ at $x = 2$. Find

$$\lim_{x \rightarrow 2} \frac{\sqrt{x} - \sqrt{2}}{x - 2}$$

assuming the result from the previous example.

Solution: The function is a fraction whose numerator and denominator vanish when $x = 2$, i.e. the limit is of the form $\frac{0}{0}$. We use the same algebra trick as before, namely we factor numerator and denominator:

$$\frac{\sqrt{x} - \sqrt{2}}{x - 2} = \frac{\sqrt{x} - \sqrt{2}}{(\sqrt{x} - \sqrt{2})(\sqrt{x} + \sqrt{2})} = \frac{1}{\sqrt{x} + \sqrt{2}}.$$

Now one can use the limit properties to compute

$$\lim_{x \rightarrow 2} \frac{\sqrt{x} - \sqrt{2}}{x - 2} = \lim_{x \rightarrow 2} \frac{1}{\sqrt{x} + \sqrt{2}} = \frac{1}{2\sqrt{2}} = \frac{\sqrt{2}}{4}.$$

6.5. Limit as $x \rightarrow \infty$ of rational functions. A rational function is the quotient of two polynomials, so

$$(11) \quad R(x) = \frac{a_n x^n + \cdots + a_1 x + a_0}{b_m x^m + \cdots + b_1 x + b_0}.$$

We have seen that

$$\lim_{x \rightarrow \infty} \frac{1}{x} = 0$$

We even proved this in example 4.7. Using this you can find the limit at ∞ for any rational function $R(x)$ as in (11). One could turn the outcome of the calculation of $\lim_{x \rightarrow \infty} R(x)$ into a recipe/formula involving the degrees n and m of the numerator and denominator, and also their coefficients a_i , b_j , which students would then memorize, but it is better to remember “the trick.”

To find $\lim_{x \rightarrow \infty} R(x)$ divide numerator and denominator by x^m (the highest power of x occurring in the denominator).

For example, let's compute

$$\lim_{x \rightarrow \infty} \frac{3x^2 + 3}{5x^2 + 7x - 39}.$$

Remember the trick and divide top and bottom by x^2 , and you get

$$\begin{aligned}\lim_{x \rightarrow \infty} \frac{3x^2 + 3}{5x^2 + 7x - 39} &= \lim_{x \rightarrow \infty} \frac{3 + 3/x^2}{5 + 7/x - 39/x^2} \\ &= \frac{\lim_{x \rightarrow \infty} 3 + 3/x^2}{\lim_{x \rightarrow \infty} 5 + 7/x - 39/x^2} \\ &= \frac{3}{5}\end{aligned}$$

Here we have used the limit properties (P_*) to break the limit down into little pieces like $\lim_{x \rightarrow \infty} 39/x^2$ which we can compute as follows

$$\lim_{x \rightarrow \infty} 39/x^2 = \lim_{x \rightarrow \infty} 39 \cdot \left(\frac{1}{x}\right)^2 = \left(\lim_{x \rightarrow \infty} 39\right) \cdot \left(\lim_{x \rightarrow \infty} \frac{1}{x}\right)^2 = 39 \cdot 0^2 = 0.$$

6.6. Another example with a rational function . Compute

$$\lim_{x \rightarrow \infty} \frac{x}{x^3 + 5}.$$

We apply “the trick” again and divide numerator and denominator by x^3 . This leads to

$$\lim_{x \rightarrow \infty} \frac{x}{x^3 + 5} = \lim_{x \rightarrow \infty} \frac{1/x^2}{1 + 5/x^3} = \frac{\lim_{x \rightarrow \infty} 1/x^2}{\lim_{x \rightarrow \infty} 1 + 5/x^3} = \frac{0}{1} = 0.$$

To show all possible ways a limit of a rational function can turn out we should do yet another example, but that one belongs in the next section (see example 7.6.)

7. When limits fail to exist

In the last couple of examples we worried about the possibility that a limit $\lim_{x \rightarrow a} g(x)$ actually might not exist. This can actually happen, and in this section we’ll see a few examples of what failed limits look like. First let’s agree on what we will call a “failed limit.”

7.1. Definition. *If there is no number L such that $\lim_{x \rightarrow a} f(x) = L$, then we say that the limit $\lim_{x \rightarrow a} f(x)$ does not exist.*

7.2. The sign function near $x = 0$. The “sign function¹” is defined by

$$\text{sign}(x) = \begin{cases} -1 & \text{for } x < 0 \\ 0 & \text{for } x = 0 \\ 1 & \text{for } x > 0 \end{cases}$$

Note that “the sign of zero” is defined to be zero. But does the sign function have a limit at $x = 0$, i.e. does $\lim_{x \rightarrow 0} \text{sign}(x)$ exist? And is it also zero? The answers are **no** and **no**, and here is why: suppose that for some number L one had

$$\lim_{x \rightarrow 0} \text{sign}(x) = L,$$

then since for arbitrary small positive values of x one has $\text{sign}(x) = +1$ one would think that $L = +1$. But for arbitrarily small negative values of x one has $\text{sign}(x) = -1$, so one would conclude that $L = -1$. But one number L can’t be both $+1$ and -1 at the same time, so there is no such L , i.e. there is no limit.

$$\lim_{x \rightarrow 0} \text{sign}(x) \text{ does not exist.}$$

¹Some people don’t like the notation $\text{sign}(x)$, and prefer to write

$$g(x) = \frac{x}{|x|}$$

instead of $g(x) = \text{sign}(x)$. If you think about this formula for a moment you’ll see that $\text{sign}(x) = x/|x|$ for all $x \neq 0$. When $x = 0$ the quotient $x/|x|$ is of course not defined.

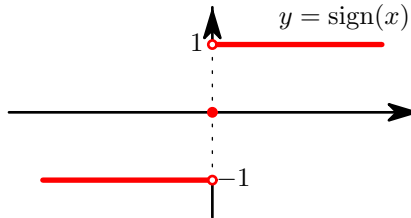


Figure 1. The sign function.

In this example the one-sided limits do exist, namely,

$$\lim_{x \searrow 0} \text{sign}(x) = 1 \text{ and } \lim_{x \nearrow 0} \text{sign}(x) = -1.$$

All this says is that when x approaches 0 through positive values, its sign approaches +1, while if x goes to 0 through negative values, then its sign approaches -1.

7.3. The example of the backward sine. Contemplate the limit as $x \rightarrow 0$ of the “backward sine,” i.e.

$$\lim_{x \rightarrow 0} \sin\left(\frac{\pi}{x}\right).$$

When $x = 0$ the function $f(x) = \sin(\pi/x)$ is not defined, because its definition involves division by x . What happens to $f(x)$ as $x \rightarrow 0$? First, π/x becomes larger and larger (“goes to infinity”) as $x \rightarrow 0$. Then, taking the sine, we see that $\sin(\pi/x)$ oscillates between +1 and -1 infinitely often as $x \rightarrow 0$. This means that $f(x)$ gets close to any number between -1 and +1 as $x \rightarrow 0$, but that the function $f(x)$ **never stays close** to any particular value because it keeps oscillating up and down.

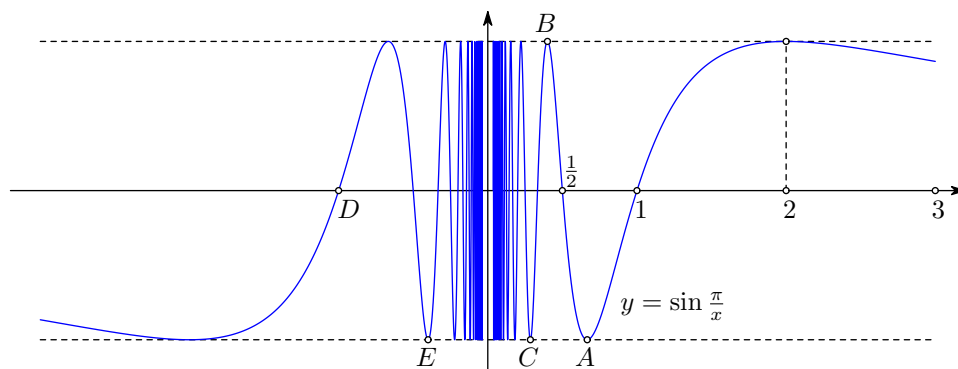


Figure 2. Graph of $y = \sin \frac{\pi}{x}$ for $-3 < x < 3$, $x \neq 0$.

Here again, the limit $\lim_{x \rightarrow 0} f(x)$ does not exist. We have arrived at this conclusion by only considering what $f(x)$ does for small positive values of x . So the limit fails to exist in a stronger way than in the example of the sign-function. There, even though the limit didn’t exist, the one-sided limits existed. In the present example we see that even the one-sided limit

$$\lim_{x \searrow 0} \sin \frac{\pi}{x}$$

does not exist.

7.4. Trying to divide by zero using a limit. The expression $1/0$ is not defined, but what about

$$\lim_{x \rightarrow 0} \frac{1}{x}?$$

This limit also does not exist. Here are two reasons:

It is common wisdom that if you divide by a small number you get a large number, so as $x \searrow 0$ the quotient $1/x$ will not be able to stay close to any particular finite number, and the limit can't exist.

“Common wisdom” is not always a reliable tool in mathematical proofs, so here is a better argument. The limit can't exist, because that would contradict the limit properties $(P_1) \cdots (P_6)$. Namely, suppose that there were an number L such that

$$\lim_{x \rightarrow 0} \frac{1}{x} = L.$$

Then the limit property (P_5) would imply that

$$\lim_{x \rightarrow 0} \left(\frac{1}{x} \cdot x \right) = \left(\lim_{x \rightarrow 0} \frac{1}{x} \right) \cdot \left(\lim_{x \rightarrow 0} x \right) = L \cdot 0 = 0.$$

On the other hand $\frac{1}{x} \cdot x = 1$ so the above limit should be 1! A number can't be both 0 and 1 at the same time, so we have a contradiction. The assumption that $\lim_{x \rightarrow 0} 1/x$ exists is to blame, so it must go.

7.5. Using limit properties to show a limit does *not* exist. The limit properties tell us how to prove that certain limits exist (and how to compute them). Although it is perhaps not so obvious at first sight, they also allow you to prove that certain limits do not exist. The previous example shows one instance of such use. Here is another.

Property (P_3) says that if both $\lim_{x \rightarrow a} g(x)$ and $\lim_{x \rightarrow a} h(x)$ exist then $\lim_{x \rightarrow a} g(x) + h(x)$ also must exist. You can turn this around and say that if $\lim_{x \rightarrow a} g(x) + h(x)$ does not exist then either $\lim_{x \rightarrow a} g(x)$ or $\lim_{x \rightarrow a} h(x)$ does not exist (or both limits fail to exist).

For instance, the limit

$$\lim_{x \rightarrow 0} \frac{1}{x} - x$$

can't exist, for if it did, then the limit

$$\lim_{x \rightarrow 0} \frac{1}{x} = \lim_{x \rightarrow 0} \left(\frac{1}{x} - x + x \right) = \lim_{x \rightarrow 0} \left(\frac{1}{x} - x \right) + \lim_{x \rightarrow 0} x$$

would also have to exist, and we know $\lim_{x \rightarrow 0} \frac{1}{x}$ doesn't exist.

7.6. Limits at ∞ which don't exist. If you let x go to ∞ , then x will not get “closer and closer” to any particular number L , so it seems reasonable to guess that

$$\lim_{x \rightarrow \infty} x \text{ does not exist.}$$

One can prove this from the limit definition (and see exercise 72).

Let's consider

$$L = \lim_{x \rightarrow \infty} \frac{x^2 + 2x - 1}{x + 2}.$$

Once again we divide numerator and denominator by the highest power in the denominator (i.e. x)

$$L = \lim_{x \rightarrow \infty} \frac{x + 2 - \frac{1}{x}}{1 + 2/x}$$

Here the denominator has a limit ('tis 1), but the numerator does not, for if $\lim_{x \rightarrow \infty} x + 2 - \frac{1}{x}$ existed then, since $\lim_{x \rightarrow \infty} (2 - 1/x) = 2$ exists,

$$\lim_{x \rightarrow \infty} x = \lim_{x \rightarrow \infty} \left[\left(x + 2 - \frac{1}{x} \right) - \left(2 - \frac{1}{x} \right) \right]$$

would also have to exist, and $\lim_{x \rightarrow \infty} x$ doesn't exist.

So we see that L is the limit of a fraction in which the denominator has a limit, but the numerator does not. In this situation the limit L itself can never exist. If it did, then

$$\lim_{x \rightarrow \infty} \left(x + 2 - \frac{1}{x} \right) = \lim_{x \rightarrow \infty} \frac{x + 2 - \frac{1}{x}}{1 + 2/x} \cdot (1 + 2/x)$$

would also have to have a limit.

8. What's in a name?

There is a big difference between the variables x and a in the formula

$$\lim_{x \rightarrow a} 2x + 1,$$

namely a is a **free variable**, while x is a **dummy variable** (or “placeholder” or a “bound variable.”)

The difference between these two kinds of variables is this:

- if you replace a dummy variable in some formula consistently by some other variable then the value of the formula does not change. On the other hand, it never makes sense to substitute a number for a dummy variable.
- the value of the formula may depend on the value of the free variable.

To understand what this means consider the example $\lim_{x \rightarrow a} 2x + 1$ again. The limit is easy to compute:

$$\lim_{x \rightarrow a} 2x + 1 = 2a + 1.$$

If we replace x by, say u (systematically) then we get

$$\lim_{u \rightarrow a} 2u + 1$$

which is again equal to $2a + 1$. This computation says that *if some number gets close to a then two times that number plus one gets close to $2a + 1$* . This is a very wordy way of expressing the formula, and you can shorten things by giving a name (like x or u) to the number which approaches a . But the result of our computation shouldn't depend on the name we choose, i.e. it doesn't matter if we call it x or u .

Since the name of the variable x doesn't matter it is called a dummy variable. Some prefer to call x a bound variable, meaning that in

$$\lim_{x \rightarrow a} 2x + 1$$

the x in the expression $2x + 1$ is bound to the x written underneath the limit – you can't change one without changing the other.

Substituting a number for a dummy variable usually leads to complete nonsense. For instance, let's try setting $x = 3$ in our limit, i.e. what is

$$\lim_{3 \rightarrow a} 2 \cdot 3 + 1?$$

Of course $2 \cdot 3 + 1 = 7$, but what does 7 do when 3 gets closer and closer to the number a ? That's a silly question, because 3 is a constant and it doesn't “get closer” to some other number like a ! If you ever see 3 get closer to another number then it's time to take a vacation.

On the other hand the variable a is free: you can assign it particular values, and its value will affect the value of the limit. For instance, if we set $a = 3$ (but leave x alone) then we get

$$\lim_{x \rightarrow 3} 2x + 1$$

and there's nothing strange about that (the limit is $2 \cdot 3 + 1 = 7$, no problem.) You could substitute other values of a and you would get a different answer. In general you get $2a + 1$.

9. Limits and Inequalities

This section has two theorems which let you compare limits of different functions. The properties in these theorems are not formulas that allow you to compute limits like the properties $(P_1) \dots (P_6)$ from §5. Instead, they allow you to **reason** about limits, i.e. they let you say that this or that limit is positive, or that it must be the same as some other limit which you find easier to think about.

The first theorem should not surprise you – all it says is that bigger functions have bigger limits.

9.1. Theorem. *Let f and g be functions whose limits for $x \rightarrow a$ exist, and assume that $f(x) \leq g(x)$ holds for all x . Then*

$$\lim_{x \rightarrow a} f(x) \leq \lim_{x \rightarrow a} g(x).$$

A useful special case arises when you set $f(x) = 0$. The theorem then says that if a function g never has negative values, then its limit will also never be negative.

The statement may seem obvious, but it still needs a proof, starting from the ε - δ definition of limit. This will be done in lecture.

Here is the second theorem about limits and inequalities.

9.2. The Sandwich Theorem. *Suppose that*

$$f(x) \leq g(x) \leq h(x)$$

(for all x) and that

$$\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} h(x).$$

Then

$$\lim_{x \rightarrow a} g(x) = \lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} h(x).$$

The theorem is useful when you want to know the limit of g , and when you can **sandwich** it between two functions f and h whose limits are easier to compute. The Sandwich Theorem looks like the first theorem of this section, but there is an important difference: in the Sandwich Theorem you don't have to assume that the limit of g exists. The inequalities $f \leq g \leq h$ combined with the circumstance that f and h have the same limit are enough to guarantee that the limit of g exists.

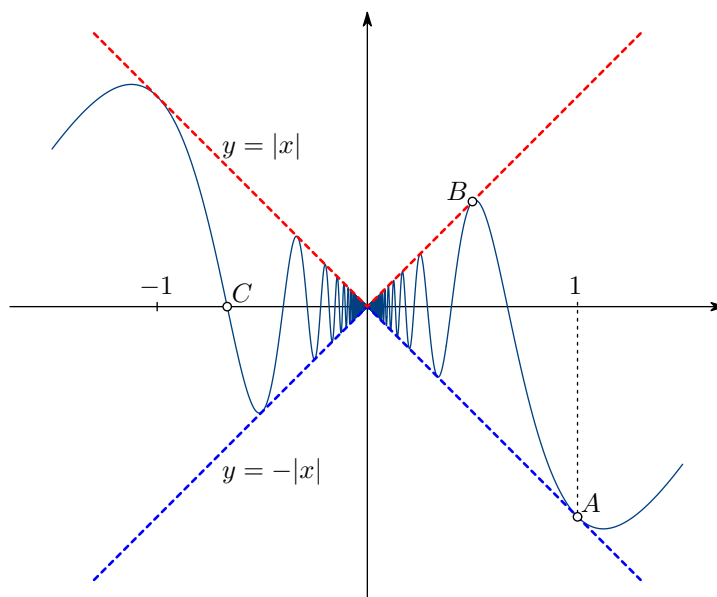


Figure 3. Graphs of $|x|$, $-|x|$ and $x \cos \frac{\pi}{x}$ for $-1.2 < x < 1.2$

9.3. Example: a Backward Cosine Sandwich. The Sandwich Theorem says that if the function $g(x)$ is sandwiched between two functions $f(x)$ and $h(x)$ and the limits of the outside functions f and h exist and are equal, then the limit of the inside function g exists and equals this common value. For example

$$-|x| \leq x \cos \frac{1}{x} \leq |x|$$

since the cosine is always between -1 and 1 . Since

$$\lim_{x \rightarrow 0} -|x| = \lim_{x \rightarrow 0} |x| = 0$$

the sandwich theorem tells us that

$$\lim_{x \rightarrow 0} x \cos \frac{1}{x} = 0.$$

Note that the limit $\lim_{x \rightarrow 0} \cos(1/x)$ does **not** exist, for the same reason that the “backward sine” did not have a limit for $x \rightarrow 0$ (see example 7.3). Multiplying with x changed that.

10. Continuity

10.1. Definition. A function g is **continuous** at a if

$$(12) \quad \lim_{x \rightarrow a} g(x) = g(a)$$

A function is continuous if it is continuous at every a in its domain.

Note that when we say that a function is continuous on some interval it is understood that the domain of the function includes that interval. For example, the function $f(x) = 1/x^2$ is continuous on the interval $1 < x < 5$ but is **not** continuous on the interval $-1 < x < 1$.

10.2. Polynomials are continuous. For instance, let us show that $P(x) = x^2 + 3x$ is continuous at $x = 2$. To show that you have to prove that

$$\lim_{x \rightarrow 2} P(x) = P(2),$$

i.e.

$$\lim_{x \rightarrow 2} x^2 + 3x = 2^2 + 3 \cdot 2.$$

You can do this two ways: using the definition with ε and δ (i.e. the hard way), or using the limit properties $(P_1) \dots (P_6)$ from §5 (just as good, and easier, even though it still takes a few lines to write it out – do both!)

10.3. Rational functions are continuous. Let $R(x) = \frac{P(x)}{Q(x)}$ be a rational function, and let a be any number in the domain of R , i.e. any number for which $Q(a) \neq 0$. Then one has

$$\begin{aligned} \lim_{x \rightarrow a} R(x) &= \lim_{x \rightarrow a} \frac{P(x)}{Q(x)} \\ &= \frac{\lim_{x \rightarrow a} P(x)}{\lim_{x \rightarrow a} Q(x)} && \text{property } (P_6) \\ &= \frac{P(a)}{Q(a)} && P \text{ and } Q \text{ are continuous} \\ &= R(a). \end{aligned}$$

This shows that R is indeed continuous at a .

10.4. Some discontinuous functions. If $\lim_{x \rightarrow a} g(x)$ does not exist, then it certainly cannot be equal to $g(a)$, and therefore any failed limit provides an example of a discontinuous function.

For instance, the sign function $g(x) = \text{sign}(x)$ from example ?? is not continuous at $x = 0$.

Is the backward sine function $g(x) = \sin(1/x)$ from example ?? also discontinuous at $x = 0$? No, it is not, for two reasons: first, the limit $\lim_{x \rightarrow 0} \sin(1/x)$ does not exist, and second, we haven’t even defined the function $g(x)$ at $x = 0$, so even if the limit existed, we would have no value $g(0)$ to compare it with.

10.5. How to make functions discontinuous. Here is a discontinuous function:

$$f(x) = \begin{cases} x^2 & \text{if } x \neq 3, \\ 47 & \text{if } x = 3. \end{cases}$$

In other words, we take a continuous function like $g(x) = x^2$, and change its value somewhere, e.g. at $x = 3$. Then

$$\lim_{x \rightarrow 3} f(x) = 9 \neq 47 = f(3).$$

The reason that the limit is 9 is that our new function $f(x)$ coincides with our old continuous function $g(x)$ for all x except $x = 3$. Therefore the limit of $f(x)$ as $x \rightarrow 3$ is the same as the limit of $g(x)$ as $x \rightarrow 3$, and since g is continuous this is $g(3) = 9$.

10.6. Sandwich in a bow tie. We return to the function from example ???. Consider

$$f(x) = \begin{cases} x \cos\left(\frac{1}{x}\right) & \text{for } x \neq 0, \\ 0 & \text{for } x = 0 \end{cases}$$

Then f is continuous at $x = 0$ by the Sandwich Theorem (see Example ??).

If we change the definition of f by picking a different value at $x = 0$ the new function will not be continuous, since changing f at $x = 0$ does not change the limit $\lim_{x \rightarrow 0} f(x)$. Since this limit is zero, $f(0) = 0$ is the only possible choice of $f(0)$ which makes f continuous at $x = 0$.

11. Substitution in Limits

Given two functions f and g one can consider their composition $h(x) = f(g(x))$. To compute the limit

$$\lim_{x \rightarrow a} f(g(x))$$

we write $u = g(x)$, so that we want to know

$$\lim_{x \rightarrow a} f(u) \text{ where } u = g(x).$$

Suppose that you can find the limits

$$L = \lim_{x \rightarrow a} g(x) \text{ and } \lim_{u \rightarrow L} f(u) = M.$$

Then it seems reasonable that as x approaches a , $u = g(x)$ will approach L , and $f(g(x))$ approaches M .

This is in fact a theorem:

11.1. Theorem. *If $\lim_{x \rightarrow a} g(x) = a$, and if the function f is continuous at $u = L$, then*

$$\lim_{x \rightarrow a} f(g(x)) = \lim_{u \rightarrow L} f(u) = f(L).$$

Another way to write this is

$$\lim_{x \rightarrow a} f(g(x)) = f\left(\lim_{x \rightarrow a} g(x)\right).$$

11.2. Example: compute $\lim_{x \rightarrow 3} \sqrt{x^3 - 3x^2 + 2}$. The given function is the composition of two functions, namely

$$\sqrt{x^3 - 3x^2 + 2} = \sqrt{u}, \text{ with } u = x^3 - 3x^2 + 2,$$

or, in function notation, we want to find $\lim_{x \rightarrow 3} h(x)$ where

$$h(x) = f(g(x)), \text{ with } g(x) = x^3 - 3x^2 + 2 \text{ and } f(u) = \sqrt{u}.$$

Either way, we have

$$\lim_{x \rightarrow 3} x^3 - 3x^2 + 2 = 2 \quad \text{and} \quad \lim_{u \rightarrow 2} \sqrt{u} = \sqrt{2}.$$

You get the first limit from the limit properties $(P_1) \dots (P_5)$. The second limit says that taking the square root is a continuous function, which it is. We have not proved that (yet), but this particular limit is the one from example 6.3. Putting these two limits together we conclude that the limit is $\sqrt{2}$.

Normally, you write this whole argument as follows:

$$\lim_{x \rightarrow 3} \sqrt{x^3 - 3x^2 + 2} = \sqrt{\lim_{x \rightarrow 3} x^3 - 3x^2 + 2} = \sqrt{2},$$

where you must point out that $f(x) = \sqrt{x}$ is a continuous function to justify the first step.

Another possible way of writing this is

$$\lim_{x \rightarrow 3} \sqrt{x^3 - 3x^2 + 2} = \lim_{u \rightarrow 2} \sqrt{u} = \sqrt{2},$$

where you must say that you have substituted $u = x^3 - 3x^2 + 2$.

12. Exercises

Find the following limits.

52. $\lim_{x \rightarrow -7} (2x + 5)$

53. $\lim_{x \rightarrow 7^-} (2x + 5)$

54. $\lim_{x \rightarrow -\infty} (2x + 5)$

55. $\lim_{x \rightarrow -4} (x + 3)^{2006}$

56. $\lim_{x \rightarrow -4} (x + 3)^{2007}$

57. $\lim_{x \rightarrow -\infty} (x + 3)^{2007}$

58. $\lim_{t \rightarrow 1} \frac{t^2 + t - 2}{t^2 - 1}$

59. $\lim_{t \nearrow 1} \frac{t^2 + t - 2}{t^2 - 1}$

60. $\lim_{t \rightarrow -1} \frac{t^2 + t - 2}{t^2 - 1}$

61. $\lim_{x \rightarrow \infty} \frac{x^2 + 3}{x^2 + 4}$

62. $\lim_{x \rightarrow \infty} \frac{x^5 + 3}{x^2 + 4}$

63. $\lim_{x \rightarrow \infty} \frac{x^2 + 1}{x^5 + 2}$

64. $\lim_{x \rightarrow \infty} \frac{(2x + 1)^4}{(3x^2 + 1)^2}$

65. $\lim_{u \rightarrow \infty} \frac{(2u + 1)^4}{(3u^2 + 1)^2}$

66. $\lim_{t \rightarrow 0} \frac{(2t + 1)^4}{(3t^2 + 1)^2}$

67. What are the coordinates of the points labeled $A, \dots, E$ in Figure 2 (the graph of $y = \sin \pi/x$).

68. If $\lim_{x \rightarrow a} f(x)$ exists then f is continuous at $x = a$. *True or false?*

69. Give two examples of functions for which $\lim_{x \searrow 0} f(x)$ does not exist.

70. Group Problem.

If $\lim_{x \rightarrow 0} f(x)$ and $\lim_{x \rightarrow 0} g(x)$ both do not exist, then $\lim_{x \rightarrow 0} (f(x) + g(x))$ also does not exist. *True or false?*

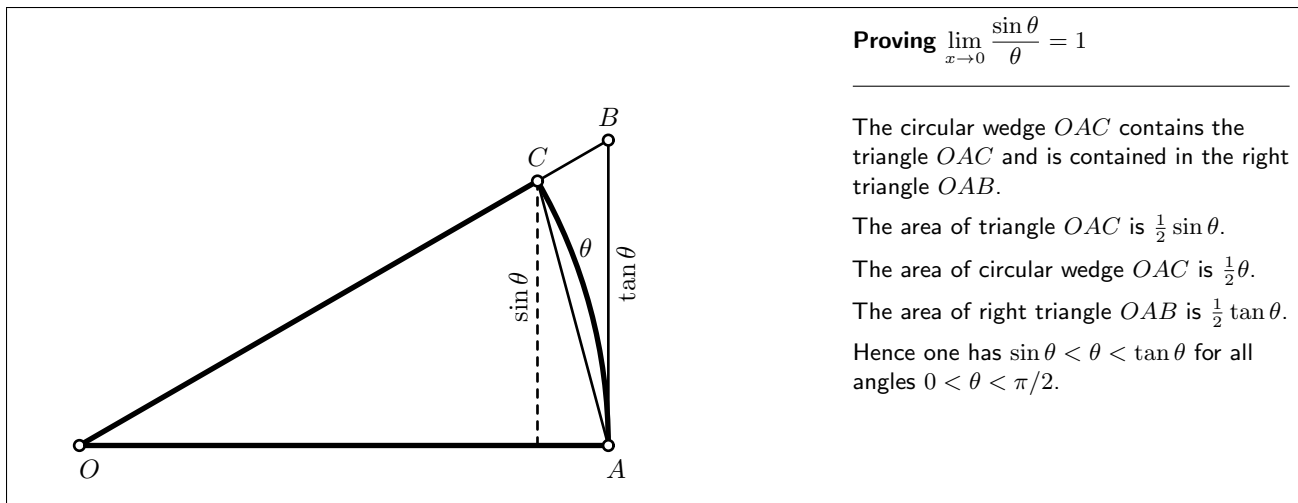
71. Group Problem.

If $\lim_{x \rightarrow 0} f(x)$ and $\lim_{x \rightarrow 0} g(x)$ both do not exist, then $\lim_{x \rightarrow 0} (f(x)/g(x))$ also does not exist. *True or false?*

72. Group Problem.

In the text we proved that $\lim_{x \rightarrow \infty} \frac{1}{x} = 0$. Show that this implies that $\lim_{x \rightarrow \infty} x$ does not exist. Hint: Suppose $\lim_{x \rightarrow \infty} x = L$ for some number L . Apply the limit properties to $\lim_{x \rightarrow \infty} x \cdot \frac{1}{x}$.

73. Evaluate $\lim_{x \rightarrow 9} \frac{\sqrt{x} - 3}{x - 9}$. Hint: Multiply top and bottom by $\sqrt{x} + 3$.



74. Evaluate $\lim_{x \rightarrow 2} \frac{\frac{1}{x} - \frac{1}{2}}{x - 2}$.

75. Evaluate $\lim_{x \rightarrow 2} \frac{\frac{1}{\sqrt{x}} - \frac{1}{\sqrt{2}}}{x - 2}$.

76. A function f is defined by

$$f(x) = \begin{cases} x^3 & \text{for } x < -1 \\ ax + b & \text{for } -1 \leq x < 1 \\ x^2 + 2 & \text{for } x \geq 1. \end{cases}$$

where a and b are constants. The function f is continuous. What are a and b ?

77. Find a constant k such that the function

$$f(x) = \begin{cases} 3x + 2 & \text{for } x < 2 \\ x^2 + k & \text{for } x \geq 2. \end{cases}$$

is continuous. Hint: Compute the one-sided limits.

78. Find constants a and c such that the function

$$f(x) = \begin{cases} x^3 + c & \text{for } x < 0 \\ ax + c^2 & \text{for } 0 \leq x < 1 \\ \arctan x & \text{for } x \geq 1. \end{cases}$$

is continuous for all x .

13. Two Limits in Trigonometry

In this section we'll derive a few limits involving the trigonometric functions. You can think of them as saying that for small angles θ one has

$$\sin \theta \approx \theta \quad \text{and} \quad \cos \theta \approx 1 - \frac{1}{2}\theta^2.$$

We will use these limits when we compute the derivatives of Sine, Cosine and Tangent.

13.1. Theorem. $\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = 1$.

PROOF. The proof requires a few sandwiches and some geometry.

We begin by only considering positive angles, and in fact we will only consider angles $0 < \theta < \pi/2$.

Since the wedge OAC contains the triangle OAC its area must be larger. The area of the wedge is $\frac{1}{2}\theta$ and the area of the triangle is $\frac{1}{2}\sin \theta$, so we find that

$$(13) \quad 0 < \sin \theta < \theta \text{ for } 0 < \theta < \frac{\pi}{2}.$$

The Sandwich Theorem implies that

$$(14) \quad \lim_{\theta \searrow 0} \sin \theta = 0.$$

Moreover, we also have

$$(15) \quad \lim_{\theta \searrow 0} \cos \theta = \lim_{\theta \searrow 0} \sqrt{1 - \sin^2 \theta} = 1.$$

Next we compare the areas of the wedge OAC and the larger triangle OAB . Since OAB has area $\frac{1}{2} \tan \theta$ we find that

$$\theta < \tan \theta$$

for $0 < \theta < \frac{\pi}{2}$. Since $\tan \theta = \frac{\sin \theta}{\cos \theta}$ we can multiply with $\cos \theta$ and divide by θ to get

$$\cos \theta < \frac{\sin \theta}{\theta} \text{ for } 0 < \theta < \frac{\pi}{2}$$

If we go back to (15) and divide by θ , then we get

$$\cos \theta < \frac{\sin \theta}{\theta} < 1$$

The Sandwich Theorem can be used once again, and now it gives

$$\lim_{\theta \searrow 0} \frac{\sin \theta}{\theta} = 1.$$

This is a one-sided limit. To get the limit in which $\theta \nearrow 0$, you use that $\sin \theta$ is an odd function. □

13.2. An example. We will show that

$$(16) \quad \lim_{\theta \rightarrow 0} \frac{1 - \cos \theta}{\theta^2} = \frac{1}{2}.$$

This follows from $\sin^2 \theta + \cos^2 \theta = 1$. Namely,

$$\begin{aligned} \frac{1 - \cos \theta}{\theta^2} &= \frac{1}{1 + \cos \theta} \frac{1 - \cos^2 \theta}{\theta^2} \\ &= \frac{1}{1 + \cos \theta} \frac{\sin^2 \theta}{\theta^2} \\ &= \frac{1}{1 + \cos \theta} \left\{ \frac{\sin \theta}{\theta} \right\}^2. \end{aligned}$$

We have just shown that $\cos \theta \rightarrow 1$ and $\frac{\sin \theta}{\theta} \rightarrow 1$ as $\theta \rightarrow 0$, so (16) follows.

14. Exercises

Find each of the following limits or show that it does not exist. Distinguish between limits which are infinite and limits which do not exist.

79. $\lim_{\alpha \rightarrow 0} \frac{\sin 2\alpha}{\sin \alpha}$ (two ways: with and without the double angle formula!)

80. $\lim_{x \rightarrow 0} \frac{\sin 3x}{2x}$.

81. $\lim_{\theta \rightarrow 0} \frac{\tan \theta}{\theta}$.

82. $\lim_{\alpha \rightarrow 0} \frac{\tan 4\alpha}{\sin 2\alpha}$.

83. $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x \sin x}$.

84. $\lim_{\theta \rightarrow \pi/2} \frac{1 - \sin \theta}{\theta - \pi/2}$

85. $\lim_{x \rightarrow \infty} \frac{2x^3 + 3x^2 \cos x}{(x+2)^3}$.

86. $\lim_{x \rightarrow 0} \frac{\sin^2 x}{1 - \cos x}$.

87. $\lim_{x \rightarrow 0} \frac{\sin(x^2)}{x^2}$.

88. $\lim_{x \rightarrow 0} \frac{x(1 - \cos x)}{\tan^3 x}$.

89. $\lim_{x \rightarrow 0} \frac{\sin(x^2)}{1 - \cos x}$.

90. $\lim_{x \rightarrow \pi/2} \frac{x - \frac{\pi}{2}}{\cos x}$.

91. $\lim_{x \rightarrow \pi/2} (x - \frac{\pi}{2}) \tan x$.

92. $\lim_{x \rightarrow 0} \frac{\cos x}{x^2 + 9}$.

93. $\lim_{x \rightarrow \pi} \frac{\sin x}{x - \pi}$.

94. $\lim_{x \rightarrow 0} \frac{\sin x}{x + \sin x}$.

95. $A = \lim_{x \rightarrow \infty} \frac{\sin x}{x}$. **(!!)**

96. $B = \lim_{x \rightarrow \infty} \frac{\cos x}{x}$. **(!! again)**

97. Is there a constant k such that the function

$$f(x) = \begin{cases} \sin(1/x) & \text{for } x \neq 0 \\ k & \text{for } x = 0. \end{cases}$$

is continuous? If so, find it; if not, say why.

98. Find a constant A so that the function

$$f(x) = \begin{cases} \frac{\sin x}{2x} & \text{for } x \neq 0 \\ A & \text{when } x = 0 \end{cases}$$

99. Compute $\lim_{x \rightarrow \infty} x \sin \frac{\pi}{x}$ and $\lim_{x \rightarrow \infty} x \tan \frac{\pi}{x}$. (Hint: substitute something).

100. Group Problem.

(*Geometry & Trig review*) Let A_n be the area of the regular n -gon inscribed in the unit circle, and let B_n be the area of the regular n -gon whose inscribed circle has radius 1.

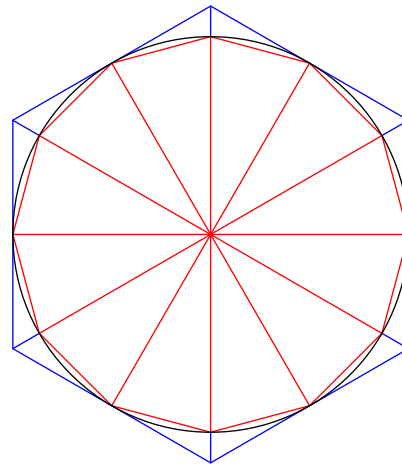
(a) Show that $A_n < \pi < B_n$.

(b) Show that

$$A_n = \frac{n}{2} \sin \frac{2\pi}{n} \text{ and } B_n = n \tan \frac{\pi}{n}$$

(c) Compute $\lim_{n \rightarrow \infty} A_n$ and $\lim_{n \rightarrow \infty} B_n$.

Here is a picture of A_{12} , B_6 and π :



On a historical note: Archimedes managed to compute A_{96} and B_{96} and by doing this got the most accurate approximation for π that was known in his time. See also:

http://www-history.mcs.st-andrews.ac.uk/HistTopics/Pi_through_the_ages.html

CHAPTER 4

Derivatives (2)

“Leibniz never thought of the derivative as a limit”

<http://www.gap-system.org/~history/Biographies/Leibniz.html>

In chapter 2 we saw two mathematical problems which led to expressions of the form $\frac{0}{0}$. Now that we know how to handle limits, we can state the definition of the derivative of a function. After computing a few derivatives using the definition we will spend most of this section developing the *differential calculus*, which is a collection of rules that allow you to compute derivatives without always having to use basic definition.

1. Derivatives Defined

1.1. Definition. Let f be a function which is defined on some interval (c, d) and let a be some number in this interval.

The **derivative of the function f at a** is the value of the limit

$$(17) \quad f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}.$$

f is said to be **differentiable at a** if this limit exists.

f is called **differentiable on the interval (c, d)** if it is differentiable at every point a in (c, d) .

1.2. Other notations. One can substitute $x = a + h$ in the limit (17) and let $h \rightarrow 0$ instead of $x \rightarrow a$. This gives the formula

$$(18) \quad f'(a) = \lim_{h \rightarrow 0} \frac{f(a + h) - f(a)}{h},$$

Often you will find this equation written with x instead of a and Δx instead of h , which makes it look like this:

$$f'(x) = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x}.$$

The interpretation is the same as in equation (6) from §4. The numerator $f(x + \Delta x) - f(x)$ represents the amount by which the function value of f changes if one increases its argument x by a (small) amount Δx . If you write $y = f(x)$ then we can call the increase in f

$$\Delta y = f(x + \Delta x) - f(x),$$

so that the derivative $f'(x)$ is

$$f'(x) = \lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x}.$$

GOTTFRIED WILHELM VON LEIBNIZ, one of the inventors of calculus, came up with the idea that one should write this limit as

$$\frac{dy}{dx} = \lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x},$$

the idea being that after letting Δx go to zero it didn't vanish, but instead became an infinitely small quantity which Leibniz called “ dx .” The result of increasing x by this infinitely small quantity dx is that $y = f(x)$ increased by another infinitely small quantity dy . The ratio of these two infinitely small quantities is what we call the derivative of $y = f(x)$.

There are no “infinitely small real numbers,” and this makes Leibniz’ notation difficult to justify. In the 20th century mathematicians have managed to create a consistent theory of “infinitesimals” which allows you to compute with “ dx and dy ” as Leibniz and his contemporaries would have done. This theory is called “non standard analysis.” We won’t mention it any further¹. Nonetheless, even though we won’t use infinitely small numbers, Leibniz’ notation is very useful and we will use it.

2. Direct computation of derivatives

2.1. Example – The derivative of $f(x) = x^2$ is $f'(x) = 2x$. We have done this computation before in §2. The result was

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{(x+h)^2 - x^2}{h} = \lim_{h \rightarrow 0} (2x+h) = 2x.$$

Leibniz would have written

$$\frac{dx^2}{dx} = 2x.$$

2.2. The derivative of $g(x) = x$ is $g'(x) = 1$. Indeed, one has

$$g'(x) = \lim_{h \rightarrow 0} \frac{g(x+h) - g(x)}{h} = \lim_{h \rightarrow 0} \frac{(x+h) - x}{h} = \lim_{h \rightarrow 0} \frac{h}{h} = 1.$$

In Leibniz’ notation:

$$\frac{dx}{dx} = 1.$$

This is an example where Leibniz’ notation is most misleading, because if you divide dx by dx then you should of course get 1. Nonetheless, this is not what is going on. The expression $\frac{dx}{dx}$ is not really a fraction since there are no two “infinitely small” quantities dx which we are dividing.

2.3. The derivative of any constant function is zero . Let $k(x) = c$ be a constant function. Then we have

$$k'(x) = \lim_{h \rightarrow 0} \frac{k(x+h) - k(x)}{h} = \lim_{h \rightarrow 0} \frac{c - c}{h} = \lim_{h \rightarrow 0} 0 = 0.$$

Leibniz would have said that if c is a constant, then

$$\frac{dc}{dx} = 0.$$

2.4. Derivative of x^n for $n = 1, 2, 3, \dots$. To differentiate $f(x) = x^n$ one proceeds as follows:

$$f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} = \lim_{x \rightarrow a} \frac{x^n - a^n}{x - a}.$$

We need to simplify the fraction $(x^n - a^n)/(x - a)$. For $n = 2$ we have

$$\frac{x^2 - a^2}{x - a} = x + a.$$

For $n = 1, 2, 3, \dots$ the geometric sum formula tells us that

$$(19) \quad \frac{x^n - a^n}{x - a} = x^{n-1} + x^{n-2}a + x^{n-3}a^2 + \dots + xa^{n-2} + a^{n-1}.$$

¹But if you want to read more on this you should see Keisler’s calculus text at

<http://www.math.wisc.edu/~keisler/calc.html>

I would not recommend using Keisler’s text and this text at the same time, but if you like math you should remember that it exists, and look at it (later, say, after you pass 221.)

If you don't remember the geometric sum formula, then you could also just verify (19) by carefully multiplying both sides with $x - a$. For instance, when $n = 3$ you would get

$$\begin{array}{rcl} x \times (x^2 + xa + a^2) & = & x^3 + ax^2 + a^2x \\ -a \times (x^2 + xa + a^2) & = & -ax^2 - a^2x - a^3 \\ \hline (x - a) \times (x^2 + xa + a^2) & = & x^3 - a^3 \end{array}$$

With formula (19) in hand we can now easily find the derivative of x^n :

$$\begin{aligned} f'(a) &= \lim_{x \rightarrow a} \frac{x^n - a^n}{x - a} \\ &= \lim_{x \rightarrow a} \{x^{n-1} + x^{n-2}a + x^{n-3}a^2 + \cdots + xa^{n-2} + a^{n-1}\} \\ &= a^{n-1} + a^{n-2}a + a^{n-3}a^2 + \cdots + a a^{n-2} + a^{n-1}. \end{aligned}$$

Here there are n terms, and they all are equal to a^{n-1} , so the final result is

$$f'(a) = na^{n-1}.$$

One could also write this as $f'(x) = nx^{n-1}$, or, in Leibniz' notation

$$\frac{dx^n}{dx} = nx^{n-1}.$$

This formula turns out to be true in general, but here we have only proved it for the case in which n is a positive integer.

3. Differentiable implies Continuous

3.1. Theorem. *If a function f is differentiable at some a in its domain, then f is also continuous at a .*

PROOF. We are given that

$$\lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$$

exists, and we must show that

$$\lim_{x \rightarrow a} f(x) = f(a).$$

This follows from the following computation

$$\begin{aligned} \lim_{x \rightarrow a} f(x) &= \lim_{x \rightarrow a} (f(x) - f(a) + f(a)) && \text{(algebra)} \\ &= \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} \cdot (x - a) + f(a) && \text{(more algebra)} \\ &= \left\{ \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} \right\} \cdot \lim_{x \rightarrow a} (x - a) + \lim_{x \rightarrow a} f(a) && \text{(Limit Properties)} \\ &= f'(a) \cdot 0 + f(a) && (f'(a) \text{ exists}) \\ &= f(a). \end{aligned}$$

□

4. Some non-differentiable functions

4.1. A graph with a corner. Consider the function

$$f(x) = |x| = \begin{cases} x & \text{for } x \geq 0, \\ -x & \text{for } x < 0. \end{cases}$$

This function is continuous at all x , but it is not differentiable at $x = 0$.

To see this try to compute the derivative at 0,

$$f'(0) = \lim_{x \rightarrow 0} \frac{|x| - |0|}{x - 0} = \lim_{x \rightarrow 0} \frac{|x|}{x} = \lim_{x \rightarrow 0} \text{sign}(x).$$

We know this limit does not exist (see §7.2)

If you look at the graph of $f(x) = |x|$ then you see what is wrong: the graph has a corner at the origin and it is not clear which line, if any, deserves to be called the tangent to the graph at the origin.

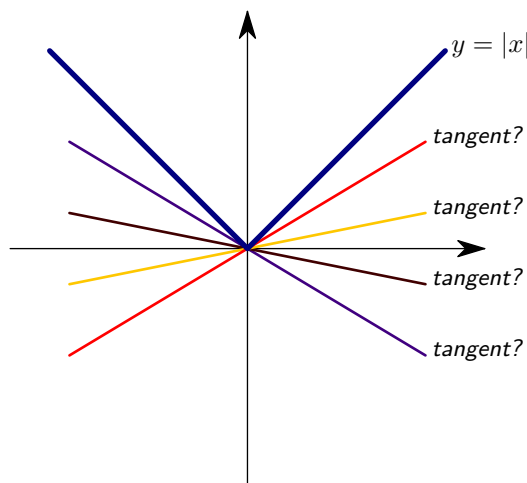


Figure 1. The graph of $y = |x|$ has no tangent at the origin.

4.2. A graph with a cusp. Another example of a function without a derivative at $x = 0$ is

$$f(x) = \sqrt{|x|}.$$

When you try to compute the derivative you get this limit

$$f'(0) = \lim_{x \rightarrow 0} \frac{\sqrt{|x|}}{x} = ?$$

The limit from the right is

$$\lim_{x \searrow 0} \frac{\sqrt{|x|}}{x} = \lim_{x \searrow 0} \frac{1}{\sqrt{x}},$$

which does not exist (it is $+\infty$). Likewise, the limit from the left also does not exist ('tis $-\infty$). Nonetheless, a drawing for the graph of f suggests an obvious tangent to the graph at $x = 0$, namely, the y -axis. That observation does not give us a derivative, because the y -axis is vertical and hence has no slope.

4.3. A graph with absolutely no tangents, *anywhere*. The previous two examples were about functions which did not have a derivative at $x = 0$. In both examples the point $x = 0$ was the only point where the function failed to have a derivative. It is easy to give examples of functions which are not differentiable at more than one value of x , but here I would like to show you a function f which doesn't have a derivative *anywhere in its domain*.

To keep things short I won't write a formula for the function, and merely show you a graph. In this graph you see a typical path of a Brownian motion, i.e. t is time, and $x(t)$ is the position of a particle which undergoes a Brownian motion – come to lecture for further explanation (see also the article on wikipedia). To see a similar graph check the Dow Jones or Nasdaq in the upper left hand corner of the web page at <http://finance.yahoo.com> in the afternoon on any weekday.

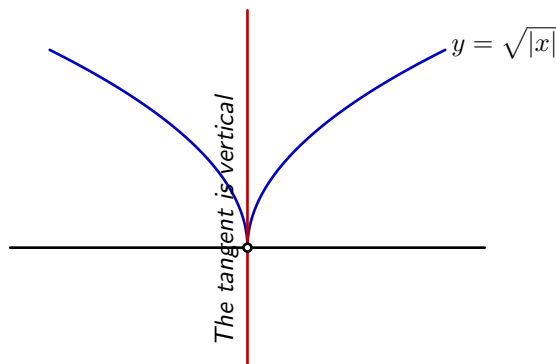


Figure 2. Tangent to the graph of $y = |x|^{1/2}$ at the origin

5. Exercises

101. Compute the derivative of the following functions

$$f(x) = x^2 - 2x$$

$$g(x) = \frac{1}{x}$$

$$k(x) = x^3 - 17x$$

$$u(x) = \frac{2}{1+x}$$

$$v(x) = \sqrt{x}$$

$$w(x) = \frac{1}{\sqrt{x}}$$

using either (17) or (18).

102. Which of the following functions is differentiable at $x = 0$?

$$f(x) = x|x|, \quad g(x) = x\sqrt{|x|},$$

$$h(x) = x + |x|, \quad k(x) = x^2 \sin \frac{\pi}{x},$$

$$\ell(x) = x \sin \frac{\pi}{x}.$$

These formulas do not define k and ℓ at $x = 0$. We define $k(0) = \ell(0) = 0$.

103. For which value(s) is the function defined by

$$f(x) = \begin{cases} ax + b & \text{for } x < 0 \\ x - x^2 & \text{for } x \geq 0 \end{cases}$$

differentiable at $x = 0$? Sketch the graph of the function f for the values a and b you found.

104. For which value(s) is the function defined by

$$f(x) = \begin{cases} ax^2 + b & \text{for } x < 1 \\ x - x^2 & \text{for } x \geq 1 \end{cases}$$

differentiable at $x = 0$? Sketch the graph of the function f for the values a and b you found.

105. For which value(s) is the function defined by

$$f(x) = \begin{cases} ax^2 & \text{for } x < 2 \\ x + b & \text{for } x \geq 2 \end{cases}$$

differentiable at $x = 0$? Sketch the graph of the function f for the values a and b you found.

106. Group Problem.

True or false: If a function f is continuous at some $x = a$ then it must also be differentiable at $x = a$?

107. Group Problem.

True or false: If a function f is differentiable at some $x = a$ then it must also be continuous at $x = a$?

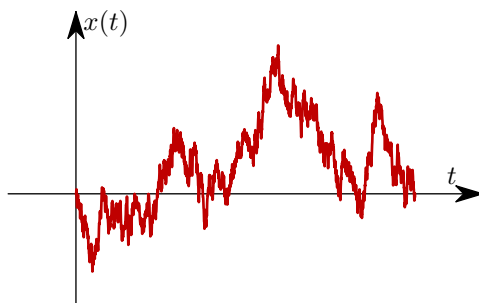


Figure 3. A Brownian motion. Note how the graph doesn't seem to have a tangent anywhere at all.

6. The Differentiation Rules

You could go on and compute more derivatives from the definition. Each time you would have to compute a new limit, and hope that there is some trick that allows you to find that limit. This is fortunately not necessary. It turns out that if you know a few basic derivatives (such as $dx^n/dx = nx^{n-1}$) the you can find derivatives of arbitrarily complicated functions by breaking them into smaller pieces. In this section we'll look at rules which tell you how to differentiate a function which is either the sum, difference, product or quotient of two other functions.

<i>Constant rule:</i>	$c' = 0$	$\frac{dc}{dx} = 0$
<i>Sum rule:</i>	$(u \pm v)' = u' \pm v'$	$\frac{du \pm v}{dx} = \frac{du}{dx} \pm \frac{dv}{dx}$
<i>Product rule:</i>	$(u \cdot v)' = u' \cdot v + u \cdot v'$	$\frac{d(uv)}{dx} = \frac{du}{dx}v + u\frac{dv}{dx}$
<i>Quotient rule:</i>	$\left(\frac{u}{v}\right)' = \frac{u' \cdot v - u \cdot v'}{v^2}$	$\frac{d\frac{u}{v}}{dx} = \frac{v\frac{du}{dx} - u\frac{dv}{dx}}{v^2}$

Table 1. The differentiation rules

The situation is analogous to that of the “limit-properties” $(P_1) \dots (P_6)$ from the previous chapter which allowed us to compute limits without always having to go back to the epsilon-delta definition.

6.1. Sum, product and quotient rules. In the following c and n are constants, u and v are functions of x , and $'$ denotes differentiation. The Differentiation Rules in function notation, and Leibniz notation, are listed in figure 1.

Note that we already proved the Constant Rule in example 2.2. We will now prove the sum, product and quotient rules.

6.2. Proof of the Sum Rule. Suppose that $f(x) = u(x) + v(x)$ for all x where u and v are differentiable. Then

$$\begin{aligned}
 f'(a) &= \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} && \text{(definition of } f') \\
 &= \lim_{x \rightarrow a} \frac{(u(x) + v(x)) - (u(a) + v(a))}{x - a} && \text{(use } f = u + v) \\
 &= \lim_{x \rightarrow a} \left(\frac{u(x) - u(a)}{x - a} + \frac{v(x) - v(a)}{x - a} \right) && \text{(algebra)} \\
 &= \lim_{x \rightarrow a} \frac{u(x) - u(a)}{x - a} + \lim_{x \rightarrow a} \frac{v(x) - v(a)}{x - a} && \text{(limit property)} \\
 &= u'(a) + v'(a) && \text{(definition of } u', v')
 \end{aligned}$$

6.3. Proof of the Product Rule. Let $f(x) = u(x)v(x)$. To find the derivative we must express the change of f in terms of the changes of u and v

$$\begin{aligned}
 f(x) - f(a) &= u(x)v(x) - u(a)v(a) \\
 &= u(x)v(x) - u(x)v(a) + u(x)v(a) - u(a)v(a) \\
 &= u(x)(v(x) - v(a)) + (u(x) - u(a))v(a)
 \end{aligned}$$

Now divide by $x - a$ and let $x \rightarrow a$:

$$\begin{aligned}\lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} &= \lim_{x \rightarrow a} u(x) \frac{v(x) - v(a)}{x - a} + \frac{u(x) - u(a)}{x - a} v(a) \\ &\quad \text{(use the limit properties)} \\ &= \left(\lim_{x \rightarrow a} u(x) \right) \left(\lim_{x \rightarrow a} \frac{v(x) - v(a)}{x - a} \right) + \left(\lim_{x \rightarrow a} \frac{u(x) - u(a)}{x - a} \right) v(a) \\ &= u(a)v'(a) + u'(a)v(a),\end{aligned}$$

as claimed. In this last step we have used that

$$\lim_{x \rightarrow a} \frac{u(x) - u(a)}{x - a} = u'(a) \quad \text{and} \quad \lim_{x \rightarrow a} \frac{v(x) - v(a)}{x - a} = v'(a)$$

and also that

$$\lim_{x \rightarrow a} u(x) = u(a)$$

This last limit follows from the fact that u is continuous, which in turn follows from the fact that u is differentiable.

6.4. Proof of the Quotient Rule . We can break the proof into two parts. First we do the special case where $f(x) = 1/v(x)$, and then we use the product rule to differentiate

$$f(x) = \frac{u(x)}{v(x)} = u(x) \cdot \frac{1}{v(x)}.$$

So let $f(x) = 1/v(x)$. We can express the change in f in terms of the change in v

$$f(x) - f(a) = \frac{1}{v(x)} - \frac{1}{v(a)} = \frac{v(x) - v(a)}{v(x)v(a)}.$$

Dividing by $x - a$ we get

$$\frac{f(x) - f(a)}{x - a} = \frac{1}{v(x)v(a)} \frac{v(x) - v(a)}{x - a}.$$

Now we want to take the limit $x \rightarrow a$. We are given the v is differentiable, so it must also be continuous and hence

$$\lim_{x \rightarrow a} v(x) = v(a).$$

Therefore we find

$$\lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} = \lim_{x \rightarrow a} \frac{1}{v(x)v(a)} \lim_{x \rightarrow a} \frac{v(x) - v(a)}{x - a} = \frac{v'(a)}{v(a)^2}.$$

That completes the first step of the proof. In the second step we use the product rule to differentiate $f = u/v$

$$f' = \left(\frac{u}{v} \right)' = \left(u \cdot \frac{1}{v} \right)' = u' \cdot \frac{1}{v} + u \cdot \left(\frac{1}{v} \right)' = \frac{u'}{v} - u \frac{v'}{v^2} = \frac{u'v - uv'}{v^2}.$$

6.5. A shorter, but not quite perfect derivation of the Quotient Rule . The Quotient Rule can be derived from the Product Rule as follows: if $w = u/v$ then

$$(20) \quad w \cdot v = u$$

By the product rule we have

$$w' \cdot v + w \cdot v' = u',$$

so that

$$w' = \frac{u' - w \cdot v'}{v} = \frac{u' - (u/v) \cdot v'}{v} = \frac{u' \cdot v - u \cdot v'}{v^2}.$$

Unlike the proof in §6.4 above, this argument does not prove that w is differentiable if u and v are. It only says that *if the derivative exists* then it must be what the Quotient Rule says it is.

The trick which is used here, is a special case of a method called “implicit differentiation.” We have an equation (20) which the quotient w satisfies, and from by differentiating this equation we find w' .

6.6. Differentiating a constant multiple of a function . Note that the rule

$$(cu)' = cu'$$

follows from the Constant Rule and the Product Rule.

6.7. Picture of the Product Rule. If u and v are quantities which depend on x , and if increasing x by Δx causes u and v to change by Δu and Δv , then the product of u and v will change by

$$(21) \quad \Delta(uv) = (u + \Delta u)(v + \Delta v) - uv = u\Delta v + v\Delta u + \Delta u\Delta v.$$

If u and v are differentiable functions of x , then the changes Δu and Δv will be of the same order of magnitude as Δx , and thus one expects $\Delta u\Delta v$ to be much smaller. One therefore ignores the last term in (21), and thus arrives at

$$\Delta(uv) = u\Delta v + v\Delta u.$$

Leibniz would now divide by Δx and replace Δ 's by d 's to get the product rule:

$$\frac{\Delta(uv)}{\Delta x} = u \frac{\Delta v}{\Delta x} + v \frac{\Delta u}{\Delta x}.$$

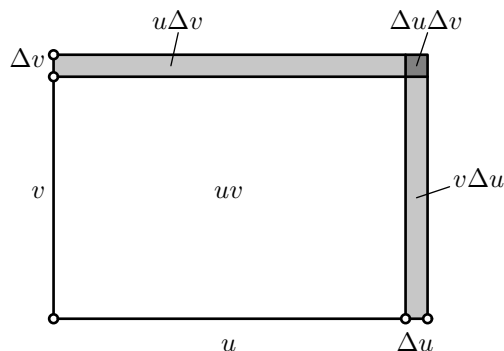


Figure 4. The Product Rule. *How much does the area of a rectangle change if its sides u and v are increased by Δu and Δv ? Most of the increase is accounted for by the two thin rectangles whose areas are $u\Delta v$ and $v\Delta u$. So the increase in area is approximately $u\Delta v + v\Delta u$, which explains why the product rule says $(uv)' = uv' + vu'$.*

7. Differentiating powers of functions

7.1. Product rule with more than one factor. If a function is given as the product of n functions, i.e.

$$f(x) = u_1(x) \times u_2(x) \times \cdots \times u_n(x),$$

then you can differentiate it by applying the product rule $n - 1$ times (there are n factors, so there are $n - 1$ multiplications.)

After the first step you would get

$$f' = u_1'(u_2 \cdots u_n) + u_1(u_2 \cdots u_n)'.$$

In the second step you apply the product rule to $(u_2 u_3 \cdots u_n)'$. This yields

$$\begin{aligned} f' &= u_1' u_2 \cdots u_n + u_1 [u_2' u_3 \cdots u_n + u_2 (u_3 \cdots u_n)'] \\ &= u_1' u_2 \cdots u_n + u_1 u_2' u_3 \cdots u_n + u_1 u_2 (u_3 \cdots u_n)'. \end{aligned}$$

Continuing this way one finds after $n - 1$ applications of the product rule that

$$(22) \quad (u_1 \cdots u_n)' = u_1' u_2 \cdots u_n + u_1 u_2' u_3 \cdots u_n + \cdots + u_1 u_2 u_3 \cdots u_n'.$$

7.2. The Power rule . If all n factors in the previous paragraph are the same, so that the function f is the n^{th} power of some other function,

$$f(x) = (u(x))^n,$$

then all terms in the right hand side of (22) are the same, and, since there are n of them, one gets

$$f'(x) = nu^{n-1}(x)u'(x),$$

or, in Leibniz' notation,

$$(23) \quad \frac{du^n}{dx} = nu^{n-1} \frac{du}{dx}.$$

7.3. The Power Rule for Negative Integer Exponents . We have just proved the power rule (23) assuming n is a positive integer. The rule actually holds for all real exponents n , but the proof is harder.

Here we prove the Power Rule for negative exponents using the Quotient Rule. Suppose $n = -m$ where m is a positive integer. Then the Quotient Rule tells us that

$$(u^n)' = (u^{-m})' = \left(\frac{1}{u^m} \right)' \stackrel{\text{Q.R.}}{=} -\frac{(u^m)'}{(u^m)^2}.$$

Since m is a positive integer, we can use (23), so $(u^m)' = mu^{m-1}$, and hence

$$(u^n)' = -\frac{mu^{m-1} \cdot u'}{u^{2m}} = -mu^{-m-1} \cdot u' = nu^{n-1}u'.$$

7.4. The Power Rule for Rational Exponents . So far we have proved that the power law holds if the exponent n is an integer.

We will now see how you can show that the power law holds even if the exponent n is any fraction, $n = p/q$. The following derivation contains the trick called *implicit differentiation* which we will study in more detail in Section 15.

So let $n = p/q$ where p and q are integers and consider the function

$$w(x) = u(x)^{p/q}.$$

Assuming that both u and w are differentiable functions, we will show that

$$(24) \quad w'(x) = \frac{p}{q} u(x)^{\frac{p}{q}-1} u'(x)$$

Raising both sides to the q th power gives

$$w(x)^q = u(x)^p.$$

Here the exponents p and q are integers, so we may apply the Power Rule to both sides. We get

$$qw^{q-1} \cdot w' = pu^{p-1} \cdot u'.$$

Dividing both sides by qw^{q-1} and substituting $u^{p/q}$ for w gives

$$w' = \frac{pu^{p-1} \cdot u'}{qw^{q-1}} = \frac{pu^{p-1} \cdot u'}{qu^{p(q-1)/q}} = \frac{pu^{p-1} \cdot u'}{qu^{p-(p/q)}} = \frac{p}{q} \cdot u^{(p/q)-1} \cdot u'$$

which is the Power Rule for $n = p/q$.

This proof is flawed because we did not show that $w(x) = u(x)^{p/q}$ is differentiable: we only showed what the derivative should be, *if it exists*.

7.5. Derivative of x^n for integer n . If you choose the function $u(x)$ in the Power Rule to be $u(x) = x$, then $u'(x) = 1$, and hence the derivative of $f(x) = u(x)^n = x^n$ is

$$f'(x) = nu(x)^{n-1}u'(x) = nx^{n-1} \cdot 1 = nx^{n-1}.$$

We already knew this of course.

7.6. Example – differentiate a polynomial. Using the Differentiation Rules you can easily differentiate any polynomial and hence any rational function. For example, using the Sum Rule, the Power Rule with $u(x) = x$, the rule $(cu)' = cu'$, the derivative of the polynomial

$$f(x) = 2x^4 - x^3 + 7$$

is

$$f'(x) = 8x^3 - 3x^2.$$

7.7. Example – differentiate a rational function. By the Quotient Rule the derivative of the function

$$g(x) = \frac{2x^4 - x^3 + 7}{1 + x^2}$$

is

$$\begin{aligned} g'(x) &= \frac{(8x^3 - 3x^2)(1 + x^2) - (2x^4 - x^3 + 7)2x}{(1 + x^2)^2} \\ &= \frac{6x^5 - x^4 + 8x^3 - 3x^2 - 14x}{(1 + x^2)^2}. \end{aligned}$$

If you compare this example with the previous then you see that polynomials simplify when you differentiate them while rational functions become more complicated.

7.8. Derivative of the square root. The derivative of $f(x) = \sqrt{x} = x^{1/2}$ is

$$f'(x) = \frac{1}{2}x^{1/2-1} = \frac{1}{2}x^{-1/2} = \frac{1}{2x^{1/2}} = \frac{1}{2\sqrt{x}}$$

where we used the power rule with $n = 1/2$ and $u(x) = x$.

8. Exercises

108. Let $f(x) = (x^2 + 1)(x^3 + 3)$. Find $f'(x)$ in two ways:

(a) by multiplying and then differentiating,

(b) by using the product rule.

Are your answers the same?

109. Let $f(x) = (1 + x^2)^4$. Find $f'(x)$ in two ways, first by expanding to get an expression for $f(x)$ as a polynomial in x and then differentiating, and then by using the power rule. Are the answers the same?

110. Prove the statement in §6.6, i.e. show that $(cu)' = c(u')$ follows from the product rule.

Compute the derivatives of the following functions.

(try to simplify your answers)

111. $f(x) = x + 1 + (x + 1)^2$

112. $f(x) = \frac{x - 2}{x^4 + 1}$

113. $f(x) = \left(\frac{1}{1 + x}\right)^{-1}$

114. $f(x) = \sqrt{1 - x^2}$

115. $f(x) = \frac{ax + b}{cx + d}$

116. $f(x) = \frac{1}{(1 + x^2)^2}$

117. $f(x) = \frac{x}{1 + \sqrt{x}}$

118. $f(x) = \sqrt{\frac{1 - x}{1 + x}}$

119. $f(x) = \sqrt[3]{x + \sqrt{x}}$

120. $\varphi(t) = \frac{t}{1 + \sqrt{t}}$

121. $g(s) = \sqrt{\frac{1 - s}{1 + s}}$

122. $h(\rho) = \sqrt[3]{\rho + \sqrt{\rho}}$

123. Group Problem.

Using derivatives to approximate numbers.

(a) Find the derivative of $f(x) = x^{4/3}$.

(b) Use (a) to estimate the number

$$\frac{127^{4/3} - 125^{4/3}}{2}$$

approximately without a calculator. Your answer should have the form p/q where p and q are integers. [Hint: Note that $5^3 = 125$ and take a good look at equation (17).]

(c) Approximate in the same way the numbers $\sqrt{143}$ and $\sqrt{145}$ (Hint: $12 \times 12 = 144$).

124. Group Problem.

(Making the product and quotient rules look nicer.)

Instead of looking at the derivative of a function you can look at the ratio of its derivative to the function itself, i.e. you can compute f'/f . This quantity is called the **logarithmic derivative of the function** f for reasons that will become clear later this semester.

(a) Compute the logarithmic derivative of these functions (i.e. find $f'(x)/f(x)$)

$$\begin{aligned} F(x) &= x, & g(x) &= 3x, & h(x) &= x^2 \\ k(x) &= -x^2, & \ell(x) &= 2007x^2, & m(x) &= x^{2007} \end{aligned}$$

(b) Show that for any pair of functions u and v one has

$$\begin{aligned} \frac{(uv)'}{uv} &= \frac{u'}{u} + \frac{v'}{v} \\ \frac{(u/v)'}{u/v} &= \frac{u'}{u} - \frac{v'}{v} \\ \frac{(u^n)'}{u^n} &= n \frac{u'}{u} \end{aligned}$$

125. (a) Find $f'(x)$ and $g'(x)$ if

$$f(x) = \frac{1+x^2}{2x^4+7}, \quad g(x) = \frac{2x^4+7}{1+x^2}.$$

Note that $f(x) = 1/g(x)$.

(b) Is it true that $f'(x) = 1/g'(x)$?

(c) Is it true that $f(x) = g^{-1}(x)$?

(d) Is it true that $f(x) = g(x)^{-1}$?

126. Group Problem. (a) Let $x(t) = (1-t^2)/(1+t^2)$, $y(t) = 2t/(1+t^2)$ and $u(t) = y(t)/x(t)$. Find dx/dt , dy/dt .

(b) Now that you've done (a) there are two different ways of finding du/dt . What are they, and use one of both to find du/dt .

9. Higher Derivatives

9.1. The derivative is a function. If the derivative $f'(a)$ of some function f exists for all a in the domain of f , then we have a new function: namely, for each number in the domain of f we compute the derivative of f at that number. This function is called the **derivative function** of f , and it is denoted by f' . Now that we have agreed that the derivative of a function is a function, we can repeat the process and try to differentiate the derivative. The result, if it exists, is called the **second derivative of f** . It is denoted f'' . The derivative of the second derivative is called the third derivative, written f''' , and so on.

The n th derivative of f is denoted $f^{(n)}$. Thus

$$f^{(0)} = f, \quad f^{(1)} = f', \quad f^{(2)} = f'', \quad f^{(3)} = f''', \dots$$

Leibniz' notation for the n th derivative of $y = f(x)$ is

$$\frac{d^n y}{dx^n} = f^{(n)}(x).$$

9.2. Example. If $f(x) = x^2 - 2x + 3$ then

$$f(x) = x^2 - 2x + 3$$

$$f'(x) = 2x - 2$$

$$f''(x) = 2$$

$$f^{(3)}(x) = 0$$

$$f^{(4)}(x) = 0$$

$\vdots$

All further derivatives of f are zero.

9.3. Operator notation. A common variation on Leibniz' notation for derivatives is the so-called *operator notation*, as in

$$\frac{d(x^3 - x)}{dx} = \frac{d}{dx}(x^3 - x) = 3x^2 - 1.$$

For higher derivatives one can write

$$\frac{d^2 y}{dx^2} = \left(\frac{d}{dx} \right)^2 y$$

Be careful to distinguish the second derivative from the square of the first derivative. Usually

$$\frac{d^2 y}{dx^2} \neq \left(\frac{dy}{dx} \right)^2 \text{ !!!!}$$

10. Exercises

127. The equation

$$(\dagger) \quad \frac{2x}{x^2 - 1} = \frac{1}{x + 1} + \frac{1}{x - 1}$$

holds for all values of x (except $x = \pm 1$), so you should get the same answer if you differentiate both sides. Check this.

Compute the third derivative of $f(x) = 2x/(x^2 - 1)$ by using either the left or right hand side (your choice) of $(\dagger)$.

128. Compute the first, second and third derivatives of the following functions

$$f(x) = (x + 1)^4$$

$$g(x) = (x^2 + 1)^4$$

$$h(x) = \sqrt{x - 2}$$

$$k(x) = \sqrt[3]{x - \frac{1}{x}}$$

129. Find the derivatives of 10th order of the functions

$$f(x) = x^{12} + x^8$$

$$g(x) = \frac{1}{x}$$

$$h(x) = \frac{12}{1 - x}$$

$$k(x) = \frac{x^2}{1 - x}$$

130. Find $f'(x)$, $f''(x)$ and $f^{(3)}(x)$ if

$$f(x) = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \frac{x^4}{24} + \frac{x^5}{120} + \frac{x^6}{720}.$$

131. Group Problem.

(a) Find the 12th derivative of the function $f(x) = \frac{1}{x + 2}$.

(b) Find the n^{th} order derivative of $f(x) = \frac{1}{x + 2}$ (i.e. find a formula for $f^{(n)}(x)$ which is valid for all $n = 0, 1, 2, 3, \dots$).

(c) Find the n^{th} order derivative of $g(x) = \frac{x}{x + 2}$.

132. (About notation.)

(a) Find dy/dx and d^2y/dx^2 if $y = x/(x + 2)$.

(b) Find du/dt and d^2u/dt^2 if $u = t/(t + 2)$. Hint: See previous problem.

(c) Find $\frac{d}{dx} \left(\frac{x}{x + 2} \right)$ and $\frac{d^2}{dx^2} \left(\frac{x}{x + 2} \right)$. Hint: See previous problem.

(d) Find $\frac{d}{dx} \left(\frac{x}{x + 2} \right) \Big|_{x=1}$ and $\frac{d}{dx} \left(\frac{1}{1 + 2} \right)$.

133. Find d^2y/dx^2 and $(dy/dx)^2$ if $y = x^3$.

11. Differentiating Trigonometric functions

The trigonometric functions Sine, Cosine and Tangent are differentiable, and their derivatives are given by the following formulas

$$(25) \quad \frac{d \sin x}{dx} = \cos x, \quad \frac{d \cos x}{dx} = -\sin x, \quad \frac{d \tan x}{dx} = \frac{1}{\cos^2 x}.$$

Note the minus sign in the derivative of the cosine!

PROOF. By definition one has

$$\sin'(x) = \lim_{h \rightarrow 0} \frac{\sin(x + h) - \sin(x)}{h}.$$

To simplify the numerator we use the trigonometric addition formula

$$\sin(\alpha + \beta) = \sin \alpha \cos \beta + \cos \alpha \sin \beta.$$

with $\alpha = x$ and $\beta = h$, which results in

$$\begin{aligned}\frac{\sin(x+h) - \sin(x)}{h} &= \frac{\sin(x)\cos(h) + \cos(x)\sin(h) - \sin(x)}{h} \\ &= \cos(x)\frac{\sin(h)}{h} + \sin(x)\frac{\cos(h) - 1}{h}\end{aligned}$$

Hence by the formulas

$$\lim_{h \rightarrow 0} \frac{\sin(h)}{h} = 1 \quad \text{and} \quad \lim_{h \rightarrow 0} \frac{\cos(h) - 1}{h} = 0$$

from Section 13 we have

$$\begin{aligned}\sin'(x) &= \lim_{h \rightarrow 0} \cos(x)\frac{\sin(h)}{h} + \sin(x)\frac{\cos(h) - 1}{h} \\ &= \cos(x) \cdot 1 + \sin(x) \cdot 0 \\ &= \cos(x).\end{aligned}$$

A similar computation leads to the stated derivative of $\cos x$.

To find the derivative of $\tan x$ we apply the quotient rule to

$$\tan x = \frac{\sin x}{\cos x} = \frac{f(x)}{g(x)}.$$

We get

$$\tan'(x) = \frac{\cos(x)\sin'(x) - \sin(x)\cos'(x)}{\cos^2(x)} = \frac{\cos^2(x) + \sin^2(x)}{\cos^2(x)} = \frac{1}{\cos^2(x)}$$

as claimed. □

12. Exercises

Find the derivatives of the following functions (try to simplify your answers)

134. $f(x) = \sin(x) + \cos(x)$

135. $f(x) = 2\sin(x) - 3\cos(x)$

136. $f(x) = 3\sin(x) + 2\cos(x)$

137. $f(x) = x\sin(x) + \cos(x)$

138. $f(x) = x\cos(x) - \sin x$

139. $f(x) = \frac{\sin x}{x}$

140. $f(x) = \cos^2(x)$

141. $f(x) = \sqrt{1 - \sin^2 x}$

142. $f(x) = \sqrt{\frac{1 - \sin x}{1 + \sin x}}$

143. $\cot(x) = \frac{\cos x}{\sin x}$.

144. Can you find a and b so that the function

$$f(x) = \begin{cases} \cos x & \text{for } x \leq \frac{\pi}{4} \\ a + bx & \text{for } x > \frac{\pi}{4} \end{cases}$$

is differentiable at $x = \pi/4$?

145. Can you find a and b so that the function

$$f(x) = \begin{cases} \tan x & \text{for } x < \frac{\pi}{6} \\ a + bx & \text{for } x \geq \frac{\pi}{6} \end{cases}$$

is differentiable at $x = \pi/4$?

146. If f is a given function, and you have another function g which satisfies $g(x) = f(x) + 12$ for all x , then f and g have the same derivatives. Prove this. [*Hint: it's a short proof – use the differentiation rules.*]

147. **Group Problem.**

Show that the functions

$$f(x) = \sin^2 x \text{ and } g(x) = -\cos^2 x$$

have the same derivative by computing $f'(x)$ and $g'(x)$.

With hindsight this was to be expected – why?

148. Find the first and second derivatives of the functions

$$f(x) = \tan^2 x \text{ and } g(x) = \frac{1}{\cos^2 x}.$$

Hint: remember your trig to reduce work!

A depends on B depends on C depends on . . .

Someone is pumping water into a balloon. Assuming that the balloon is spherical you can say how large it is by specifying its radius R . For a growing balloon this radius will change with time t .

The volume of the balloon is a function of its radius, since the volume of a sphere of radius r is given by

$$V = \frac{4}{3}\pi r^3.$$

We now have two functions, the first f turns tells you the radius r of the balloon at time t ,

$$r = f(t)$$

and the second tells you the volume of the balloon given its radius

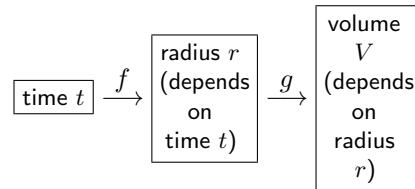
$$V = g(r).$$

The volume of the balloon at time t is then given by

$$V = g(f(t)) = g \circ f(t),$$

i.e. the function which tells you the volume of the balloon at time t is the composition of first f and then g .

Schematically we can summarize this chain of cause-and-effect relations as follows: you could either say that V depends on r , and r depends on t ,



or you could say that V depends directly on t :

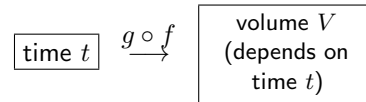


Figure 5. A “real world example” of a composition of functions.

13. The Chain Rule

13.1. Composition of functions. Given two functions f and g , one can define a new function called the **composition of f and g** . The notation for the composition is $f \circ g$, and it is defined by the formula

$$f \circ g(x) = f(g(x)).$$

The domain of the composition is the set of all numbers x for which this formula gives you something well-defined.

For instance, if $f(x) = x^2 + x$ and $g(x) = 2x + 1$ then

$$f \circ g(x) = f(2x + 1) = (2x + 1)^2 + (2x + 1)$$

$$\text{and } g \circ f(x) = g(x^2 + x) = 2(x^2 + x) + 1$$

Note that $f \circ g$ and $g \circ f$ are not the same function in this example (they hardly ever are the same).

If you think of functions as expressing dependence of one quantity on another, then the composition of functions arises as follows. If a quantity z is a function of another quantity y , and if y itself depends on x , then z depends on x via y .

To get $f \circ g$ from the previous example, we could say $z = f(y)$ and $y = g(x)$, so that

$$z = f(y) = y^2 + y \text{ and } y = 2x + 1.$$

Given x one can compute y , and from y one can then compute z . The result will be

$$z = y^2 + y = (2x + 1)^2 + (2x + 1),$$

in other notation,

$$z = f(y) = f(g(x)) = f \circ g(x).$$

One says that **the composition of f and g is the result of substituting g in f** .

13.2. Theorem (Chain Rule). *If f and g are differentiable, so is the composition $f \circ g$.*

The derivative of $f \circ g$ is given by

$$(f \circ g)'(x) = f'(g(x)) g'(x).$$

The chain rule tells you how to find the derivative of the composition $f \circ g$ of two functions f and g provided you now how to differentiate the two functions f and g .

When written in Leibniz' notation the chain rule looks particularly easy. Suppose that $y = g(x)$ and $z = f(y)$, then $z = f \circ g(x)$, and the derivative of z with respect to x is the derivative of the function $f \circ g$. The derivative of z with respect to y is the derivative of the function f , and the derivative of y with respect to x is the derivative of the function g . In short,

$$\frac{dz}{dx} = (f \circ g)'(x), \quad \frac{dz}{dy} = f'(y) \quad \text{and} \quad \frac{dy}{dx} = g'(x)$$

so that the chain rule says

$$(26) \quad \frac{dz}{dx} = \frac{dz}{dy} \frac{dy}{dx}.$$

FIRST PROOF OF THE CHAIN RULE (USING LEIBNIZ' NOTATION). We first consider difference quotients instead of derivatives, i.e. using the same notation as above, we consider the effect of an increase of x by an amount Δx on the quantity z .

If x increases by Δx , then $y = g(x)$ will increase by

$$\Delta y = g(x + \Delta x) - g(x),$$

and $z = f(y)$ will increase by

$$\Delta z = f(y + \Delta y) - f(y).$$

The ratio of the increase in $z = f(g(x))$ to the increase in x is

$$\frac{\Delta z}{\Delta x} = \frac{\Delta z}{\Delta y} \cdot \frac{\Delta y}{\Delta x}.$$

In contrast to dx , dy and dz in equation (26), the Δx , etc. here are finite quantities, so this equation is just algebra: you can cancel the two Δy s. If you let the increase Δx go to zero, then the increase Δy will also go to zero, and the difference quotients converge to the derivatives,

$$\frac{\Delta z}{\Delta x} \longrightarrow \frac{dz}{dx}, \quad \frac{\Delta z}{\Delta y} \longrightarrow \frac{dz}{dy}, \quad \frac{\Delta y}{\Delta x} \longrightarrow \frac{dy}{dx}$$

which immediately leads to Leibniz' form of the quotient rule. □

PROOF OF THE CHAIN RULE. We verify the formula in Theorem 13.2 at some arbitrary value $x = a$, i.e. we will show that

$$(f \circ g)'(a) = f'(g(a)) g'(a).$$

By definition the left hand side is

$$(f \circ g)'(a) = \lim_{x \rightarrow a} \frac{(f \circ g)(x) - (f \circ g)(a)}{x - a} = \lim_{x \rightarrow a} \frac{f(g(x)) - f(g(a))}{x - a}.$$

The two derivatives on the right hand side are given by

$$g'(a) = \lim_{x \rightarrow a} \frac{g(x) - g(a)}{x - a}$$

and

$$f'(g(a)) = \lim_{y \rightarrow a} \frac{f(y) - f(g(a))}{y - g(a)}.$$

Since g is a differentiable function it must also be a continuous function, and hence $\lim_{x \rightarrow a} g(x) = g(a)$. So we can substitute $y = g(x)$ in the limit defining $f'(g(a))$

$$(27) \quad f'(g(a)) = \lim_{y \rightarrow a} \frac{f(y) - f(g(a))}{y - g(a)} = \lim_{x \rightarrow a} \frac{f(g(x)) - f(g(a))}{g(x) - g(a)}.$$

Put all this together and you get

$$\begin{aligned} (f \circ g)'(a) &= \lim_{x \rightarrow a} \frac{f(g(x)) - f(g(a))}{x - a} \\ &= \lim_{x \rightarrow a} \frac{f(g(x)) - f(g(a))}{g(x) - g(a)} \cdot \frac{g(x) - g(a)}{x - a} \\ &= \lim_{x \rightarrow a} \frac{f(g(x)) - f(g(a))}{g(x) - g(a)} \cdot \lim_{x \rightarrow a} \frac{g(x) - g(a)}{x - a} \\ &= f'(g(a)) \cdot g'(a) \end{aligned}$$

which is what we were supposed to prove – the proof seems complete.

There is one flaw in this proof, namely, we have divided by $g(x) - g(a)$, which is not allowed when $g(x) - g(a) = 0$. This flaw can be fixed but we will not go into the details here.² $\square$

13.3. First example. We go back to the functions

$$z = f(y) = y^2 + y \text{ and } y = g(x) = 2x + 1$$

from the beginning of this section. The composition of these two functions is

$$z = f(g(x)) = (2x + 1)^2 + (2x + 1) = 4x^2 + 6x + 2.$$

We can compute the derivative of this composed function, i.e. the derivative of z with respect to x in two ways. First, you simply differentiate the last formula we have:

$$(28) \quad \frac{dz}{dx} = \frac{d(4x^2 + 6x + 2)}{dx} = 8x + 6.$$

The other approach is to use the chain rule:

$$\frac{dz}{dy} = \frac{d(y^2 + y)}{dy} = 2y + 1,$$

and

$$\frac{dy}{dx} = \frac{d(2x + 1)}{dx} = 2.$$

Hence, by the chain rule one has

$$(29) \quad \frac{dz}{dx} = \frac{dz}{dy} \frac{dy}{dx} = (2y + 1) \cdot 2 = 4y + 2.$$

The two answers (28) and (29) should be the same. Once you remember that $y = 2x + 1$ you see that this is indeed true:

$$y = 2x + 1 \implies 4y + 2 = 4(2x + 1) + 2 = 8x + 6.$$

The two computations of dz/dx therefore lead to the same answer. In this example there was no clear advantage in using the chain rule. The chain rule becomes useful when the functions f and g become more complicated.

² Briefly, you have to show that the function

$$h(y) = \begin{cases} \{f(y) - f(g(a))\}/(y - g(a)) & y \neq a \\ f'(g(a)) & y = a \end{cases}$$

is continuous.

13.4. Example where you really need the Chain Rule. We know what the derivative of $\sin x$ with respect to x is, but none of the rules we have found so far tell us how to differentiate $f(x) = \sin(2x)$.

The function $f(x) = \sin 2x$ is the composition of two simpler functions, namely

$$f(x) = g(h(x)) \text{ where } g(u) = \sin u \text{ and } h(x) = 2x.$$

We know how to differentiate each of the two functions g and h :

$$g'(u) = \cos u, \quad h'(x) = 2.$$

Therefore the chain rule implies that

$$f'(x) = g'(h(x))h'(x) = \cos(2x) \cdot 2 = 2 \cos 2x.$$

Leibniz would have decomposed the relation $y = \sin 2x$ between y and x as

$$y = \sin u, \quad u = 2x$$

and then computed the derivative of $\sin 2x$ with respect to x as follows

$$\frac{d \sin 2x}{dx} \stackrel{u=2x}{=} \frac{d \sin u}{dx} = \frac{d \sin u}{du} \cdot \frac{du}{dx} = \cos u \cdot 2 = 2 \cos 2x.$$

13.5. The Power Rule and the Chain Rule. The Power Rule, which says that for any function f and any rational number n one has

$$\frac{d}{dx}(f(x)^n) = n f(x)^{n-1} f'(x),$$

is a special case of the Chain Rule, for one can regard $y = f(x)^n$ as the composition of two functions

$$y = g(u), \quad u = f(x)$$

where $g(u) = u^n$. Since $g'(u) = nu^{n-1}$ the Chain Rule implies that

$$\frac{du^n}{dx} = \frac{du^n}{du} \cdot \frac{du}{dx} = nu^{n-1} \frac{du}{dx}.$$

Setting $u = f(x)$ and $\frac{du}{dx} = f'(x)$ then gives you the Power Rule.

13.6. The volume of an inflating balloon. Consider the “real world example” from page 53 again. There we considered a growing water balloon of radius

$$r = f(t).$$

The volume of this balloon is

$$V = \frac{4}{3}\pi r^3 = \frac{4}{3}\pi f(t)^3.$$

We can regard this as the composition of two functions, $V = g(r) = \frac{4}{3}\pi r^3$ and $r = f(t)$.

According to the chain rule the rate of change of the volume with time is now

$$\frac{dV}{dt} = \frac{dV}{dr} \frac{dr}{dt}$$

i.e. it is the product of the rate of change of the volume with the radius of the balloon and the rate of change of the balloon’s radius with time. From

$$\frac{dV}{dr} = \frac{d\frac{4}{3}\pi r^3}{dr} = 4\pi r^2$$

we see that

$$\frac{dV}{dr} = 4\pi r^2 \frac{dr}{dt}.$$

For instance, if the radius of the balloon is growing at 0.5inch/sec, and if its radius is $r = 3.0$ inch, then the volume is growing at a rate of

$$\frac{dV}{dt} = 4\pi(3.0\text{inch})^2 \times 0.5\text{inch/sec} \approx 57\text{inch}^3/\text{sec}.$$

13.7. A more complicated example. Suppose you needed to find the derivative of

$$y = h(x) = \frac{\sqrt{x+1}}{(\sqrt{x+1}+1)^2}$$

We can write this function as a composition of two simpler functions, namely,

$$y = f(u), \quad u = g(x),$$

with

$$f(u) = \frac{u}{(u+1)^2} \text{ and } g(x) = \sqrt{x+1}.$$

The derivatives of f and g are

$$f'(u) = \frac{1 \cdot (u+1)^2 - u \cdot 2(u+1)}{(u+1)^4} = \frac{u+1-2}{(u+1)^3} = \frac{u-1}{(u+1)^3},$$

and

$$g'(x) = \frac{1}{2\sqrt{x+1}}.$$

Hence the derivative of the composition is

$$h'(x) = \frac{d}{dx} \left\{ \frac{\sqrt{x+1}}{(\sqrt{x+1}+1)^2} \right\} = f'(u)g'(x) = \frac{u-1}{(u+1)^3} \cdot \frac{1}{2\sqrt{x+1}}.$$

The result should be a function of x , and we achieve this by replacing all u 's with $u = \sqrt{x+1}$:

$$\frac{d}{dx} \left\{ \frac{\sqrt{x+1}}{(\sqrt{x+1}+1)^2} \right\} = \frac{\sqrt{x+1}-1}{(\sqrt{x+1}+1)^3} \cdot \frac{1}{2\sqrt{x+1}}.$$

The last step (where you replace u by its definition in terms of x) is important because the problem was presented to you with only x and y as variables while u was a variable you introduced yourself to do the problem.

Sometimes it is possible to apply the Chain Rule without introducing new letters, and you will simply think “the derivative is the derivative of the outside with respect to the inside times the derivative of the inside.” For instance, to compute

$$\frac{d}{dx} \frac{4 + \sqrt{7+x^3}}{dx}$$

you could set $u = 7 + x^3$, and compute

$$\frac{d}{dx} \frac{4 + \sqrt{7+x^3}}{dx} = \frac{d}{du} \frac{4 + \sqrt{u}}{du} \cdot \frac{du}{dx}.$$

Instead of writing all this explicitly, you could think of $u = 7 + x^3$ as the function “inside the square root,” and think of $4 + \sqrt{u}$ as “the outside function.” You would then immediately write

$$\frac{d}{dx} (4 + \sqrt{7+x^3}) = \frac{1}{2\sqrt{7+x^3}} \cdot 3x^2.$$

13.8. The Chain Rule and composing more than two functions. Often we have to apply the Chain Rule more than once to compute a derivative. Thus if $y = f(u)$, $u = g(v)$, and $v = h(x)$ we have

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dv} \cdot \frac{dv}{dx}.$$

In functional notation this is

$$(f \circ g \circ h)'(x) = f'(g(h(x))) \cdot g'(h(x)) \cdot h'(x).$$

Note that each of the three derivatives on the right is evaluated at a different point. Thus if $b = h(a)$ and $c = g(b)$ the Chain Rule is

$$\left. \frac{dy}{dx} \right|_{x=a} = \left. \frac{dy}{du} \right|_{u=c} \cdot \left. \frac{du}{dv} \right|_{v=b} \cdot \left. \frac{dv}{dx} \right|_{x=a}.$$

For example, if $y = \frac{1}{1 + \sqrt{9 + x^2}}$, then $y = 1/(1 + u)$ where $u = 1 + \sqrt{v}$ and $v = 9 + x^2$ so

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dv} \cdot \frac{dv}{dx} = -\frac{1}{(1+u)^2} \cdot \frac{1}{2\sqrt{v}} \cdot 2x.$$

so

$$\left. \frac{dy}{dx} \right|_{x=4} = \left. \frac{dy}{du} \right|_{u=6} \cdot \left. \frac{du}{dv} \right|_{v=25} \cdot \left. \frac{dv}{dx} \right|_{x=4} = -\frac{1}{7} \cdot \frac{1}{10} \cdot 8.$$

14. Exercises

- 149.** Let $y = \sqrt{1 + x^3}$ and find dy/dx using the Chain Rule. Say what plays the role of $y = f(u)$ and $u = g(x)$.

- 150.** Repeat the previous exercise with

$$y = (1 + \sqrt{1 + x})^3.$$

- 151.** Alice and Bob differentiated $y = \sqrt{1 + x^3}$ with respect to x differently. Alice wrote $y = \sqrt{u}$ and $u = 1 + x^3$ while Bob wrote $y = \sqrt{1 + v}$ and $v = x^3$. Assuming neither one made a mistake, did they get the same answer?

- 152.** Let $y = u^3 + 1$ and $u = 3x + 7$. Find $\frac{dy}{dx}$ and $\frac{dy}{du}$. Express the former in terms of x and the latter in terms of u .

- 153.** Suppose that $f(x) = \sqrt{x}$, $g(x) = 1 + x^2$, $v(x) = f \circ g(x)$, $w(x) = g \circ f(x)$. Find formulas for $v(x)$, $w(x)$, $v'(x)$, and $w'(x)$.

Compute the following derivatives

154. $f(x) = \sin 2x - \cos 3x$

155. $f(x) = \sin \frac{\pi}{x}$

156. $f(x) = \sin(\cos 3x)$

157. $f(x) = \frac{\sin x^2}{x^2}$

158. $f(x) = \tan \sqrt{1 + x^2}$

159. $f(x) = \cos^2 x - \cos x^2$

160. Group Problem.

Moe is pouring water into a glass. At time t (seconds) the height of the water in the glass is $h(t)$ (inch). The ACME glass company, which made the glass, says that the volume in the glass to height h is $V = 1.2 h^2$ (fluid ounces).

(a) The water height in the glass is rising at 2 inch per second at the moment that the height is 2 inch. How fast is Moe pouring water into the glass?

(b) If Moe pours water at a rate of 1 ounce per second, then how fast is the water level in the glass going up when it is 3 inches?

(c) Moe pours water at 1 ounce per second, and at some moment the water level is going up at 0.5 inch per second. What is the water level at that moment?

- 161.** Find the derivative of $f(x) = x \cos \frac{\pi}{x}$ at the point C in Figure 3.

- 162.** Suppose that $f(x) = x^2 + 1$, $g(x) = x + 5$, and

$$v = f \circ g, \quad w = g \circ f, \quad p = f \cdot g, \quad q = g \cdot f.$$

Find $v(x)$, $w(x)$, $p(x)$, and $q(x)$.

163. Group Problem.

Suppose that the functions f and g and their derivatives with respect to x have the following values at $x = 0$ and $x = 1$.

x	$f(x)$	$g(x)$	$f'(x)$	$g'(x)$
0	1	1	5	1/3
1	3	-4	-1/3	-8/3

Define

$$v(x) = f(g(x)), \quad w(x) = g(f(x)),$$

$$p(x) = f(x)g(x), \quad q(x) = g(x)f(x).$$

Evaluate $v(0)$, $w(0)$, $p(0)$, $q(0)$, $v'(0)$ and $w'(0)$, $p'(0)$, $q'(0)$. If there is insufficient information to answer the question, so indicate.

- 164.** A differentiable function f satisfies $f(3) = 5$, $f(9) = 7$, $f'(3) = 11$ and $f'(9) = 13$. Find an equation for the tangent line to the curve $y = f(x^2)$ at the point $(x, y) = (3, 7)$.

- 165.** There is a function f whose second derivative satisfies

$$(\dagger) \quad f''(x) = -64f(x).$$

(a) One such function is $f(x) = \sin ax$, provided you choose the right constant a : Which value should a have?

(b) For which choices of the constants A , a and b does the function $f(x) = A \sin(ax + b)$ satisfy $(\dagger)$?

166. Group Problem.

A cubical sponge, hereafter referred to as 'Bob', is absorbing water, which causes him to expand. His side at time t is $S(t)$. His volume is $V(t)$.

(a) What is the relation between $S(t)$ and $V(t)$, i.e. can you find a function f so that $V(t) = f(S(t))$?

(b) Describe the meaning of the derivatives $S'(t)$ and $V'(t)$ in one plain english sentence each. If we measure lengths in inches and time in minutes, then what units do t , $S(t)$, $V(t)$, $S'(t)$ and $V'(t)$ have?

(c) What is the relation between $S'(t)$ and $V'(t)$?

(d) At the moment that Bob's volume is 8 cubic inches, he is absorbing water at a rate of 2 cubic inch per minute. How fast is his side $S(t)$ growing?

15. Implicit differentiation

15.1. The recipe. Recall that an implicitly defined function is a function $y = f(x)$ which is defined by an equation of the form

$$F(x, y) = 0.$$

We call this equation the **defining equation** for the function $y = f(x)$. To find $y = f(x)$ for a given value of x you must solve the defining equation $F(x, y) = 0$ for y .

Here is a recipe for computing the derivative of an implicitly defined function.

- (1) Differentiate the equation $F(x, y) = 0$; you may need the chain rule to deal with the occurrences of y in $F(x, y)$;
- (2) You can rearrange the terms in the result of step 1 so as to get an equation of the form

$$(30) \quad G(x, y) \frac{dy}{dx} + H(x, y) = 0,$$

where G and H are expressions containing x and y but not the derivative.

- (3) Solve the equation in step 2 for $\frac{dy}{dx}$:

$$(31) \quad \frac{dy}{dx} = -\frac{H(x, y)}{G(x, y)}$$

- (4) If you also have an explicit description of the function (i.e. a formula expressing $y = f(x)$ in terms of x) then you can substitute $y = f(x)$ in the expression (31) to get a formula for dy/dx in terms of x only.

Often no explicit formula for y is available and you can't take this last step. In that case (31) is as far as you can go.

Observe that by following this procedure you will get a formula for the derivative $\frac{dy}{dx}$ which contains both x and y .

15.2. Dealing with equations of the form $F_1(x, y) = F_2(x, y)$. If the implicit definition of the function is not of the form $F(x, y) = 0$ but rather of the form $F_1(x, y) = F_2(x, y)$ then you move all terms to the left hand side, and proceed as above. E.g. to deal with a function $y = f(x)$ which satisfies

$$y^2 + x = xy$$

you rewrite this equation as

$$y^2 + x - xy = 0$$

and set $F(x, y) = y^2 + x - xy$.

15.3. Example – Derivative of $\sqrt[4]{1-x^4}$. Consider the function

$$f(x) = \sqrt[4]{1-x^4}, \quad -1 \leq x \leq 1.$$

We will compute its derivative in two ways: first the direct method, and then using the method of implicit differentiation (i.e. the recipe above).

The direct approach goes like this:

$$\begin{aligned} f'(x) &= \frac{d(1-x^4)^{1/4}}{dx} \\ &= \frac{1}{4}(1-x^4)^{-3/4} \frac{d(1-x^4)}{dx} \\ &= \frac{1}{4}(1-x^4)^{-3/4} (-4x^3) \\ &= -\frac{x^3}{(1-x^4)^{3/4}} \end{aligned}$$

To find the derivative using implicit differentiation we must first find a nice implicit description of the function. For instance, we could decide to get rid of all roots or fractional exponents in the function and point out that $y = \sqrt[4]{1-x^4}$ satisfies the equation $y^4 = 1-x^4$. So our implicit description of the function $y = f(x) = \sqrt[4]{1-x^4}$ is

$$x^4 + y^4 - 1 = 0; \quad \text{The defining function is therefore } F(x, y) = x^4 + y^4 - 1$$

Differentiate both sides with respect to x (and remember that $y = f(x)$, so y here is a function of x), and you get

$$\frac{dx^4}{dx} + \frac{dy^4}{dx} - \frac{d1}{dx} = 0 \implies 4x^3 + 4y^3 \frac{dy}{dx} = 0.$$

The expressions G and H from equation (30) in the recipe are $G(x, y) = 4y^3$ and $H(x, y) = 4x^3$.

This last equation can be solved for dy/dx :

$$\frac{dy}{dx} = -\frac{x^3}{y^3}.$$

This is a nice and short form of the derivative, but it contains y as well as x . To express dy/dx in terms of x only, and remove the y dependency we use $y = \sqrt[4]{1-x^4}$. The result is

$$f'(x) = \frac{dy}{dx} = -\frac{x^3}{y^3} = -\frac{x^3}{(1-x^4)^{3/4}}.$$

15.4. Another example. Let f be a function defined by

$$y = f(x) \iff 2y + \sin y = x, \text{ i.e. } 2y + \sin y - x = 0.$$

For instance, if $x = 2\pi$ then $y = \pi$, i.e. $f(2\pi) = \pi$.

To find the derivative dy/dx we differentiate the defining equation

$$\frac{d(2y + \sin y - x)}{dx} = \frac{d0}{dx} \implies 2\frac{dy}{dx} + \cos y \frac{dy}{dx} - \frac{dx}{dx} = 0 \implies (2 + \cos y) \frac{dy}{dx} - 1 = 0.$$

Solve for $\frac{dy}{dx}$ and you get

$$f'(x) = \frac{1}{2 + \cos y} = \frac{1}{2 + \cos f(x)}.$$

If we were asked to find $f'(2\pi)$ then, since we know $f(2\pi) = \pi$, we could answer

$$f'(2\pi) = \frac{1}{2 + \cos \pi} = \frac{1}{2 - 1} = 1.$$

If we were asked $f'(\pi/2)$, then all we would be able to say is

$$f'(\pi/2) = \frac{1}{2 + \cos f(\pi/2)}.$$

To say more we would first have to find $y = f(\pi/2)$, which one does by solving

$$2y + \sin y = \frac{\pi}{2}.$$

15.5. Derivatives of Arc Sine and Arc Tangent. Recall that

$$y = \arcsin x \iff x = \sin y \text{ and } -\frac{\pi}{2} \leq y \leq \frac{\pi}{2},$$

and

$$y = \arctan x \iff x = \tan y \text{ and } -\frac{\pi}{2} < y < \frac{\pi}{2}.$$

15.6. Theorem.

$$\frac{d \arcsin x}{dx} = \frac{1}{\sqrt{1-x^2}}$$

$$\frac{d \arctan x}{dx} = \frac{1}{1+x^2}$$

PROOF. If $y = \arcsin x$ then $x = \sin y$. Differentiate this relation

$$\frac{dx}{dx} = \frac{d \sin y}{dx}$$

and apply the chain rule. You get

$$1 = (\cos y) \frac{dy}{dx},$$

and hence

$$\frac{dy}{dx} = \frac{1}{\cos y}.$$

How do we get rid of the y on the right hand side? We know $x = \sin y$, and also $-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$. Therefore

$$\sin^2 y + \cos^2 y = 1 \implies \cos y = \pm \sqrt{1 - \sin^2 y} = \pm \sqrt{1 - x^2}.$$

Since $-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$ we know that $\cos y \geq 0$, so we must choose the positive square root. This leaves us with $\cos y = \sqrt{1 - x^2}$, and hence

$$\frac{dy}{dx} = \frac{1}{\sqrt{1-x^2}}.$$

The derivative of $\arctan x$ is found in the same way, and you should really do this yourself. $\square$

16. Exercises

For each of the following problems find the derivative $f'(x)$ if $y = f(x)$ satisfies the given equation. State what the expressions $F(x, y)$, $G(x, y)$ and $H(x, y)$ from the recipe in the beginning of this section are.

If you can find an explicit description of the function $y = f(x)$, say what it is.

167. $xy = \frac{\pi}{6}$

168. $\sin(xy) = \frac{1}{2}$

169. $\frac{xy}{x+y} = 1$

170. $x + y = xy$

171. $(y-1)^2 + x = 0$

172. $(y+1)^2 + y - x = 0$

173. $(y-x)^2 + x = 0$

174. $(y+x)^2 + 2y - x = 0$

175. $(y^2 - 1)^2 + x = 0$

176. $(y^2 + 1)^2 - x = 0$

177. $x^3 + xy + y^3 = 3$

178. $\sin x + \sin y = 1$

179. $\sin x + xy + y^5 = \pi$

180. $\tan x + \tan y = 1$

For each of the following explicitly defined functions find an implicit definition which does not involve taking roots. Then use this description to find the derivative dy/dx .

181. $y = f(x) = \sqrt{1-x}$

182. $y = f(x) = \sqrt[4]{x+x^2}$

183. $y = f(x) = \sqrt{1-\sqrt{x}}$

184. $y = f(x) = \sqrt[4]{x-\sqrt{x}}$

185. $y = f(x) = \sqrt[3]{\sqrt{2x+1} - x^2}$

186. $y = f(x) = \sqrt[4]{x+x^2}$

187. $y = f(x) = \sqrt[3]{x - \sqrt{2x + 1}}$

188. $y = f(x) = \sqrt[4]{\sqrt[3]{x}}$

189. Group Problem.

(Inverse trig review) Simplify the following expressions, and indicate for which values of x (or θ , or ...) your simplification is valid. In case of doubt, try plotting the function on a graphing calculator.

- | | |
|----------------------------|----------------------------|
| (a) $\sin \arcsin x$ | (e) $\tan \arctan z$ |
| (b) $\cos \arcsin x$ | (f) $\tan \arcsin \theta$ |
| (c) $\arctan(\tan \theta)$ | (g) $\arcsin(\sin \theta)$ |
| (d) $\cot \arctan x$ | (h) $\cot \arcsin x$ |

Now that you know the derivatives of $\arcsin$ and $\arctan$, you can find the derivatives of the following functions. What are they?

190. $f(x) = \arcsin(2x)$

191. $f(x) = \arcsin \sqrt{x}$

192. $f(x) = \arctan(\sin x)$

193. $f(x) = \sin \arctan x$

194. $f(x) = (\arcsin x)^2$

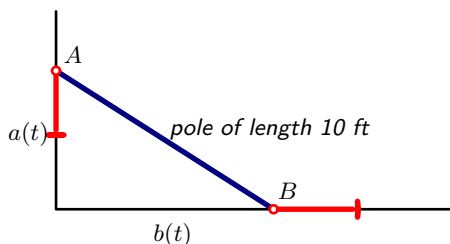
195. $f(x) = \frac{1}{1 + (\arctan x)^2}$

196. $f(x) = \sqrt{1 - (\arcsin x)^2}$

197. $f(x) = \frac{\arctan x}{\arcsin x}$

**PROBLEMS ON
RELATED RATES**

198. A 10 foot long pole has one end (B) on the floor and another (A) against a wall. If the bottom of the pole is 8 feet away from the wall, and if it is sliding away from the wall at 7 feet per second, then with what speed is the top (A) going down?



199. A pole 10 feet long rests against a vertical wall. If the bottom of the pole slides away from the wall at a speed of 2 ft/s, how fast is the angle between the top of the pole and the wall changing when the angle is $\pi/4$ radians?

200. A pole 13 meters long is leaning against a wall. The bottom of the pole is pulled along the ground away from the wall at the rate of 2 m/s. How fast is its height on the wall decreasing when the foot of the pole is 5 m away from the wall?

201. Group Problem.

A television camera is positioned 4000 ft from the base of a rocket launching pad. A rocket rises vertically and its speed is 600 ft/s when it has risen 3000 feet.

(a) How fast is the distance from the television camera to the rocket changing at that moment?

(b) How fast is the camera's angle of elevation changing at that same moment? (Assume that the television camera points toward the rocket.)

202. Group Problem.

A 2-foot tall dog is walking away from a streetlight which is on a 10-foot pole. At a certain moment, the tip of the dog's shadow is moving away from the streetlight at 5 feet per second. How fast is the dog walking at that moment?

203. An isosceles triangle is changing its shape: the lengths of the two equal sides remain fixed at 2 inch, but the angle $\theta(t)$ between them changes.

Let $A(t)$ be the area of the triangle at time t . If the area increases at a constant rate of $0.5 \text{ inch}^2/\text{sec}$, then how fast is the angle increasing or decreasing when $\theta = 60^\circ$?

204. A point P is moving in the first quadrant of the plane. Its motion is parallel to the x -axis; its distance to the x -axis is always 10 (feet). Its velocity is 3 feet per second to the left. We write θ for the angle between the positive x -axis and the line segment from the origin to P .

(a) Make a drawing of the point P .

(b) Where is the point when $\theta = \pi/3$?

(c) Compute the rate of change of the angle θ at the moment that $\theta = \frac{\pi}{3}$.

205. The point Q is moving on the line $y = x$ with velocity 3 m/sec. Find the rate of change of the following quantities at the moment in which Q is at the point $(1, 1)$:

(a) the distance from Q to the origin,

(b) the distance from Q to the point $R(2, 0)$,

(c) the angle $\angle ORQ$ where R is again the point $R(2, 0)$.

206. A point P is sliding on the parabola with equation $y = x^2$. Its x -coordinate is increasing at a constant rate of 2 feet/minute.

Find the rate of change of the following quantities at the moment that P is at $(3, 9)$:

(a) the distance from P to the origin,

(b) the area of the rectangle whose lower left corner is the origin and whose upper right corner is P ,

- (c) the slope of the tangent to the parabola at P ,
- (d) the angle $\angle OPQ$ where Q is the point $(0, 3)$.

207. Group Problem.

A certain amount of gas is trapped in a cylinder with a piston. The **ideal gas law** from thermodynamics says that if the cylinder is not heated, and if the piston moves slowly, then one has

$$pV = CT$$

where p is the pressure in the gas, V is its volume, T its temperature (in degrees Kelvin) and C is a constant depending on the amount of gas trapped in the cylinder.

(a) If the pressure is 10psi (pounds per square inch), if the volume is 25inch^3 , and if the piston is moving so

that the gas volume is expanding at a rate of 2inch^3 per minute, then what is the rate of change of the pressure?

(b) The ideal gas law turns out to be only approximately true. A more accurate description of gases is given by **van der Waals' equation of state**, which says that

$$\left(p + \frac{a}{V^2}\right)(V - b) = C$$

where a, b, C are constants depending on the temperature and the amount and type of gas in the cylinder.

Suppose that the cylinder contains fictitious gas for which one has $a = 12$ and $b = 3$. Suppose that at some moment the volume of gas is 12in^3 , the pressure is 25psi and suppose the gas is expanding at 2inch^3 per minute. Then how fast is the pressure changing?

Graph Sketching and Max-Min Problems

The signs of the first and second derivatives of a function tell us something about the shape of its graph. In this chapter we learn how to find that information.

1. Tangent and Normal lines to a graph

The slope of the tangent the tangent to the graph of f at the point $(a, f(a))$ is

$$(32) \quad m = f'(a)$$

and hence the equation for the tangent is

$$(33) \quad y = f(a) + f'(a)(x - a).$$

The slope of the normal line to the graph is $-1/m$ and thus one could write the equation for the normal as

$$(34) \quad y = f(a) - \frac{x - a}{f'(a)}.$$

When $f'(a) = 0$ the tangent is horizontal, and hence the normal is vertical. In this case the equation for the normal cannot be written as in (34), but instead one gets the simpler equation

$$y = f(a).$$

Both cases are covered by this form of the equation for the normal

$$(35) \quad x = a + f'(a)(f(a) - y).$$

Both (35) and (34) are formulas that you shouldn't try to remember. It is easier to remember that if the slope of the tangent is $m = f'(a)$, then the slope of the normal is $-1/m$.

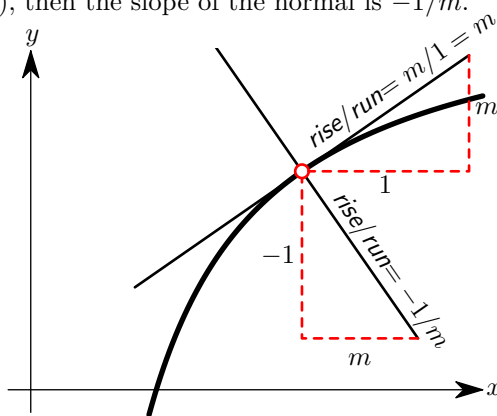


Figure 1. Why “slope of normal = $\frac{-1}{\text{slope of tangent}}$.”

2. The Intermediate Value Theorem

It is said that a function is continuous if you can draw its graph without taking your pencil off the paper. A more precise version of this statement is the *Intermediate Value Theorem*:

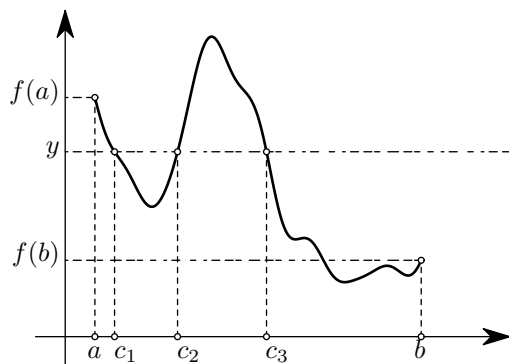


Figure 2. The Intermediate Value Theorem says that a continuous function must attain any given value y between $f(a)$ and $f(b)$ at least once. In this example there are three values of c for which $f(c) = y$ holds.

2.1. Intermediate Value Theorem. If f is a continuous function on an interval $a \leq x \leq b$, and if y is some number between $f(a)$ and $f(b)$, then there is a number c with $a \leq c \leq b$ such that $f(c) = y$.

Here “ y between $f(a)$ and $f(b)$ ” means that $f(a) \leq y \leq f(b)$ if $f(a) \leq f(b)$, and $f(b) \leq y \leq f(a)$ if $f(b) \leq f(a)$.

2.2. Example – Square root of 2. Consider the function $f(x) = x^2$. Since $f(1) < 2$ and $f(2) = 4 > 2$ the intermediate value theorem with $a = 1$, $b = 2$, $y = 2$ tells us that there is a number c between 1 and 2 such that $f(c) = 2$, i.e. for which $c^2 = 2$. So the theorem tells us that the square root of 2 exists.

2.3. Example – The equation $\theta + \sin \theta = \frac{\pi}{2}$. Consider the function $f(x) = x + \sin x$. It is a continuous function at all x , so from $f(0) = 0$ and $f(\pi) = \pi$ it follows that there is a number θ between 0 and π such that $f(\theta) = \pi/2$. In other words, the equation

$$(36) \quad \theta + \sin \theta = \frac{\pi}{2}$$

has a solution θ with $0 \leq \theta \leq \pi/2$. Unlike the previous example, where we knew the solution was $\sqrt{2}$, there is no simple formula for the solution to (36).

2.4. Example – Solving $1/x = 0$. If we apply the intermediate value theorem to the function $f(x) = 1/x$ on the interval $[a, b] = [-1, 1]$, then we see that for any y between $f(a) = f(-1) = -1$ and $f(b) = f(1) = 1$ there is a number c in the interval $[-1, 1]$ such that $1/c = y$. For instance, we could choose $y = 0$ (that’s between -1 and $+1$), and conclude that there is some c with $-1 \leq c \leq 1$ and $1/c = 0$.

But there is no such c , because $1/c$ is never zero! So we have done something wrong, and the mistake we made is that we overlooked that our function $f(x) = 1/x$ is not defined on the **whole** interval $-1 \leq x \leq 1$ because it is not defined at $x = 0$. *The moral: always check the hypotheses of a theorem before you use it!*

3. Exercises

208. Where does the normal to the graph of $y = x^2$ at the point $(1, 1)$ intersect the x -axis?

209. Where does the tangent to the graph of $y = x^2$ at the point (a, a^2) intersect the x -axis?

210. Where does the normal to the graph of $y = x^2$ at the point (a, a^2) intersect the x -axis?

211. Where does the normal to the graph of $y = \sqrt{x}$ at the point $(a, \sqrt{a})$ intersect the x -axis?

212. Does the graph of $y = x^4 - 2x^2 + 2$ have any horizontal tangents? If so, where?

Does the graph of the same function have any vertical tangents?

Does it have vertical normals?

Does it have horizontal normals?

213. At some point $(a, f(a))$ on the graph of $f(x) = -1 + 2x - x^2$ the tangent to this graph goes through the origin. Which point is it?

214. Find equations for the tangent and normal lines

to the curve ... at the point...

- | | |
|----------------------|--------|
| (a) $y = 4x/(1+x^2)$ | (1, 2) |
| (b) $y = 8/(4+x^2)$ | (2, 1) |
| (c) $y^2 = 2x + x^2$ | (2, 2) |
| (d) $xy = 3$ | (1, 3) |

satisfies $f(-1) = -2$ and $f(+1) = +2$, so, by the Intermediate Value Theorem, there should be some value c between -1 and $+1$ such that $f(c) = 0$. **True or False?**

215. Group Problem.

The function

$$f(x) = \frac{x^2 + |x|}{x}$$

216. Find the equation for the tangents to the graph of the Backward Sine at the points $x = 1$, $x = \frac{1}{2}$ and at D (see Figure 2 in §7.3.)

217. Find the equation for the tangent to the graph of the Backward Cosine in a Bow Tie at the point C (see Figure 3 in §9.3)

4. Finding sign changes of a function

The intermediate value theorem implies the following very useful fact.

4.1. Theorem. *If f is continuous function on some interval $a < x < b$, and if $f(x) \neq 0$ for all x in this interval, then $f(x)$ is either positive for all $a < x < b$ or else it is negative for all $a < x < b$.*

PROOF. The theorem says that there can't be two numbers $a < x_1 < x_2 < b$ such that $f(x_1)$ and $f(x_2)$ have opposite signs. If there were two such numbers then the intermediate value theorem would imply that somewhere between x_1 and x_2 there was a c with $f(c) = 0$. But we are assuming that $f(c) \neq 0$ whenever $a < c < b$. $\square$

4.2. Example. Consider

$$f(x) = (x-3)(x-1)^2(2x+1)^3.$$

The zeros of f (i.e. the solutions of $f(x) = 0$) are $-\frac{1}{2}, 1, 3$. These numbers split the real line into four intervals

$$(-\infty, -\frac{1}{2}), \quad (-\frac{1}{2}, 1), \quad (1, 3), \quad (3, \infty).$$

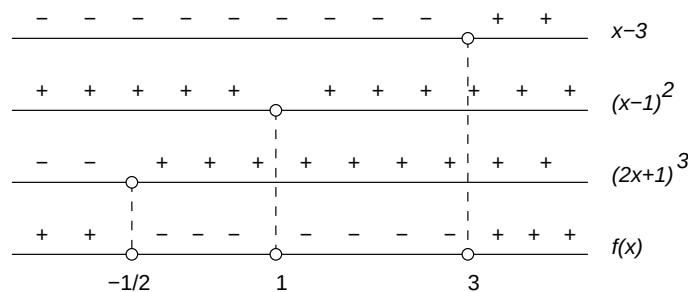
Theorem 4.1 tells us that $f(x)$ cannot change its sign in any of these intervals. For instance, $f(x)$ has the same sign for all x in the first interval $(-\infty, -\frac{1}{2})$. Now we choose a number we like from this interval (e.g. -1) and find the sign of $f(-1)$: $f(-1) = (-4)(-2)^2(-3)^3$ is positive. Therefore $f(x) > 0$ for all x in the interval $(-\infty, -\frac{1}{2})$. In the same we find

$$\begin{aligned} f(-1) &= (-4)(-2)^2(-3)^3 > 0 &\implies f(x) > 0 \text{ for } x < -\frac{1}{2} \\ f(0) &= (-3)(-1)^2(1)^3 < 0 &\implies f(x) < 0 \text{ for } -\frac{1}{2} < x < 1 \\ f(2) &= (-1)(1)^2(5)^3 < 0 &\implies f(x) < 0 \text{ for } 1 < x < 3 \\ f(4) &= (1)(3)^2(9)^3 > 0 &\implies f(x) > 0 \text{ for } x > 3. \end{aligned}$$

If you know all the zeroes of a continuous function, then this method allows you to decide where the function is positive or negative. However, when the given function is factored into easy functions, as in this example, there is a different way of finding the signs of f . For each of the factors $x-3$, $(x-1)^2$ and $(2x+1)^3$ it is easy to determine the sign, for any given x . These signs can only change at a zero of the factor. Thus we have

- $x-3$ is positive for $x > 3$ and negative for $x < 3$;
- $(x-1)^2$ is always positive (except at $x = 1$);
- $(2x+1)^3$ is positive for $x > -\frac{1}{2}$ and negative for $x < -\frac{1}{2}$.

Multiplying these signs we get the same conclusions as above. We can summarize this computation in the following diagram:



5. Increasing and decreasing functions

Here are four very similar definitions – look closely to see how they differ.

- A function is called **increasing** if $a < b$ implies $f(a) < f(b)$ for all numbers a and b in the domain of f .
- A function is called **decreasing** if $a < b$ implies $f(a) > f(b)$ for all numbers a and b in the domain of f .
- The function f is called **non-decreasing** if $a < b$ implies $f(a) \leq f(b)$ for all numbers a and b in the domain of f .
- The function f is called **non-increasing** if $a < b$ implies $f(a) \geq f(b)$ for all numbers a and b in the domain of f .

You can summarize these definitions as follows:

f is ...	if for all a and b one has...
Increasing:	$a < b \implies f(a) < f(b)$
Decreasing:	$a < b \implies f(a) > f(b)$
Non-increasing:	$a < b \implies f(a) \geq f(b)$
Non-decreasing:	$a < b \implies f(a) \leq f(b)$

The sign of the derivative of f tells you if f is increasing or not. More precisely:

5.1. Theorem. *If a function is non-decreasing on an interval $a < x < b$ then $f'(x) \geq 0$ for all x in that interval.*

If a function is non-increasing on an interval $a < x < b$ then $f'(x) \leq 0$ for all x in that interval.

For instance, if f is non-decreasing, then for any given x and any positive Δx one has $f(x + \Delta x) \geq f(x)$ and hence

$$\frac{f(x + \Delta x) - f(x)}{\Delta x} \geq 0.$$

Now let $\Delta x \searrow 0$ and you find that

$$f'(x) = \lim_{\Delta x \searrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x} \geq 0.$$

What about the converse, i.e. if you know the sign of f' then what can you say about f ? For this we have the following

5.2. Theorem. *Suppose f is a differentiable function on an interval (a, b) .*

If $f'(x) > 0$ for all $a < x < b$, then f is increasing.

If $f'(x) < 0$ for all $a < x < b$, then f is decreasing.

The proof is based on the Mean Value theorem which also finds use in many other situations:

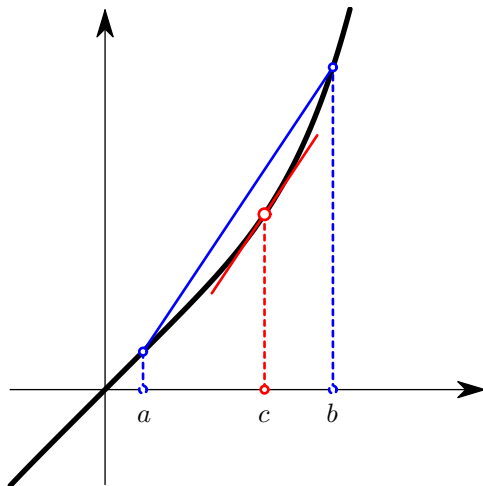


Figure 3. According to the Mean Value Theorem there always is some number c between a and b such that the tangent to the graph of f is parallel to the line segment connecting the two points $(a, f(a))$ and $(b, f(b))$. This is true for any choice of a and b ; c depends on a and b of course.

5.3. The Mean Value Theorem. *If f is a differentiable function on the interval $a \leq x \leq b$, then there is some number c , with $a < c < b$ such that*

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

PROOF OF THEOREM 5.2. We show that $f'(x) > 0$ for all x implies that f is increasing. Let $x_1 < x_2$ be two numbers between a and b . Then the Mean Value Theorem implies that there is some c between x_1 and x_2 such that

$$f'(c) = \frac{f(x_2) - f(x_1)}{x_2 - x_1},$$

or

$$f(x_2) - f(x_1) = f'(c)(x_2 - x_1).$$

Since we know that $f'(c) > 0$ and $x_2 - x_1 > 0$ it follows that $f(x_2) - f(x_1) > 0$, i.e. $f(x_2) > f(x_1)$. □

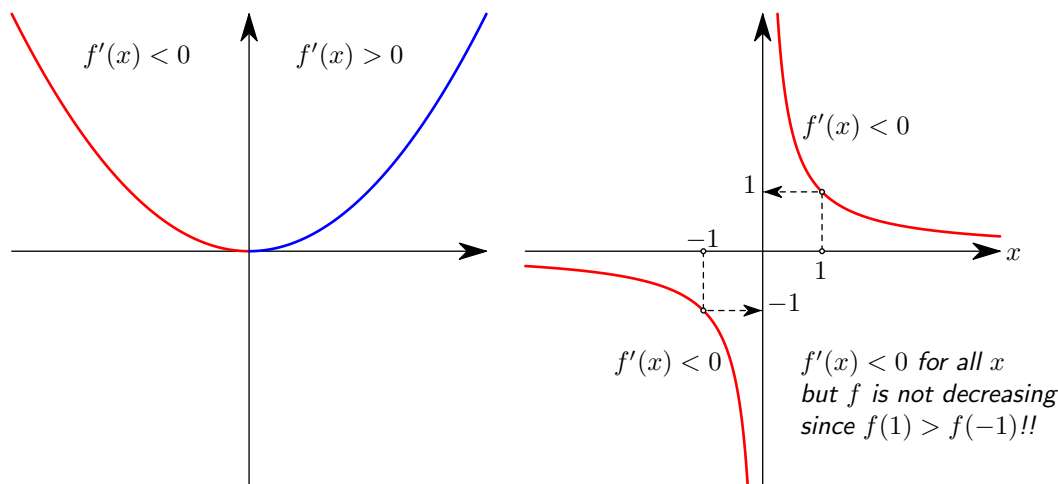
6. Examples

Armed with these theorems we can now split the graph of any function into increasing and decreasing parts simply by computing the derivative $f'(x)$ and finding out where $f'(x) > 0$ and where $f'(x) < 0$ – i.e. we apply the method from the previous section to f' rather than f .

6.1. Example: the parabola $y = x^2$. The familiar graph of $f(x) = x^2$ consists of two parts, one decreasing and one increasing. You can see this from the derivative which is

$$f'(x) = 2x \begin{cases} > 0 & \text{for } x > 0 \\ < 0 & \text{for } x < 0. \end{cases}$$

Therefore the function $f(x) = x^2$ is **decreasing for** $x < 0$ and **increasing for** $x > 0$.



6.2. Example: the hyperbola $y = 1/x$. The derivative of the function $f(x) = 1/x = x^{-1}$ is

$$f'(x) = -\frac{1}{x^2}$$

which is always negative. You would therefore think that this function is decreasing, or at least non-increasing: if $a < b$ then $1/a \geq 1/b$. But this isn't true if you take $a = -1$ and $b = 1$:

$$a = -1 < 1 = b, \text{ but } \frac{1}{a} = -1 < 1 = \frac{1}{b} !!$$

The problem is that we used theorem 5.2, but if you carefully read that theorem then you see that it applies to functions **that are defined on an interval**. The function in this example, $f(x) = 1/x$, is not defined on the interval $-1 < x < 1$ because it isn't defined at $x = 0$. That's why you can't conclude that the $f(x) = 1/x$ is increasing from $x = -1$ to $x = +1$.

On the other hand, the function is defined and differentiable on the interval $0 < x < \infty$, so theorem 5.2 tells us that $f(x) = 1/x$ is decreasing for $x > 0$. This means, that as long as x is positive, increasing x will decrease $1/x$.

6.3. Graph of a cubic function. Consider the function

$$y = f(x) = x^3 - x.$$

Its derivative is

$$f'(x) = 3x^2 - 1.$$

We try to find out where f' is positive, and where it is negative by factoring $f'(x)$

$$f'(x) = 3\left(x^2 - \frac{1}{3}\right) = 3\left(x + \frac{1}{3}\sqrt{3}\right)\left(x - \frac{1}{3}\sqrt{3}\right)$$

from which you see that

$$f'(x) > 0 \text{ for } x < -\frac{1}{3}\sqrt{3}$$

$$f'(x) < 0 \text{ for } -\frac{1}{3}\sqrt{3} < x < \frac{1}{3}\sqrt{3}$$

$$f'(x) > 0 \text{ for } x > \frac{1}{3}\sqrt{3}$$

Therefore the function f is

$$\text{increasing on } \left(-\infty, -\frac{1}{3}\sqrt{3}\right), \quad \text{decreasing on } \left(-\frac{1}{3}\sqrt{3}, \frac{1}{3}\sqrt{3}\right), \quad \text{increasing on } \left(\frac{1}{3}\sqrt{3}, \infty\right).$$

At the two points $x = \pm\frac{1}{3}\sqrt{3}$ one has $f'(x) = 0$ so there the tangent will be horizontal. This leads us to the following picture of the graph of f :

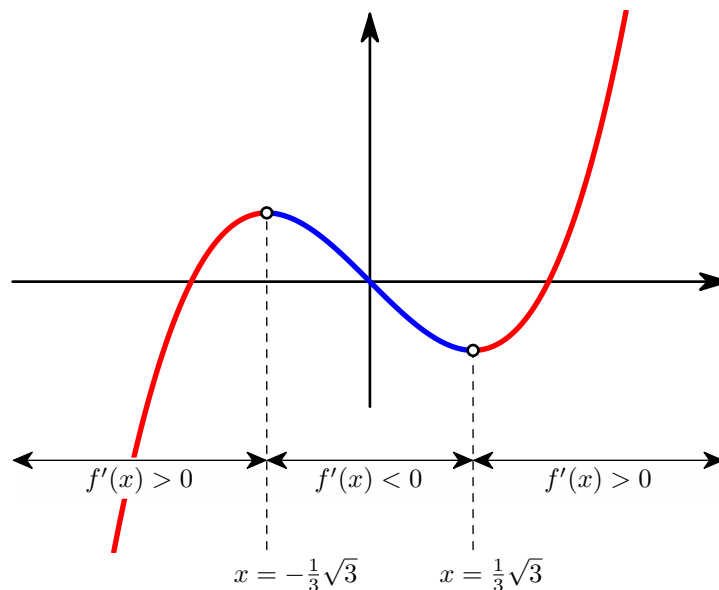


Figure 4. The graph of $f(x) = x^3 - x$.

6.4. A function whose tangent turns up and down infinitely often near the origin. We end with a weird example. Somewhere in the mathematician's zoo of curious functions the following will be on exhibit. Consider the function

$$f(x) = \frac{x}{2} + x^2 \sin \frac{\pi}{x}.$$

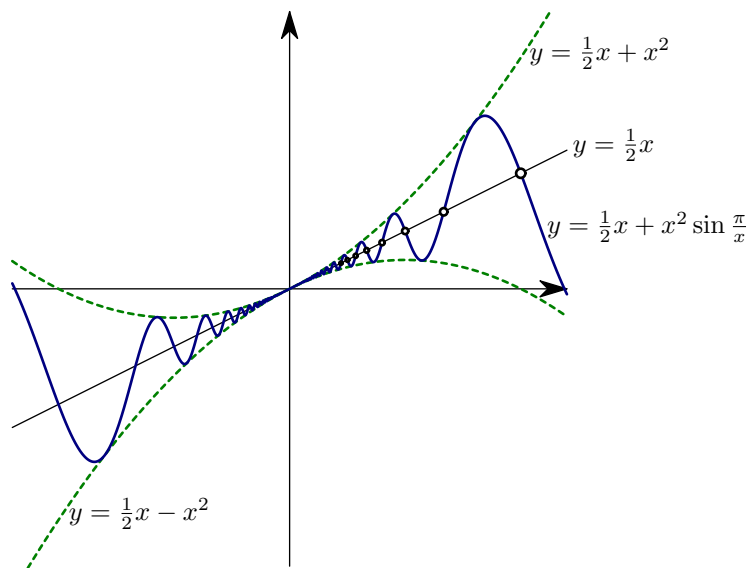


Figure 5. Positive derivative at a point ($x = 0$) does not mean that the function is “increasing near that point.” The slopes at the intersection points alternate between $\frac{1}{2} - \pi$ and $\frac{1}{2} + \pi$.

For $x = 0$ this formula is undefined, and we are free to define $f(0) = 0$. This makes the function continuous at $x = 0$. In fact, this function is differentiable at $x = 0$, with derivative given by

$$f'(0) = \lim_{x \rightarrow 0} \frac{f(x) - f(0)}{x - 0} = \lim_{x \rightarrow 0} \frac{1}{2} + x \sin \frac{\pi}{x} = \frac{1}{2}.$$

(To find the limit apply the sandwich theorem to $-|x| \leq x \sin \frac{\pi}{x} \leq |x|$.)

So the slope of the tangent to the graph at the origin is positive ($\frac{1}{2}$), and one would *think* that the function should be increasing near $x = 0$ (i.e. bigger x gives bigger $f(x)$.) The point of this example is that this turns out not to be true.

To explain why not, we must compute the derivative of this function for $x \neq 0$. It is given by

$$f'(x) = \frac{1}{2} - \pi \cos \frac{\pi}{x} + 2x \sin \frac{\pi}{x}.$$

Now consider the sequence of intersection points $P_1, P_2, \dots$ of the graph with the line $y = x/2$. They are

$$P_k(x_k, y_k), \quad x_k = \frac{1}{k}, \quad y_k = f(x_k).$$

For larger and larger k the points P_k tend to the origin (the x coordinate is $\frac{1}{k}$ which goes to 0 as $k \rightarrow \infty$). The slope of the tangent at P_k is given by

$$\begin{aligned} f'(x_k) &= \frac{1}{2} - \pi \cos \frac{\pi}{1/k} + 2 \frac{1}{k} \sin \frac{\pi}{1/k} \\ &= \frac{1}{2} - \pi \underbrace{\cos k\pi}_{=(-1)^k} + \frac{2}{k} \underbrace{\sin k\pi}_{=0} \\ &= \begin{cases} -\frac{1}{2} - \pi \approx -2.64159265358979 \dots & \text{for } k \text{ even} \\ \frac{1}{2} + \pi \approx +3.64159265358979 \dots & \text{for } k \text{ odd} \end{cases} \end{aligned}$$

In other words, along the sequence of points P_k the slope of the tangent flip-flops between $\frac{1}{2} - \pi$ and $\frac{1}{2} + \pi$, i.e. between a positive and a negative number.

In particular, the slope of the tangent at the odd intersection points is negative, and so you would expect the function to be decreasing there. In other words we see that ***even though the derivative at $x = 0$ of this function is positive, there are points on the graph arbitrarily close to the origin where the tangent has negative slope.***

7. Maxima and Minima

A function has a **global maximum** at some a in its domain if $f(x) \leq f(a)$ for all other x in the domain of f . Global maxima are sometimes also called “absolute maxima.”

A function has a **local maximum** at some a in its domain if there is a small $\delta > 0$ such that $f(x) \leq f(a)$ for all x with $a - \delta < x < a + \delta$ which lie in the domain of f .

Every global maximum is a local maximum, but a local maximum doesn't have to be a global maximum.

7.1. Where to find local maxima and minima. Any x value for which $f'(x) = 0$ is called a **stationary point** for the function f .

7.2. Theorem. Suppose f is a differentiable function on some interval $[a, b]$.

Every local maximum or minimum of f is either one of the end points of the interval $[a, b]$, or else it is a stationary point for the function f .

PROOF. Suppose that f has a local maximum at x and suppose that x is not a or b . By assumption the left and right hand limits

$$f'(x) = \lim_{\Delta x \nearrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x} \quad \text{and} \quad f'(x) = \lim_{\Delta x \searrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

both exist and they are equal.

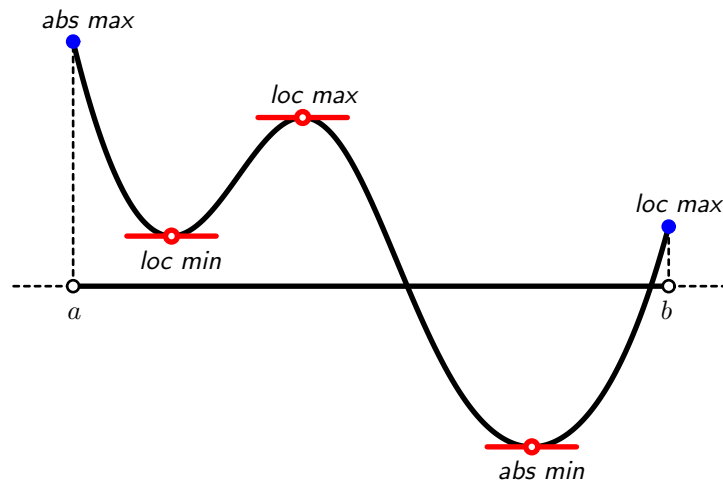


Figure 6. A function defined on an interval $[a, b]$ with one interior absolute minimum, another interior local minimum, an interior local maximum, and two local maxima on the boundary, one of which is in fact an absolute maximum.

Since f has a local maximum at x we have $f(x + \Delta x) - f(x) \leq 0$ if $-\delta < \Delta x < \delta$. In the first limit we also have $\Delta x < 0$, so that

$$\lim_{\Delta x \nearrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x} \leq 0$$

Hence $f'(x) \leq 0$.

In the second limit we have $\Delta x > 0$, so

$$\lim_{\Delta x \searrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x} \geq 0$$

which implies $f'(x) \geq 0$.

Thus we have shown that $f'(x) \leq 0$ and $f'(x) \geq 0$ at the same time. This can only be true if $f'(x) = 0$. $\square$

7.3. How to tell if a stationary point is a maximum, a minimum, or neither. If $f'(c) = 0$ then c is a stationary point (by definition), and it might be local maximum or a local minimum. You can tell what kind of stationary point c is by looking at the signs of $f'(x)$ for x near c .

7.4. Theorem. *If in some small interval $(c - \delta, c + \delta)$ you have $f'(x) < 0$ for $x < c$ and $f'(x) > 0$ for $x > c$ then f has a local minimum at $x = c$.*

If in some small interval $(c - \delta, c + \delta)$ you have $f'(x) > 0$ for $x < c$ and $f'(x) < 0$ for $x > c$ then f has a local maximum at $x = c$. The reason is simple: if f increases to the left of c and decreases to the right of c then it has a maximum at c . More precisely:

if $f'(x) > 0$ for x between $c - \delta$ and c , then f is increasing for $c - \delta < x < c$ and therefore $f(x) < f(c)$ for x between $c - \delta$ and c .

If in addition $f'(x) < 0$ for $x > c$ then f is decreasing for x between c and $c + \delta$, so that $f(x) < f(c)$ for those x .

Combine these two facts and you get $f(x) \leq f(c)$ for $c - \delta < x < c + \delta$.

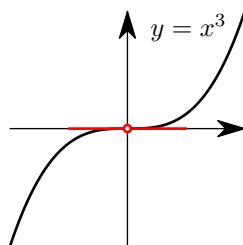
7.5. Example – local maxima and minima of $f(x) = x^3 - x$. In §6.3 we had found that the function $f(x) = x^3 - x$ is decreasing when $-\infty < x < -\frac{1}{3}\sqrt{3}$, and also when $\frac{1}{3}\sqrt{3} < x < \infty$, while it is increasing when $-\frac{1}{3}\sqrt{3} < x < \frac{1}{3}\sqrt{3}$. It follows that the function has a local minimum at $x = -\frac{1}{3}\sqrt{3}$, and a local maximum at $x = \frac{1}{3}\sqrt{3}$.

Neither the local maximum nor the local minimum are global max or min since

$$\lim_{x \rightarrow -\infty} f(x) = +\infty \text{ and } \lim_{x \rightarrow \infty} f(x) = -\infty.$$

7.6. A stationary point that is neither a maximum nor a minimum. If you look for stationary points of the function $f(x) = x^3$ you find that there's only one, namely $x = 0$. The derivative $f'(x) = 3x^2$ does not change sign at $x = 0$, so the test in Theorem 7.4 does not tell us anything.

And in fact, $x = 0$ is neither a local maximum nor a local minimum since $f(x) < f(0)$ for $x < 0$ and $f(x) > 0$ for $x > 0$.



8. Must there always be a maximum?

Theorem 7.2 is very useful since it tells you how to find (local) maxima and minima. The following theorem is also useful, but in a different way. It doesn't say how to find maxima or minima, but it tells you that they do exist, and hence that you are not wasting your time trying to find a maximum or minimum.

8.1. Theorem. *Let f be continuous function defined on the **closed** interval $a \leq x \leq b$. Then f attains its maximum and also its minimum somewhere in this interval. In other words there exist real numbers c and d such that*

$$f(c) \leq f(x) \leq f(d)$$

whenever $a \leq x \leq b$. The proof of this theorem requires a more careful definition of the real numbers than we have given in Chapter 1, and we will take the theorem for granted.

9. Examples – functions with and without maxima or minima

In the following three example we explore what can happen if some of the hypotheses in Theorem 8.1 are not met.

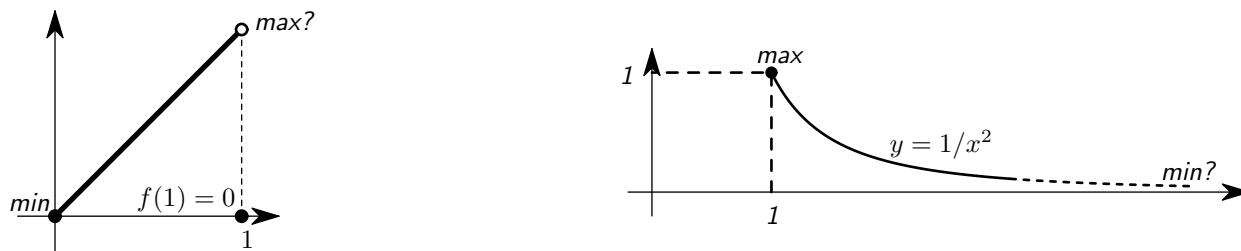


Figure 7. The function on the left has no maximum, and the one on the right has no minimum.

9.1. Question: Does the function

$$f(x) = \begin{cases} x & \text{for } 0 \leq x < 1 \\ 0 & \text{for } x = 1. \end{cases}$$

have a maximum on the interval $0 \leq x \leq 1$?

Answer: No. What would the maximal value be? Since

$$\lim_{x \nearrow 1} f(x) = \lim_{x \nearrow 1} x = 1$$

The maximal value cannot be less than 1. On the other hand the function is never larger than 1. So if there were a number a in the interval $[0, 1]$ such that $f(a)$ was the maximal value of f , then we would have $f(a) = 1$. If you now search the interval for numbers a with $f(a) = 1$, then you notice that such an a does not exist. Conclusion: this function does **not** attain its maximum on the interval $[0, 1]$.

What about Theorem 8.1? That theorem only applies to continuous functions, and the function f in this example is not continuous at $x = 1$. For at $x = 1$ one has

$$f(1) = 0 \neq 1 = \lim_{x \nearrow 1} f(x).$$

So all it takes for the Theorem to fail is that the function f be discontinuous at just one point.

9.2. Question: Does the function

$$f(x) = \frac{1}{x^2}, \quad 1 \leq x < \infty$$

have a maximum or minimum?

Answer: The function has a maximum at $x = 1$, but it has no minimum.

Concerning the maximum: if $x \geq 1$ then $f(x) = 1/x^2 \leq 1$, while $f(1) = 1$. Hence $f(x) \leq f(1)$ for all x in the interval $[1, \infty)$ and that is why f attains its maximum at $x = 1$.

If we look for a minimal value of f then we note that $f(x) \geq 0$ for all x in the interval $[1, \infty)$, and also that

$$\lim_{x \rightarrow \infty} f(x) = 0,$$

so that **if** f attains a minimum at some a with $1 \leq a < \infty$, then the minimal value $f(a)$ must be zero. However, the equation $f(a) = 0$ has no solution – f does not attain its minimum.

Why does Theorem 8.1 not apply? In this example the function f is continuous on the whole interval $[1, \infty)$, but this interval is not a closed interval, i.e. it is not of the form $[a, b]$ (it does not include its endpoints).

10. General method for sketching the graph of a function

Given a differentiable function f defined on some interval $a \leq x \leq b$, you can find the increasing and decreasing parts of the graph, as well as all the local maxima and minima by following this procedure:

- (1) find all solutions of $f'(x) = 0$ in the interval $[a, b]$: these are called the *critical* or *stationary* points for f .
- (2) find the sign of $f'(x)$ at all other points
- (3) each stationary point at which $f'(x)$ actually changes sign is a local maximum or local minimum. Compute the function value $f(x)$ at each stationary point.
- (4) compute the function values at the endpoints of the interval, i.e. compute $f(a)$ and $f(b)$.
- (5) the absolute maximum is attained at the stationary point or the boundary point with the highest function value; the absolute minimum occurs at the boundary or stationary point with the smallest function value.

If the interval is unbounded, i.e. if the function is defined for $-\infty < x < \infty$ then you can't compute the values $f(a)$ and $f(b)$, but instead you should compute $\lim_{x \rightarrow \pm\infty} f(x)$.

10.1. Example – the graph of a rational function. Let's “sketch the graph” of the function

$$f(x) = \frac{x(1-x)}{1+x^2}.$$

By looking at the signs of numerator and denominator we see that

$$\begin{aligned} f(x) &> 0 \text{ for } 0 < x < 1 \\ f(x) &< 0 \text{ for } x < 0 \text{ and also for } x > 1. \end{aligned}$$

We compute the derivative of f

$$f'(x) = \frac{1-2x-x^2}{(1+x^2)^2}.$$

Hence $f'(x) = 0$ holds if and only if

$$1-2x-x^2=0$$

and the solutions to this quadratic equation are $-1 \pm \sqrt{2}$. These two roots will appear several times and it will shorten our formulas if we abbreviate

$$A = -1 - \sqrt{2} \text{ and } B = -1 + \sqrt{2}.$$

To see if the derivative changes sign we factor the numerator and denominator. The denominator is always positive, and the numerator is

$$-x^2 - 2x + 1 = -(x^2 + 2x - 1) = -(x - A)(x - B).$$

Therefore

$$f'(x) \begin{cases} < 0 & \text{for } x < A \\ > 0 & \text{for } A < x < B \\ < 0 & \text{for } x > B \end{cases}$$

It follows that f is decreasing on the interval $(-\infty, A)$, increasing on the interval (A, B) and decreasing again on the interval (B, ∞) . Therefore

A is a local minimum, and B is a local maximum.

Are these global maxima and minima?

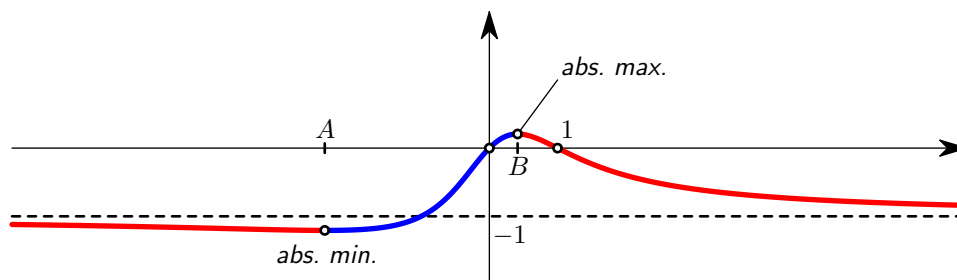


Figure 8. The graph of $f(x) = (x - x^2)/(1 + x^2)$

Since we are dealing with an unbounded interval we must compute the limits of $f(x)$ as $x \rightarrow \pm\infty$. You find

$$\lim_{x \rightarrow \infty} f(x) = \lim_{x \rightarrow -\infty} f(x) = -1.$$

Since f is decreasing between $-\infty$ and A , it follows that

$$f(A) \leq f(x) < -1 \text{ for } -\infty < x \leq A.$$

Similarly, f is decreasing from B to $+\infty$, so

$$-1 < f(x) \leq f(-1 + \sqrt{2}) \text{ for } B < x < \infty.$$



Figure 9. If a graph is convex then all chords lie above the graph. If it is not convex then some chords will cross the graph or lie below it.

Between the two stationary points the function is increasing, so

$$f(-1 - \sqrt{2}) \leq f(x) \leq f(B) \text{ for } A \leq x \leq B.$$

From this it follows that $f(x)$ is the smallest it can be when $x = A = -1 - \sqrt{2}$ and at its largest when $x = B = -1 + \sqrt{2}$: the local maximum and minimum which we found are in fact a global maximum and minimum.

11. Convexity, Concavity and the Second Derivative

By definition, a function f is **convex** on some interval $a < x < b$ if the line segment connecting any pair of points on the graph lies *above* the piece of the graph between those two points.

The function is called **concave** if the line segment connecting any pair of points on the graph lies *below* the piece of the graph between those two points.

A point on the graph of f where $f''(x)$ changes sign is called an **inflection point**.

Instead of “convex” and “concave” one often says “curved upwards” or “curved downwards.”

You can use the second derivative to tell if a function is concave or convex.

11.1. Theorem. A function f is convex on some interval $a < x < b$ if and only if $f''(x) \geq 0$ for all x on that interval.

11.2. Theorem. A function f is concave on some interval $a < x < b$ if and only if the derivative $f'(x)$ is a nondecreasing function on that interval.

A proof using the Mean Value Theorem will be given in class.

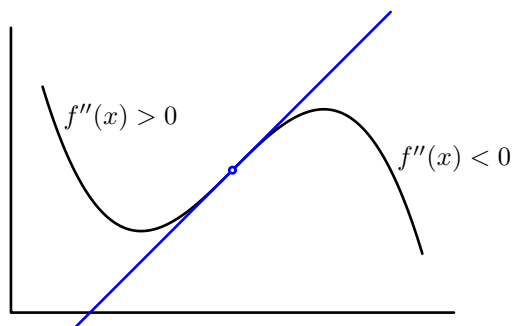


Figure 10. At an inflection point the tangent crosses the graph.

11.3. Example – the cubic function $f(x) = x^3 - x$. The second derivative of the function $f(x) = x^3 - x$ is

$$f''(x) = 6x$$

which is positive for $x > 0$ and negative for $x < 0$. Hence, in the graph in §6.3, the origin is an inflection point, and the piece of the graph where $x > 0$ is convex, while the piece where $x < 0$ is concave.

11.4. The second derivative test. In §7.3 we saw how you can tell if a stationary point is a local maximum or minimum by looking at the sign changes of $f'(x)$. There is another way of distinguishing between local maxima and minima which involves computing the second derivative.

11.5. Theorem. *If c is a stationary point for a function f , and if $f''(c) < 0$ then f has a local maximum at $x = c$.*

If $f''(c) > 0$ then f has a local minimum at c . The theorem doesn't say what happens when $f''(c) = 0$. In that case you must go back to checking the signs of the first derivative near the stationary point.

The basic reason why this theorem is true is that if c is a stationary point with $f''(c) > 0$ then " $f'(x)$ is increasing near $x = c$ " and hence $f'(x) < 0$ for $x < c$ and $f'(x) > 0$ for $x > c$. So the function f is decreasing for $x < c$ and increasing for $x > c$, and therefore it reaches a local minimum at $x = c$.

11.6. Example – that cubic function again. Consider the function $f(x) = x^3 - x$ from §6.3 and §11.3. We had found that this function has two stationary points, namely at $x = \pm\frac{1}{3}\sqrt{3}$. By looking at the sign of $f'(x) = 3x^2 - 1$ we concluded that $-\frac{1}{3}\sqrt{3}$ is a local maximum while $+\frac{1}{3}\sqrt{3}$ is a local minimum. Instead of looking at $f'(x)$ we could also have computed $f''(x)$ at $x = \pm\frac{1}{3}\sqrt{3}$ and applied the second derivative test. Here is how it goes:

Since $f''(x) = 6x$ we have

$$f''(-\frac{1}{3}\sqrt{3}) = -2\sqrt{3} < 0 \text{ and } f''(\frac{1}{3}\sqrt{3}) = 2\sqrt{3} > 0.$$

Therefore f has a local maximum at $-\frac{1}{3}\sqrt{3}$ and a local minimum at $\frac{1}{3}\sqrt{3}$.

11.7. When the second derivative test doesn't work. Usually the second derivative test will work, but sometimes a stationary point c has $f''(c) = 0$. In this case the second derivative test gives no information at all. The figure below shows you the graphs of three functions, all three of which have a stationary point at $x = 0$. In all three cases the second derivative vanishes at $x = 0$ so the second derivative test says nothing. As you can see, the stationary point can be a local maximum, a local minimum, or neither.

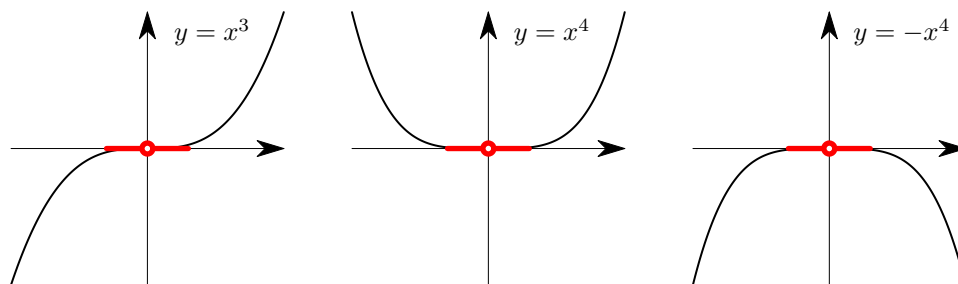


Figure 11. Three functions for which the second derivative test doesn't work.

12. Proofs of some of the theorems

12.1. Proof of the Mean Value Theorem. Let m be the slope of the chord connecting the points $(a, f(a))$ and $(b, f(b))$, i.e.

$$m = \frac{f(b) - f(a)}{b - a},$$

and consider the function

$$g(x) = f(x) - f(a) - m(x - a).$$

This function is continuous (since f is continuous), and g attains its maximum and minimum at two numbers $c_{\min}$ and $c_{\max}$.

There are now two possibilities: either at least one of $c_{\min}$ or $c_{\max}$ is an interior point, or else both $c_{\min}$ and $c_{\max}$ are endpoints of the interval $a \leq x \leq b$.

Consider the first case: one of these two numbers is an interior point, i.e. if $a < c_{\min} < b$ or $a < c_{\max} < b$, then the derivative of g must vanish at $c_{\min}$ or $c_{\max}$. If one has $g'(c_{\min}) = 0$, then one has

$$0 = g'(c_{\min}) = f'(c_{\min}) - m, \text{ i.e. } m = f'(c_{\min}).$$

The definition of m implies that one gets

$$f'(c_{\min}) = \frac{f(b) - f(a)}{b - a}.$$

If $g'(c_{\max}) = 0$ then one gets $m = f'(c_{\max})$ and hence

$$f'(c_{\max}) = \frac{f(b) - f(a)}{b - a}.$$

We are left with the remaining case, in which both $c_{\min}$ and $c_{\max}$ are end points. To deal with this case note that at the endpoints one has

$$g(a) = 0 \text{ and } g(b) = 0.$$

Thus the maximal and minimal values of g are both zero! This means that $g(x) = 0$ for all x , and thus that $g'(x) = 0$ for all x . Therefore we get $f'(x) = m$ for **all** x , and not just for some c .

12.2. Proof of Theorem 5.1. If f is a non-increasing function and if it is differentiable at some interior point a , then we must show that $f'(a) \geq 0$.

Since f is non-decreasing, one has $f(x) \geq f(a)$ for all $x > a$. Hence one also has

$$\frac{f(x) - f(a)}{x - a} \geq 0$$

for all $x > a$. Let $x \searrow a$, and you get

$$f'(a) = \lim_{x \searrow a} \frac{f(x) - f(a)}{x - a} \geq 0.$$

12.3. Proof of Theorem 5.2. Suppose f is a differentiable function on an interval $a < x < b$, and suppose that $f'(x) \geq 0$ on that interval. We must show that f is non-decreasing on that interval, i.e. we have to show that if $x_1 < x_2$ are two numbers in the interval (a, b) , then $f(x_1) \leq f(x_2)$. To prove this we use the Mean Value Theorem: given x_1 and x_2 the Mean Value Theorem hands us a number c with $x_1 < c < x_2$, and

$$f'(c) = \frac{f(x_2) - f(x_1)}{x_2 - x_1}.$$

We don't know where c is exactly, but it doesn't matter because we do know that wherever c is we have $f'(c) \geq 0$. Hence

$$\frac{f(x_2) - f(x_1)}{x_2 - x_1} \geq 0.$$

Multiply with $x_2 - x_1$ (which we are allowed to do since $x_2 > x_1$ so $x_2 - x_1 > 0$) and you get

$$f(x_2) - f(x_1) \geq 0,$$

as claimed.

13. Exercises

218. What does the Intermediate Value Theorem say?

219. What does the Mean Value Theorem say?

220. **Group Problem.**

If $f(a) = 0$ and $f(b) = 0$ then there is a c between a and b such that $f'(c) = 0$. Show that this follows from the Mean Value Theorem. (Help! A proof! Relax: this one is not difficult. Make a drawing of the situation, then read the Mean Value Theorem again.)

221. What is a stationary point?

222. **Group Problem.**

How can you tell if a local maximum is a global maximum?

223. **Group Problem.**

If $f''(a) = 0$ then the graph of f has an inflection point at $x = a$. **True or False?**

224. What is an inflection point?

225. Give an example of a function for which $f'(0) = 0$ even though the function f has neither a local maximum or a local minimum at $x = 0$.

226. **Group Problem.**

Draw four graphs of functions, one for each of the following four combinations

$$\begin{array}{ll} f' > 0 \text{ and } f'' > 0 & f' > 0 \text{ and } f'' < 0 \\ f' < 0 \text{ and } f'' > 0 & f' < 0 \text{ and } f'' < 0 \end{array}$$

227. **Group Problem.**

Which of the following combinations are possible:

$$\begin{array}{l} f'(x) > 0 \text{ and } f''(x) = 0 \text{ for all } x \\ f'(x) = 0 \text{ and } f''(x) > 0 \text{ for all } x \end{array}$$

Sketch the graph of the following functions. You should

- (1) find where f , f' and f'' are positive or negative
- (2) find all stationary points
- (3) decide which stationary points are local maxima or minima
- (4) decide which local max/minima are in fact global max/minima
- (5) find all inflection points
- (6) find "horizontal asymptotes," i.e. compute the limits $\lim_{x \rightarrow \pm\infty} f(x)$ when appropriate.

228. $y = x^3 + 2x^2$

229. $y = x^3 - 4x^2$

230. $y = x^4 + 27x$

231. $y = x^4 - 27x$

232. $y = x^4 + 2x^2 - 3$

233. $y = x^4 - 5x^2 + 4$

234. $y = x^5 + 16x$

235. $y = x^5 - 16x$

236. $y = \frac{x}{x+1}$

237. $y = \frac{x}{1+x^2}$

238. $y = \frac{x^2}{1+x^2}$

239. $y = \frac{1+x^2}{1+x}$

240. $y = x + \frac{1}{x}$

241. $y = x - \frac{1}{x}$

242. $y = x^3 + 2x^2 + x$

243. $y = x^3 + 2x^2 - x$

244. $y = x^4 - x^3 - x$

245. $y = x^4 - 2x^3 + 2x$

246. $y = \sqrt{1+x^2}$

247. $y = \sqrt{1-x^2}$

248. $y = \sqrt[4]{1+x^2}$

249. $y = \frac{1}{1+x^4}$

The following functions are periodic, i.e. they satisfy $f(x+L) = f(x)$ for all x , where the constant L is called the period of the function. The graph of a periodic function repeats itself indefinitely to the left and to the right. It therefore has infinitely many (local) minima and maxima, and infinitely many inflections points. Sketch the graphs of the following functions as in the previous problem, but only list those "interesting points" that lie in the interval $0 \leq x < 2\pi$.

250. $y = \sin x$

251. $y = \sin x + \cos x$

252. $y = \sin x + \sin^2 x$

253. $y = 2 \sin x + \sin^2 x$

254. $y = 4 \sin x + \sin^2 x$

255. $y = 2 \cos x + \cos^2 x$

256. $y = \frac{4}{2 + \sin x}$

257. $y = (2 + \sin x)^2$

Find the domain and sketch the graphs of each of the following functions

258. $y = \arcsin x$

259. $y = \arctan x$

260. $y = 2 \arctan x - x$

261. $y = \arctan(x^2)$

262. $y = 3 \arcsin(x) - 5x$

263. $y = 6 \arcsin(x) - 10x^2$

264. In the following two problems it is not possible to solve the equation $f'(x) = 0$, but you can tell something from the second derivative.

(a) Show that the function $f(x) = x \arctan x$ is convex. Then sketch the graph of f .

(b) Show that the function $g(x) = x \arcsin x$ is convex. Then sketch the graph of g .

For each of the following functions use the derivative to decide if they are increasing, decreasing or neither on the indicated intervals

265. $f(x) = \frac{x}{1+x^2} \quad 10 < x < \infty$

266. $f(x) = \frac{2+x^2}{x^3-x} \quad 1 < x < \infty$

267. $f(x) = \frac{2+x^2}{x^3-x} \quad 0 < x < 1$

268. $f(x) = \frac{2+x^2}{x^3-x} \quad 0 < x < \infty$

14. Optimization Problems

Often a problem can be phrased as

For which value of x in the interval $a \leq x \leq b$ is $f(x)$ the largest?

In other words you are given a function f on an interval $[a, b]$ and you must find all global maxima of f on this interval.

If the function is continuous then according to theorem 8.1 there always is at least one x in the interval $[a, b]$ which maximizes $f(x)$.

If f is differentiable then we know what to do: any local maximum is either a stationary point or one of the end points a and b . Therefore you can find the global maxima by following this recipe:

- (1) Find all stationary points of f ;
- (2) Compute $f(x)$ at each stationary point you found in step (1);
- (3) Compute $f(a)$ and $f(b)$;
- (4) The global maxima are those stationary- or endpoints from steps (2) and (3) which have the largest function value.

Usually there is only one global maximum, but sometimes there can be more.

If you have to *minimize* rather than *maximize* a function, then you must look for global minima. The same recipe works (of course you should look for the smallest function value instead of the largest in step 4.)

The difficulty in optimization problems frequently lies not with the calculus part, but rather with setting up the problem. Choosing which quantity to call x and finding the function f is half the job.

14.1. Example – The rectangle with largest area and given perimeter. Which rectangle has the largest area, among all those rectangles for which the total length of the sides is 1?

Solution: If the sides of the rectangle have lengths x and y , then the total length of the sides is

$$L = x + x + y + y = 2(x + y)$$

and the area of the rectangle is

$$A = xy.$$

So are asked to find the largest possible value of $A = xy$ provided $2(x + y) = 1$. The lengths of the sides can also not be negative, so x and y must satisfy $x \geq 0$, $y \geq 0$.

We now want to turn this problem into a question of the form “maximize a function over some interval.” The quantity which we are asked to maximize is A , but it depends on two variables x and y instead of just one variable. However, the variables x and y are not independent since we are only allowed to consider rectangles with $L = 1$. From this equation we get

$$L = 1 \implies y = \frac{1}{2} - x.$$

Hence we must find the maximum of the quantity

$$A = xy = x\left(\frac{1}{2} - x\right)$$

The values of x which we are allowed to consider are only limited by the requirements $x \geq 0$ and $y \geq 0$, i.e. $x \leq \frac{1}{2}$. So we end up with this problem:

Find the maximum of the function $f(x) = x(\frac{1}{2} - x)$ on the interval $0 \leq x \leq \frac{1}{2}$.

Before we start computing anything we note that the function f is a polynomial so that it is differentiable, and hence continuous, and also that the interval $0 \leq x \leq \frac{1}{2}$ is closed. Therefore the theory guarantees that there is a maximum and our recipe will show us where it is.

The derivative is given by

$$f'(x) = \frac{1}{2} - 2x,$$

and hence the only stationary point is $x = \frac{1}{4}$. The function value at this point is

$$f\left(\frac{1}{4}\right) = \frac{1}{4}\left(\frac{1}{2} - \frac{1}{4}\right) = \frac{1}{16}.$$

At the endpoints one has $x = 0$ or $x = \frac{1}{2}$, which corresponds to a rectangle one of whose sides has length zero. The area of such rectangles is zero, and so this is not the maximal value we are looking for.

We conclude that the largest area is attained by the rectangle whose sides have lengths

$$x = \frac{1}{4}, \text{ and } y = \frac{1}{2} - \frac{1}{4} = \frac{1}{4},$$

i.e. by a square with sides $\frac{1}{4}$.

15. Exercises

269. By definition, the perimeter of a rectangle is the sum of the lengths of its four sides. Which rectangle, of all those whose perimeter is 1, has the smallest area? Which one has the largest area?

270. Which rectangle of area 100in^2 minimizes its height plus two times its length?

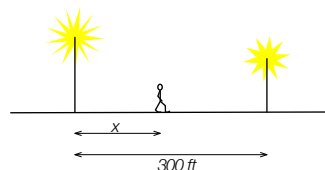
271. You have 1 yard of string from which you make a circular wedge with radius R and opening angle θ . Which choice of θ and R will give you the wedge with the largest area? Which choice leads to the smallest area?

[A circular wedge is the figure consisting of two radii of a circle and the arc connecting them. So the yard of string is used to form the two radii and the arc.]

272. Group Problem.

(The lamp post problem)

In a street two lamp posts are 300 feet apart. The light intensity at a distance d from the first lamp post is $1000/d^2$, the light intensity at distance d from the second (weaker) lamp post is $125/d^2$ (in both cases the light intensity is inversely proportional to the square of the distance to the light source).



The *combined light intensity* is the sum of the two light intensities coming from both lamp posts.

(a) If you are in between the lamp posts, at distance x feet from the stronger light, then give a formula for the combined light intensity coming from both lamp posts as a function of x .

(b) What is the darkest spot between the two lights, i.e. where is the combined light intensity the smallest?

273. (a) You have a sheet of metal with area 100 in^2 from which you are to make a cylindrical soup can. If r is the radius of the can and h its height, then which h and r will give you the can with the largest volume?

(b) If instead of making a plain cylinder you replaced the flat top and bottom of the cylinder with two spherical

caps, then (using the same 100in^2 of sheet metal), then which choice of radius and height of the cylinder give you the container with the largest volume?

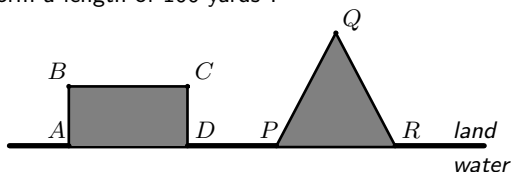
(c) Suppose you only replace the top of the cylinder with a spherical cap, and leave the bottom flat, then which choice of height and radius of the cylinder result in the largest volume?

274. A triangle has one vertex at the origin $O(0,0)$, another at the point $A(2a,0)$ and the third at $(a, a/(1+a^3))$. What are the largest and smallest areas this triangle can have if $0 \leq a < \infty$?

275. Group Problem.

Queen Dido's problem

According to tradition Dido was the founder and first Queen of Carthage. When she arrived on the north coast of Africa (~800BC) the locals allowed her to take as much land as could be enclosed with the hide of one ox. She cut the hide into thin strips and put these together to form a length of 100 yards¹.



(a) If Dido wanted a rectangular region, then how wide should she choose it to enclose as much area as possible (the coastal edge of the boundary doesn't count, so in this problem the length $AB + BC + CD$ is 100 yards.)

(b) If Dido chose a region in the shape of an isosceles triangle PQR , then how wide should she make it to maximize its area (again, don't include the coast in the perimeter: $PQ + QR$ is 100 yards long, and $PQ = QR$.)

276. The product of two numbers x, y is 16. We know $x \geq 1$ and $y \geq 1$. What is the greatest possible sum of the two numbers?

277. What are the smallest and largest values that $(\sin x)(\sin y)$ can have if $x + y = \pi$ and if x and y are both nonnegative?

278. What are the smallest and largest values that $(\cos x)(\cos y)$ can have if $x + y = \frac{\pi}{2}$ and if x and y are both nonnegative?

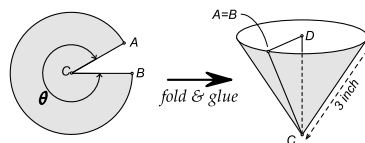
279. (a) What are the smallest and largest values that $\tan x + \tan y$ can have if $x + y = \frac{\pi}{2}$ and if x and y are both nonnegative?

(b) What are the smallest and largest values that $\tan x + 2 \tan y$ can have if $x + y = \frac{\pi}{2}$ and if x and y are both nonnegative?

280. The cost per hour of fuel to run a locomotive is $v^2/25$ dollars, where v is speed (in miles per hour), and other costs are \$100 per hour regardless of speed. What is the speed that minimizes cost per mile?

281. Group Problem.

Josh is in need of coffee. He has a circular filter with 3 inch radius. He cuts out a wedge and glues the two edges AC and BC together to make a conical filter to hold the ground coffee. The volume V of the coffee cone depends the angle θ of the piece of filter paper Josh made.



(a) Find the volume in terms of the angle θ . (Hint: how long is the circular arc AB on the left? How long is the circular top of the cone on the right? If you know that you can find the radius $AD = BD$ of the top of the cone, and also the height CD of the cone.)

(b) Which angle θ maximizes the volume V ?

¹I made that number up. For the rest start at <http://en.wikipedia.org/wiki/Dido>

Exponentials and Logarithms (naturally)

In this chapter we first recall some facts about exponentials (x^y with $x > 0$ and y arbitrary): they should be familiar from algebra, or “precalculus.” What is new is perhaps the definition of x^y when y is not a fraction: e.g., $2^{3/4}$ is the 4th root of the third power of 2 ($\sqrt[4]{2^3}$), but what is $2^{\sqrt{2}}$?

Then we ask “what is the derivative of $f(x) = a^x$?” The answer leads us to the famous number $e \approx 2.718\,281\,828\,459\,045\,235\,360\,287\,471\,352\,662\,497\,757\,247\,093\,699\,95\dots$.

Finally, we compute the derivative of $f(x) = \log_a x$, and we look at things that “grow exponentially.”

1. Exponents

Here we go over the definition of x^y when x and y are arbitrary real numbers, with $x > 0$.

For any real number x and any positive integer $n = 1, 2, 3, \dots$ one defines

$$x^n = \overbrace{x \cdot x \cdot \dots \cdot x}^{n \text{ times}}$$

and, if $x \neq 0$,

$$x^{-n} = \frac{1}{x^n}.$$

One defines $x^0 = 1$ for any $x \neq 0$.

To define $x^{p/q}$ for a general fraction $\frac{p}{q}$ one must assume that the number x is positive. One then defines

$$(37) \quad x^{p/q} = \sqrt[q]{x^p}.$$

This does not tell us how to define x^a if the exponent a is not a fraction. One can define x^a for irrational numbers a by taking limits. For example, to define $2^{\sqrt{2}}$, we look at the sequence of numbers you get by truncating the decimal expansion of $\sqrt{2}$, i.e.

$$a_1 = 1, \quad a_2 = 1.4 = \frac{14}{10}, \quad a_3 = 1.41 = \frac{141}{100}, \quad a_4 = 1.414 = \frac{1414}{1000}, \quad \dots$$

Each a_n is a fraction, so that we know what 2^{a_n} is, e.g. $2^{a_4} = \sqrt[1000]{2^{1414}}$. Our definition of $2^{\sqrt{2}}$ then is

$$2^{\sqrt{2}} = \lim_{n \rightarrow \infty} 2^{a_n},$$

i.e. we define $2^{\sqrt{2}}$ as the limit of the sequence of numbers

$$2, \sqrt[10]{2^{14}}, \sqrt[100]{2^{141}}, \sqrt[1000]{2^{1414}}, \dots$$

(See table 1.)

Here one ought to prove that this limit exists, and that its value does not depend on the particular choice of numbers a_n tending to a . We will not go into these details in this course.

It is shown in precalculus texts that the exponential functions satisfy the following properties:

$$(38) \quad \boxed{x^a x^b = x^{a+b}, \quad \frac{x^a}{x^b} = x^{a-b}, \quad (x^a)^b = x^{ab}}$$

provided a and b are fractions. One can show that these properties still hold if a and b are real numbers (not necessarily fractions.) Again, we won’t go through the proofs here.

x	2^x
1.0000000000	2.000000000000
1.4000000000	2.639015821546
1.4100000000	2.657371628193
1.4140000000	2.664749650184
1.4142000000	2.665119088532
1.4142100000	2.665137561794
1.4142130000	2.665143103798
1.4142135000	2.665144027466
$\vdots$	$\vdots$

Table 1. Approximating $2^{\sqrt{2}}$. Note that as x gets closer to $\sqrt{2}$ the quantity 2^x appears to converge to some number. This limit is our definition of $2^{\sqrt{2}}$.

Now instead of considering x^a as a function of x we can pick a positive number a and consider the function $f(x) = a^x$. This function is defined for all real numbers x (as long as the base a is positive.).

1.1. The trouble with powers of negative numbers. The cube root of a negative number is well defined. For instance $\sqrt[3]{-8} = -2$ because $(-2)^3 = -8$. In view of the definition (37) of $x^{p/q}$ we can write this as

$$(-8)^{1/3} = \sqrt[3]{(-8)^1} = \sqrt[3]{-8} = -2.$$

But there is a problem: since $\frac{2}{6} = \frac{1}{3}$ you would think that $(-8)^{2/6} = (-8)^{1/3}$. However our definition (37) tells us that

$$(-8)^{2/6} = \sqrt[6]{(-8)^2} = \sqrt[6]{+64} = +2.$$

Another example:

$$(-4)^{1/2} = \sqrt{-4} \text{ is not defined}$$

but, even though $\frac{1}{2} = \frac{2}{4}$,

$$(-4)^{2/4} = \sqrt[4]{(-4)^2} = \sqrt[4]{+16} = 2 \text{ is defined.}$$

There are two ways out of this mess:

- (1) avoid taking fractional powers of negative numbers
- (2) when you compute $x^{p/q}$ first simplify the fraction by removing common divisors of p and q .

The safest is just not to take fractional powers of negative numbers.

Given that fractional powers of negative numbers cause all these headaches it is not surprising that we didn't try to define x^a for negative x if a is irrational. For example, $(-8)^\pi$ is not defined¹.

2. Logarithms

Briefly, $y = \log_a x$ is the inverse function to $y = a^x$. This means that, by definition,

$$y = \log_a x \iff x = a^y.$$

In other words, $\log_a x$ is the answer to the question "for which number y does one have $x = a^y$?" The number $\log_a x$ is called **the logarithm with base a of x** . In this definition both a and x must be positive.

For instance,

$$2^3 = 8, \quad 2^{1/2} = \sqrt{2}, \quad 2^{-1} = \frac{1}{2}$$

so

$$\log_2 8 = 3, \quad \log_2(\sqrt{2}) = \frac{1}{2}, \quad \log_2 \frac{1}{2} = -1.$$

¹There is a definition of $(-8)^\pi$ which uses complex numbers. You will see this next semester if you take math 222.

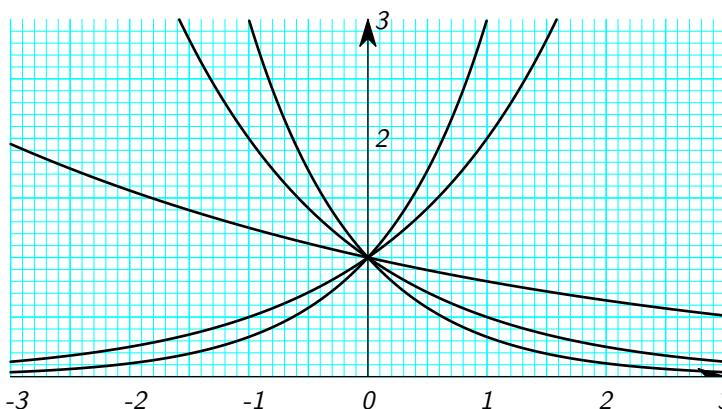


Figure 1. The graphs of $y = 2^x, 3^x, (1/2)^x, (0.1)^x$ and $y = (4/5)^x$. The graphs are purposely not labeled: can you figure out which is which?

Also:

$\log_2(-3)$ doesn't exist

because there is no number y for which $2^y = -3$ (2^y is always positive) and

$\log_{-3} 2$ doesn't exist either

because $y = \log_{-3} 2$ would have to be some real number which satisfies $(-3)^y = 2$, and we don't take non-integer powers of negative numbers.

3. Properties of logarithms

In general one has

$$\log_a a^x = x, \text{ and } a^{\log_a x} = x.$$

There is a subtle difference between these formulas: the first one holds for all real numbers x , but the second only holds for $x > 0$, since $\log_a x$ doesn't make sense for $x \leq 0$.

Again, one finds the following formulas in precalculus texts:

$$(39) \quad \begin{array}{l} \log_a xy = \log_a x + \log_a y \\ \log_a \frac{x}{y} = \log_a x - \log_a y \\ \log_a x^y = y \log_a x \\ \log_a x = \frac{\log_b x}{\log_b a} \end{array}$$

They follow from (38).

4. Graphs of exponential functions and logarithms

Figure 1 shows the graphs of some exponential functions $y = a^x$ with different values of a , and figure 2 shows the graphs of $y = \log_2 x$, $y = \log_3 x$, $\log_{1/2} x$, $\log_{1/3}(x)$ and $y = \log_{10} x$. Can you tell which is which? (Yes, you can.)

From algebra/precalc recall:

If $a > 1$ then $f(x) = a^x$ is an increasing function.

and

If $0 < a < 1$ then $f(x) = a^x$ is a decreasing function.

In other words, for $a > 1$ it follows from $x_1 < x_2$ that $a^{x_1} < a^{x_2}$; if $0 < a < 1$, then $x_1 < x_2$ implies $a^{x_1} > a^{x_2}$.

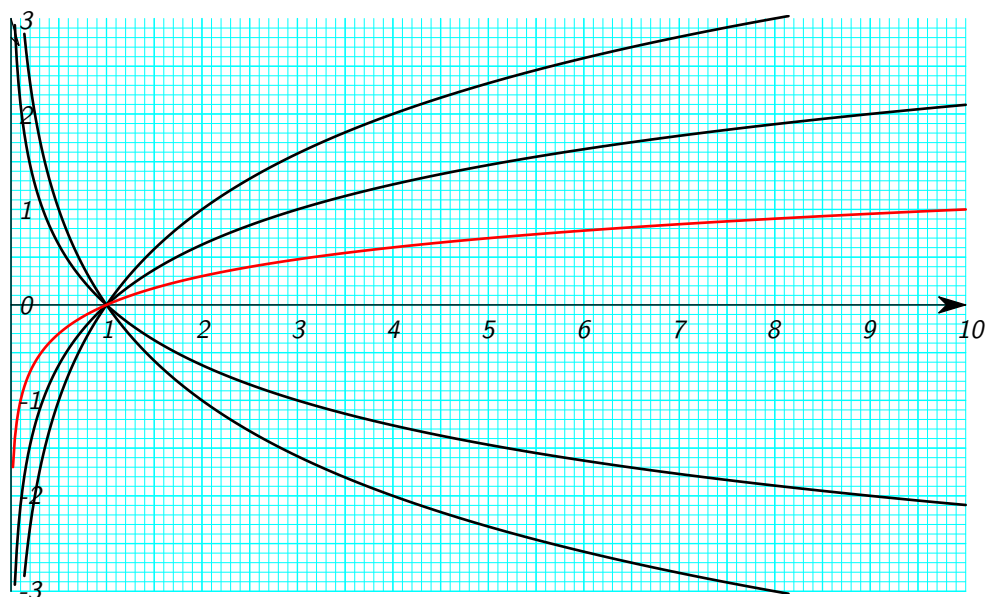


Figure 2. Graphs of some logarithms. Each curve is the graph of a function $y = \log_a x$ for various values of $a > 0$. Can you tell what a is for each graph?

5. The derivative of a^x and the definition of e

To begin, we try to differentiate the function $y = 2^x$:

$$\begin{aligned} \frac{d2^x}{dx} &= \lim_{\Delta x \rightarrow 0} \frac{2^{x+\Delta x} - 2^x}{\Delta x} \\ &= \lim_{\Delta x \rightarrow 0} \frac{2^x 2^{\Delta x} - 2^x}{\Delta x} \\ &= \lim_{\Delta x \rightarrow 0} 2^x \frac{2^{\Delta x} - 1}{\Delta x} \\ &= 2^x \lim_{\Delta x \rightarrow 0} \frac{2^{\Delta x} - 1}{\Delta x}. \end{aligned}$$

So if we assume that the limit

$$\lim_{\Delta x \rightarrow 0} \frac{2^{\Delta x} - 1}{\Delta x} = C$$

exists then we have

$$(40) \quad \frac{d2^x}{dx} = C2^x.$$

On your calculator you can compute $\frac{2^{\Delta x} - 1}{\Delta x}$ for smaller and smaller values of Δx , which leads you to suspect that the limit actually exists, and that $C \approx 0.693147 \dots$. One can in fact prove that the limit exists, but we will not do this here.

Once we know (40) we can compute the derivative of a^x for any other positive number a . To do this we write $a = 2^{\log_2 a}$, and hence

$$a^x = (2^{\log_2 a})^x = 2^{x \cdot \log_2 a}.$$

By the chain rule we therefore get

$$\begin{aligned}\frac{da^x}{dx} &= \frac{d2^{x \cdot \log_2 a}}{dx} \\ &= C 2^{x \cdot \log_2 a} \frac{dx \cdot \log_2 a}{dx} \\ &= (C \log_2 a) 2^{x \cdot \log_2 a} \\ &= (C \log_2 a) a^x.\end{aligned}$$

So the derivative of a^x is just some constant times a^x , the constant being $C \log_2 a$. This is essentially our formula for the derivative of a^x , but one can make the formula look nicer by introducing a special number, namely, we define

$$e = 2^{1/C} \text{ where } C = \lim_{\Delta x \rightarrow 0} \frac{2^{\Delta x} - 1}{\Delta x}.$$

One has

$$e \approx 2.718\,281\,818\,459 \dots$$

This number is special because if you set $a = e$, then

$$C \log_2 a = C \log_2 e = C \log_2 2^{1/C} = C \cdot \frac{1}{C} = 1,$$

and therefore the derivative of the function $y = e^x$ is

$$(41) \quad \boxed{\frac{de^x}{dx} = e^x}.$$

Read that again: the function e^x is its own derivative!

The logarithm with base e is called the **Natural Logarithm**, and is written

$$\ln x = \log_e x.$$

Thus we have

$$(42) \quad \boxed{e^{\ln x} = x \quad \ln e^x = x}$$

where the second formula holds for all real numbers x but the first one only makes sense for $x > 0$.

For any positive number a we have $a = e^{\ln a}$, and also

$$a^x = e^{x \ln a}.$$

By the chain rule you then get

$$(43) \quad \boxed{\frac{da^x}{dx} = a^x \ln a}.$$

6. Derivatives of Logarithms

Since the natural logarithm is the inverse function of $f(x) = e^x$ we can find its derivative by implicit differentiation. Here is the computation (which you should do yourself)

The function $f(x) = \log_a x$ satisfies

$$a^{f(x)} = x$$

Differentiate both sides, and use the chain rule on the left,

$$(\ln a) a^{f(x)} f'(x) = 1.$$

Then solve for $f'(x)$ to get

$$f'(x) = \frac{1}{(\ln a) a^{f(x)}}.$$

Finally we remember that $a^{f(x)} = x$ which gives us the derivative of a^x

$$\frac{da^x}{dx} = \frac{1}{x \ln a}.$$

In particular, the natural logarithm has a very simple derivative, namely, since $\ln e = 1$ we have

$$(44) \quad \boxed{\frac{d \ln x}{dx} = \frac{1}{x}}.$$

7. Limits involving exponentials and logarithms

7.1. Theorem. *Let r be any real number. Then, if $a > 1$,*

$$\lim_{x \rightarrow \infty} x^r a^{-x} = 0,$$

i.e.

$$\lim_{x \rightarrow \infty} \frac{x^r}{a^x} = 0.$$

This theorem says that *any exponential will beat any power of x as $x \rightarrow \infty$* . For instance, as $x \rightarrow \infty$ both x^{1000} and $(1.001)^x$ go to infinity, but

$$\lim_{x \rightarrow \infty} \frac{x^{1000}}{(1.001)^x} = 0,$$

so, in the long run, for very large x , 1.001^x will be much larger than 1000^x .

PROOF WHEN $a = e$. We want to show $\lim_{x \rightarrow \infty} x^r e^{-x} = 0$. To do this consider the function $f(x) = x^{r+1} e^{-x}$. Its derivative is

$$f'(x) = \frac{dx^{r+1}e^{-x}}{dx} = ((r+1)x^r - x^{r+1})e^{-x} = (r+1-x)x^r e^{-x}.$$

Therefore $f'(x) < 0$ for $x > r+1$, i.e. $f(x)$ is decreasing for $x > r+1$. It follows that $f(x) < f(r+1)$ for all $x > r+1$, i.e.

$$x^{r+1} e^{-x} < (r+1)^{r+1} e^{-(r+1)} \text{ for } x > r+1.$$

Divide by x , abbreviate $A = (r+1)^{r+1} e^{-(r+1)}$, and we get

$$0 < x^r e^{-x} < \frac{A}{x} \text{ for all } x > r+1.$$

The Sandwich Theorem implies that $\lim_{x \rightarrow \infty} x^r e^{-x} = 0$, which is what we had promised to show. □

Here are some related limits:

$$a > 1 \implies \lim_{x \rightarrow \infty} \frac{a^x}{x^r} = \infty \quad (D.N.E.)$$

$$m > 0 \implies \lim_{x \rightarrow \infty} \frac{\ln x}{x^m} = 0$$

$$m > 0 \implies \lim_{x \rightarrow 0} x^m \ln x = 0$$

The second limit says that even though $\ln x$ becomes infinitely large as $x \rightarrow \infty$, it is always much less than any power x^m with $m > 0$ real. To prove it you set $x = e^t$ and then $t = s/m$, which leads to

$$\lim_{x \rightarrow \infty} \frac{\ln x}{x^m} \stackrel{x=e^t}{=} \lim_{t \rightarrow \infty} \frac{t}{e^{mt}} \stackrel{t=s/m}{=} \frac{1}{m} \lim_{t \rightarrow \infty} \frac{s}{e^s} = 0.$$

The third limit follows from the second by substituting $x = 1/y$ and using $\ln \frac{1}{x} = -\ln x$.

8. Exponential growth and decay

A quantity X which depends on time t is said to grow or decay exponentially if it is given by

$$(45) \quad X(t) = X_0 e^{kt}.$$

The constant X_0 is the value of $X(t)$ at time $t = 0$ (sometimes called “the initial value of X ”).

The derivative of an exponentially growing quantity, i.e. its rate of change with time, is given by $X'(t) = X_0 k e^{kt}$ so that

$$(46) \quad \frac{dX(t)}{dt} = kX(t).$$

In words, *for an exponentially growing quantity the rate of change is always proportional to the quantity itself*. The proportionality constant is k and is sometimes called “the relative growth rate.”

This property of exponential functions completely describes them, by which I mean that any function which satisfies (46) automatically satisfies (45). To see that this is true, suppose you have a function $X(t)$ for which $X'(t) = kX(t)$ holds at all times t . Then

$$\begin{aligned} \frac{dX(t)e^{-kt}}{dt} &= X(t) \frac{de^{-kt}}{dt} + \frac{dX(t)}{dt} e^{-kt} \\ &= -kX(t)e^{-kt} + X'(t)e^{-kt} \\ &= (X'(t) - kX(t))e^{-kt} \\ &= 0. \end{aligned}$$

It follows that $X(t)e^{-kt}$ does not depend on t . At $t = 0$ one has

$$X(t)e^{-kt} = X(0)e^0 = X_0$$

and therefore we have

$$X(t)e^{-kt} = X_0 \text{ for all } t.$$

Multiply with e^{kt} and we end up with

$$X(t) = X_0 e^{kt}.$$

8.1. Half time and doubling time. If $X(t) = X_0 e^{kt}$ then one has

$$X(t+T) = X_0 e^{kt+ktT} = X_0 e^{kt} e^{kT} = e^{kT} X(t).$$

In words, after time T goes by an exponentially growing (decaying) quantity changes by a factor e^{kT} . If $k > 0$, so that the quantity is actually growing, then one calls

$$T = \frac{\ln 2}{k}$$

the **doubling time** for X because $X(t)$ changes by a factor $e^{kT} = e^{\ln 2} = 2$ every T time units: $X(t)$ doubles every T time units.

If $k < 0$ then $X(t)$ is decaying and one calls

$$T = \frac{\ln 2}{-k}$$

the **half life** because $X(t)$ is reduced by a factor $e^{kT} = e^{-\ln 2} = \frac{1}{2}$ every T time units.

8.2. Determining X_0 and k . The general exponential growth/decay function (45) contains only two constants, X_0 and k , and if you know the values of $X(t)$ at two different times then you can compute these constants.

Suppose that you know

$$X_1 = X(t_1) \text{ and } X_2 = X(t_2).$$

Then we have

$$X_0 e^{kt_1} = X_1 \text{ and } X_2 = X_0 e^{kt_2}$$

in which t_1, t_2, X_1, X_2 are given and k and X_0 are unknown. One first finds k from

$$\frac{X_1}{X_2} = \frac{X_0 e^{kt_1}}{X_0 e^{kt_2}} = e^{k(t_1 - t_2)} \implies \ln \frac{X_1}{X_2} = k(t_1 - t_2)$$

which implies

$$k = \frac{\ln X_1 - \ln X_2}{t_1 - t_2}.$$

Once you have computed k you can find X_0 from

$$X_0 = \frac{X_1}{e^{kt_1}} = \frac{X_2}{e^{kt_2}}.$$

(both expressions should give the same result.)

9. Exercises

Sketch the graphs of the following functions.

(Hint for some of these: if you have to solve something like $e^{4x} - 3e^{3x} + e^x = 0$, then call $w = e^x$, and you get a polynomial equation for w , namely $w^4 - 3w^3 + w = 0$.)

282. $y = e^x$

283. $y = e^{-x}$

284. $y = e^x + e^{-2x}$

285. $y = e^{3x} - 4e^x$

286. $y = \frac{e^x}{1 + e^x}$

287. $y = \frac{2e^x}{1 + e^{2x}}$

288. $y = xe^{-x}$

289. $y = \sqrt{x}e^{-x/4}$

290. $y = x^2 e^{x+2}$

291. $y = e^{x/2} - x$

292. $y = \ln \sqrt{x}$

293. $y = \ln \frac{1}{x}$

294. $y = x \ln x$

295. $y = \frac{-1}{\ln x} \quad (0 < x < \infty, x \neq 1)$

296. $y = (\ln x)^2 \quad (x > 0)$

297. $y = \frac{\ln x}{x} \quad (x > 0)$

298. $y = \ln \sqrt{\frac{1+x}{1-x}} \quad (|x| < 1)$

299. $y = \ln(1 + x^2)$

300. $y = \ln(x^2 - 3x + 2) \quad (x > 2)$

301. $y = \ln \cos x \quad (|x| < \frac{\pi}{2})$

302. The function $f(x) = e^{-x^2}$ plays a central role in statistics and its graph is called **the bell curve** (because of its shape). Sketch the graph of f .

303. Sketch the part of the graph of the function

$$f(x) = e^{-\frac{1}{x}}$$

with $x > 0$.

Find the limits

$$\lim_{x \searrow 0} \frac{f(x)}{x^n} \text{ and } \lim_{x \rightarrow \infty} f(x)$$

where n can be any positive integer (hint: substitute $x = \dots$?)

304. A **damped oscillation** is a function of the form

$$f(x) = e^{-ax} \cos bx \text{ or } f(x) = e^{-ax} \sin bx$$

where a and b are constants.

Sketch the graph of $f(x) = e^{-x} \sin 10x$ (i.e. find zeroes, local max and mins, inflection points) and draw (with pencil on paper) the piece of the graph with $0 \leq x \leq 2\pi$.

This function has many local maxima and minima. What is the ratio between the function values at two consecutive local maxima? (Hint: the answer does not depend on which pair of consecutive local maxima you consider.)

305. Find the inflection points on the graph of $f(x) = (1+x)\ln x$ ($x > 0$).

306. (a) If x is large, which is bigger: 2^x or x^2 ?
 (b) The graphs of $f(x) = x^2$ and $g(x) = 2^x$ intersect at $x = 2$ (since $2^2 = 2^2$). How many more intersections do these graphs have (with $-\infty < x < \infty$)?

Find the following limits.

307. $\lim_{x \rightarrow \infty} \frac{e^x - 1}{e^x + 1}$

308. $\lim_{x \rightarrow \infty} \frac{e^x - x^2}{e^x + x}$

309. $\lim_{x \rightarrow \infty} \frac{2^x}{3^x - 2^x}$

310. $\lim_{x \rightarrow \infty} \frac{e^x - x^2}{e^{2x} + e^{-x}}$

311. $\lim_{x \rightarrow \infty} \frac{e^{-x} - e^{-x/2}}{\sqrt{e^x + 1}}$

312. $\lim_{x \rightarrow \infty} \frac{\sqrt{x + e^{4x}}}{e^{2x} + x}$

313. $\lim_{x \rightarrow \infty} \frac{e^{\sqrt{x}}}{\sqrt{e^x + 1}}$

314. $\lim_{x \rightarrow \infty} \ln(1+x) - \ln x$

315. $\lim_{x \rightarrow \infty} \frac{\ln x}{\ln x^2}$

316. $\lim_{x \rightarrow 0} x \ln x$

317. $\lim_{x \rightarrow \infty} \frac{\ln x}{\sqrt{x} + \ln x}$

318. $\lim_{x \rightarrow 0} \frac{\ln x}{\sqrt{x} + \ln x}$

319. Find the tenth derivative of xe^x .

320. For which real number x is $2^x - 3^x$ the largest?

321. Find $\frac{dx^x}{dx}$, $\frac{dx^{x^x}}{dx}$, and $\frac{d(x^x)^x}{dx}$.

Hint: $x^x = e^{x \ln x}$.

322. Group Problem.

About logarithmic differentiation:

(a) Let $y = (x+1)^2(x+3)^4(x+5)^6$ and $u = \ln y$. Find du/dx . Hint: Use the fact that $\ln$ converts multiplication to addition before you differentiate. It will simplify the calculation.

(b) Check that the derivative of $\ln u(x)$ is the logarithmic derivative of the function u (as defined in the exercises following §25, chapter 4.)

323. After 3 days a sample of radon-222 decayed to 58% of its original amount.

(a) What is the half life of radon-222?

(b) How long would it take the sample to decay to 10% of its original amount?

324. Polonium-210 has a half life of 140 days.

(a) If a sample has a mass of 200 mg find a formula for the mass that remains after t days.

(b) Find the mass after 100 days.

(c) When will the mass be reduced to 10 mg?

(d) Sketch the graph of the mass as a function of time.

325. Current agricultural experts believe that the world's farms can feed about 10 billion people. The 1950 world population was 2.517 billion and the 1992 world population was 5.4 billion. When can we expect to run out of food?

326. Group Problem.

The ACME company runs two ads on Sunday mornings. One says that "when this baby is old enough to vote, the world will have one billion new mouths to feed" and the other says "in thirty six years, the world will have to set eight billion places at the table." What does ACME think the population of the world is at present? How fast does ACME think the population is increasing? Use units of billions of people so you can write 8 instead of 8,000,000,000. (Hint: $36 = 2 \times 18$.)

327. The population of California grows exponentially at an instantaneous rate of 2% per year. The population of California on January 1, 2000 was 20,000,000.

(a) Write a formula for the population $N(t)$ of California t years after January 1, 2000.

(b) Each Californian consumes pizzas at the rate of 70 pizzas per year. At what rate is California consuming pizzas t years after 1990?

(c) How many pizzas were consumed in California from January 1, 2005 to January 1, 2009?

328. The population of the country of Farfarawaï grows exponentially.

(a) If its population in the year 1980 was 1,980,000 and its population in the year 1990 was 1,990,000, what is its population in the year 2000?

(b) How long will it take the population to double? (Your answer may be expressed in terms of exponentials and natural logarithms.)

329. The hyperbolic functions are defined by

$$\sinh x = \frac{e^x - e^{-x}}{2},$$

$$\cosh x = \frac{e^x + e^{-x}}{2},$$

$$\tanh x = \frac{\sinh x}{\cosh x}.$$

(a) Prove the following identities

$$\cosh^2 x - \sinh^2 x = 1$$

$$\cosh 2x = \cosh^2 x + \sinh^2 x$$

$$\sinh 2x = 2 \sinh x \cosh x.$$

(b) Show that

$$\begin{aligned}\frac{d \sinh x}{dx} &= \cosh x, \\ \frac{d \cosh x}{dx} &= \sinh x, \\ \frac{d \tanh x}{dx} &= \frac{1}{\cosh^2 x}.\end{aligned}$$

(c) Sketch the graphs of the three hyperbolic functions.